



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

Subject: SECR 1213 Network Communication

Section: 01

Task 6: Documentation and Reflection

Lecturer: Dr. Muhammad Zafran bin Muhammad Zaly Shah

Date: 14/01/2024

Group Name: Beauties

Members' Name & Matric Number:

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REPORT ABSTRACT

The Group Report for Task 6 encapsulates the comprehensive efforts of our team in designing and implementing a scalable and efficient network infrastructure for the Faculty of Computing (FC). This work combines the theoretical understanding obtained in class with practical implications. The project is centered on the contemporary and possible update of the FC structure to meet current and future technological needs. Starting from the first task, our team worked together to develop a functional floor plan to integrate laboratories, video conferencing room, and common areas in line with the development strategy of FC.

In task 2, a detailed preliminary analysis was conducted and included a set of well-developed and highly determined questions that addressed the required user analysis and the consideration of the feasibility and relevance of the project relative to the organizational goals. In task 3, we focused on the identification of LAN equipment to come up with the most affordable, reliable and scalable products for use in an academic environment. Task 4 covered the physical topology of the different cables used and the layout of the equipment and its interconnection to provide neat functionality for the organization. In task 5, the design of the IP addressing system that will ensure the identification and unique connection of each host in the network was determined.

The final section of the report contains the thoughts and experiences of each team member, the health work of any nuisances observed and itemizes the applicability of improvements. Additional documentation supporting our conference and research papers establishes the depth and authority of our work. This project demonstrates the company's focus on providing FC with a sound and future-proof network solution, balancing high technicality with teamwork to adapt to the challenges of knowledge advancement in academic institutions.

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1.0 **INTRODUCTION**

As we move through this project, we still need to review the initial goals, objectives and targets of the project when defining the strategic direction of FC's future development. Therefore, our main goal is to preemptively deal with the expected 15% increase in students and academic staff in the next four years. Based on this vision, we faced the great challenge of designing a modern 2-storey building; which should accommodate 4 new labs, a video conferencing room, a hybrid classroom and a student lounge area. The scope involved not only the building planning, but also the detailed specifications of each laboratory to meet the tasks of the fourth industrial revolution in the education field.

Basically, one of the main goals we set was to create a solid, efficient and sustainable network in the long term, which is consistent with FC's philosophy of "always ready" for its infrastructure. This means designing and implementing secure, high-performance and easily scalable smart environment systems by integrating new technologies. The goal of the project is to improve the overall network architecture, guarantee efficient management, and be ready to implement new wireless links while dealing with existing and emerging information security threats.

This brings us back to the task, which is more than just a simple task, it has various goals, from designing a learning and innovation environment to preparing facilities for further development and embracing the potential of technological advancement. Our project is more than a plan, it is a commitment to build a facility that will meet FC's current needs as well as future opportunities and challenges.

2.0 PROJECT BACKGROUND AND OVERVIEW

The Faculty of Computing (FC) is currently undergoing significant growth, with 1,800 students, 100 academic staff and 40 supporting staff. Projections indicate a 15% increase in both student and academic staff populations over the next four years, creating a pressing need to expand and upgrade its facilities. To address these challenges, FC plans to construct a new two-storey building that will house key facilities to support its academic, research, and administrative functions.

The proposed building will include four new laboratories which are two general-purpose labs, a Cisco Network lab and an Embedded lab. These labs are essential for providing hands-on learning experiences in line with the latest technological advancements. Additionally, the building will feature a student lounge, a video conferencing room and a hybrid classroom to enhance student engagement, collaborative learning and virtual connectivity.

Beyond the physical infrastructure, FC recognizes the necessity of establishing a scalable, reliable and secure network environment. This infrastructure must offer high-performance internet connectivity to support the growing demand for digital resources, cloud-based applications, remote learning and research activities. A modern network system is essential not only for meeting current needs but also for ensuring the scalability and adaptability required to handle future technological advancements and the increasing user base.

This case highlights the urgent need for a well-planned solution to accommodate FC's expanding community and ensure the provision of state-of-the-art facilities that promote innovation, collaboration and technological growth.

3.0 **COMPILED SOLUTION OF TASK 1-5**

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REVIEW ON EACH TASK

TASK 1: PROJECT PROPOSAL

Task 1 is a crucial stage in the whole project as the project proposal serves as the foundation of the entire case study. In task 1, it involves the identification of case study objectives, scope and requirements in the case study. This phase in task 1 serves as a guide for the subsequent tasks by ensuring that the project meets the requirements within the allocated budget which is RM 1.5M. The main concern in this task 1 is to suggest a floor plan with the design of a new building with the facilities required and able to meet the future 15% growth in the next 20 years with the limited budget.

TASK 2: INITIAL DESIGN AND PRELIMINARY ANALYSIS

Task 2 is about the initial design on the whole network layout within the building which requires us to create the network layout for the work areas. Before creating an actual design, we are required to analyze and understand the network requirements. Our team came out with a series of questions regarding the network connections in the case study. We also undergo surveys to obtain an accurate answer for the questions to understand the situation in a better perspective. In addition, our team interviewed our lecturer to get a professional opinion in guiding us completing the task. Our team also had feasibility analysis to analyze the case study.

TASK 3: CHOOSING THE DEVICES

Task 3 is a process of selecting suitable devices for the network connections within the building for the case study. The research is done to get the most cost-effective devices which is able to meet the requirement with the maximum performance. There are several challenges in this task as it requires ample of surveys to list the devices required. Our team also conducted surveys and researches to compare and contrast between the devices to obtain suitable brand and model of devices that has the best performance within the allocated budget.

TASK 4: MAKING THE CONNECTION

Task 4 is about connecting the network within the building. Our team first identified the work areas within the building to identify the required network. The network distribution within the building to ensure that the work areas have the required network connections. Then, identification of number of connections, patch cords and switch cords are then identified. This task is crucial to ensure the network connection for the case study is well defined to ensure that all the work areas have the best network performance. This step is important as our team has better understanding towards the network connection after this task.

TASK 5: IP ADDRESSING SCHEME

Task 5 is about developing the IP addressing scheme. This stage applies the knowledge from the class in this task. The concept of subnetting the network is applied in this network IP addressing. Task 5 also involves multiple calculations in identifying subnets, configuring default gateway, identify the network address, broadcast address, network address at the router and assigning IP addresses. This task requires understanding of network concepts in the network communication.

TASK 6: DOCUMENTATION AND REFLECTION

Task 6 is about documentation and reflection on the entire project. This requires our team to complete all the previous tasks. By completing all the tasks, our team requires a strong teamwork, technical skills, strategic planning and time management. As each task requires the completion of the previous task, the project is completed with a structured and systematic approach. This project exposes us towards the future network job scope in the future working environment.



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UNIVERSITI TEKNOLOGI MALAYSIA

Subject : SECR 1213 Network Communications

Section: 01

Task 1: Project Setup

Lecturer : Dr. Muhammad Zafran bin Muhammad Zaly Shah

Date: 19/10/2024

Group Name: Beauties

Members' Name & Matric Number:

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INTRODUCTION ON CASE STUDY

This case study is about the task on upgrade needs of Faculty of Computing (FC). Faculty of Computing (FC) currently has 1800 students which consist of both undergraduate and postgraduate, 100 academic and 40 supporting staffs with the projections indicating there will be a 15% increase in both students and academic staffs over the next four years. To be able to accommodate with this expansion, FC plans to build a new 2 -storey building that will house 4 new labs which are 2 general purpose labs, 1 Cisco Network lab and 1 Embedded lab), 1 student lounge, 1 video conferencing room and 1 hybrid classroom. These facilities are intended to be able to support the faculty's need to the technology. The new building aiming not only to enhance learning but also to provide a scalable, reliable and secure network environment that offers high performance internet connectivity to meet both current and future need of FC.

SUGGESTED FLOOR PLAN

GROUND FLOOR

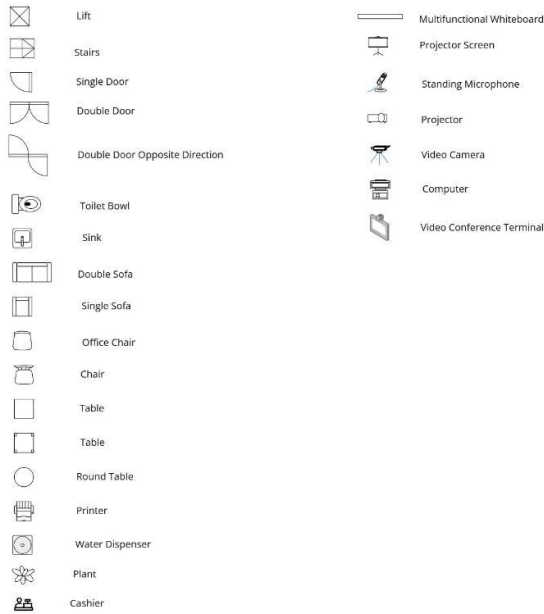
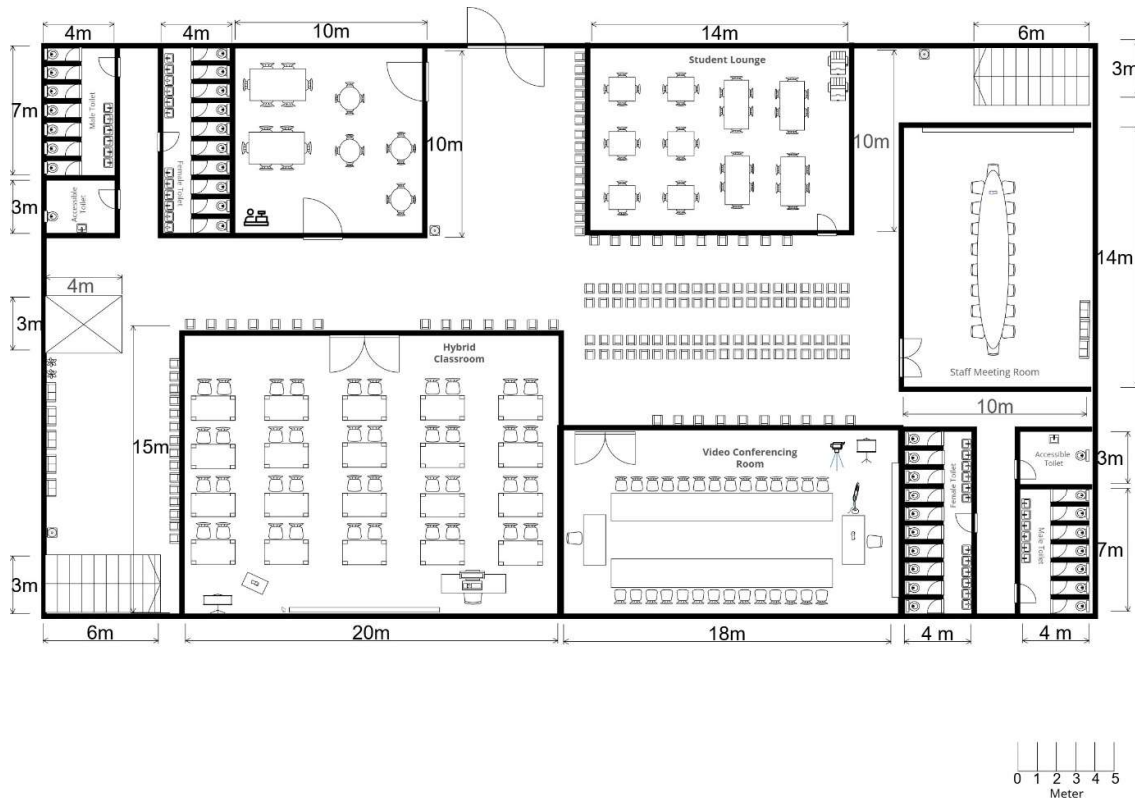


Figure 1.1: Floor Plan of Ground Floor

FIRST FLOOR

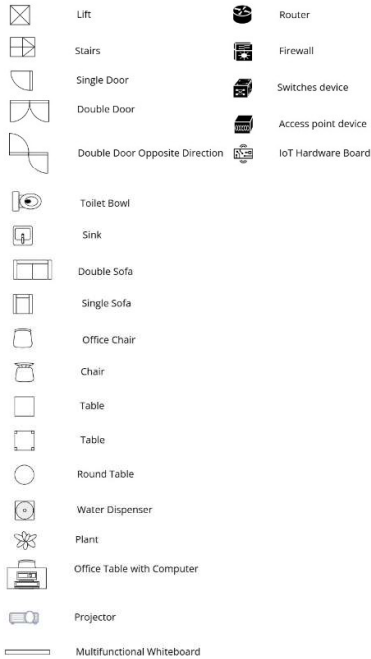
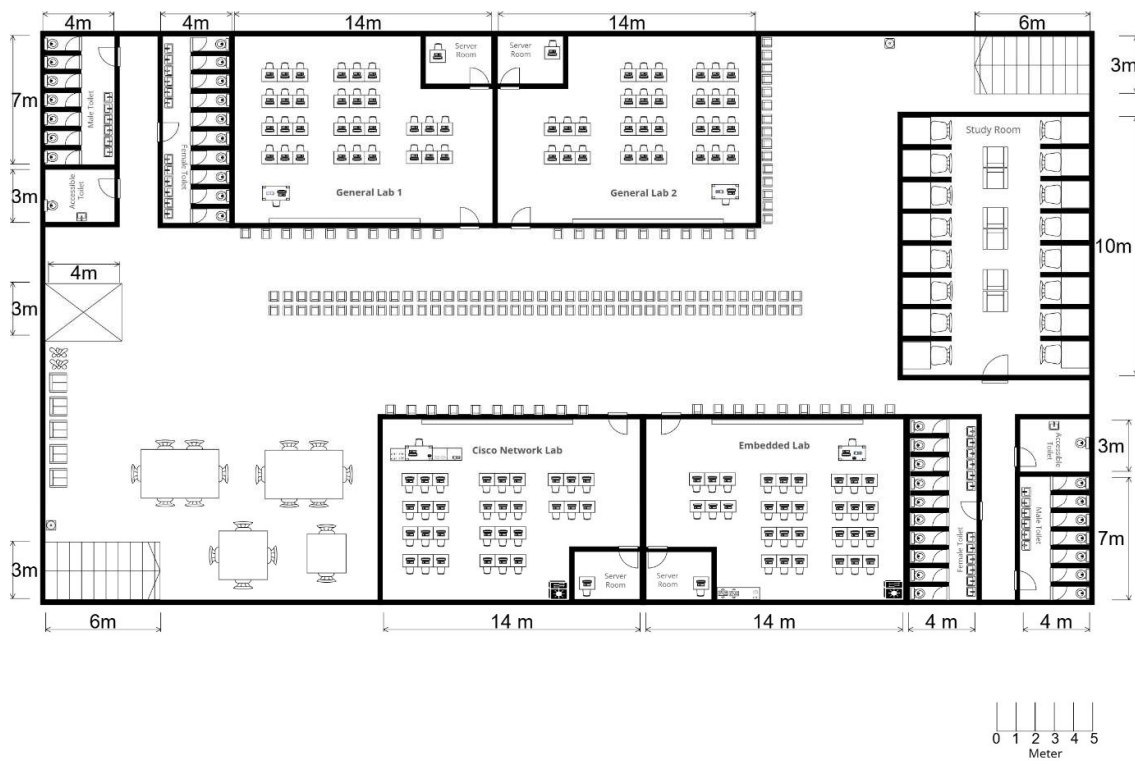


Figure 1.2: Floor Plan of First Floor

INTRODUCTION ON FLOOR PLAN

Based on the task given, the new building of Faculty of Computing is designed to be a 2-storey block. Our team basically designed to place the video conferencing room, hybrid classroom and student lounge at the ground floor. While the four labs which are 2 general labs, 1 Cisco Network lab and 1 Embedded lab are located at the first floor. There are some facilities that we included in our design such as lift, stairs and toilet. These facilities have the exactly same design for both floors. Our team have the idea that the stairs should be located at both ends of the building for safety purposes. The toilet should have the same planning to be convenience for the users. We also have the idea of having accessible toilet and more female toilet to meet the user needs.

For the ground floor, the student lounge is located at the left of the main entrance. The video conferencing room and hybrid classroom are just right in front of the main entrance. Our team decided to have the hybrid classroom larger to have the best performance and efficiency for teaching and learning. Our team also added a staff meeting room for the future needs. Considering students and staffs' welfare, our team have a café and water dispensers in the building. To cope with the increment of students and staffs in the future, we installed a number of sofas at the open area.

For the ground floor, the hybrid classroom one of the main functioning rooms which consist of some technology devices for better performance. The hybrid classroom is designed mainly to facilitate the learning for the students to create a seamless and interactive experience for students regardless of location. It can allow the lecturer to teach the students physically and also engaging with students online. Therefore, the hybrid classroom is equipped with video conference terminal which is specialized in the hybrid room. The hybrid classroom is also equipped with the projector and projector screen for better teaching environment.

Besides, another functioning room in the ground floor is the video conferencing room. This room is also equipped with projector and projector screen same as the hybrid room. The video conferencing room is also equipped with a video camera and standing microphone. The multifunctional whiteboard is installed in both functioning rooms as it can perform better than the traditional whiteboard. The video camera can facilitate the conference progress in the video conferencing room. The standing microphone can ensure that the voice of the speaker can be transmitted throughout the room for better sound quality.

For the first floor, the 4 new labs are arranged 2 in a row. Two general labs are placed together while the Cisco Network lab and Embedded lab are placed side by side. To fulfill the requirement of having high-speed internet connection in preparation for education in line with 4IR (4th Industrial Revolution), our team decided that each lab should be equipped with 1 server room. Our team also designed a study room for the students who wants to study alone without others interrupting them. For those students who wish to have group discussion can either have it in the student lounge at ground floor or the discussion table at the first floor. There are also a large amount of sofas located at the open area for students and staff to have a seat either to have their own study or chat with others.

The Cisco Network Lab and the Embedded Lab are the labs which need to have some specialized devices compared to the two general labs. The Cisco Network Lab needs to have a better internet performance. Cisco Network Lab is equipped with router, switches device, access point device and firewall. The amount of switched device and access point device in the Cisco Network Lab is more than another lab is to ensure that the lab has the best internet performance. The same goes to the Embedded Lab as the switches device, access point device and firewall is also installed in it. The Embedded Lab is equipped with the IoT hardware board as the testing and debugging process of hardware can be done in the Embedded Lab.

MEETING MINUTE

MEETING MINUTE

DATE/TIME	16 October 2024		
LOCATION	MA7		
AGENDA	Discussion on Floor Plan		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Angela Lee Su Ing		1930	-
Bernice Lou Min Yun		1930	-
Karen Yam Vei Xin		1930	-
Seow Yen Zhi		1930	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion on the case study given. Understanding the requirement in the case study.	-All members go through the case study. -Bernice listed out the important user needs.	All members (16/10)
2	Discussion on the basic facilities to be inserted in the design.	-Karen suggested that besides meeting the user needs, a building must have some basic facilities such as stairs and toilet.	All members (16/10)
3	Discussion on the allocation of the rooms.	-Angela suggested to place the student lounge should be placed at the ground floor. -Seow suggested the 4 labs should all be located at the first floor.	All members (16/10)
4	Sketching on floorplan	-All members give their opinion and discussed on the sketched floor plan. -Bernice sketch the layout using ipad. -Karen suggested that the toilet should be located at	All members (16/10)

		two ends of the building to be user friendly. -Seow then continue to suggest that the stairs should also have that concept for safety purposes. -Angela suggested that the empty space in the open area should be equipped with tables, chairs and sofa for the lecturers, students and staffs. -Bernice then suggested to place a staff meeting room at the ground floor. -Karen then suggested to only build one lift to save the budget. -Seow suggested to place create a study room at first floor. -Angela suggested to have a café at the ground floor.	
5	Enhancement on floor plan sketching	-After the basic sketching, the team proceed with discussion on the size of the rooms and facilities in the building. -Bernice and Seow have the idea that the hybrid classroom and video conference room should be designed larger to fit more users in the future. -Angela suggested the size of the lift to be able to fit a certain amount of people. -Karen then suggested the size of the stairs that are able to meet the safety requirement. -Bernice suggested that there should be more	All members (16/10)

		female toilet compared to male toilet. -Seow suggested that the lift can be located near to the toilet. -Karen suggested should have server room. -Angela then suggested to share one server room for two labs to ensure the speed of the network.	
6	Designing floorplan in software	-All members explore on the tools available online to design the floorplan. -Seow suggested to get a software that all members can access the design at the same time to improve the efficiency in designing. -All members spent some time testing on different tools. -Angela and Bernice manage to get the online tools that meets our requirement. -Karen suggested to calculate the scale before starting to design. -Seow suggested scale 1m to 10mm.	All members (16/10)
7	Distribution of task in designing floor plan	-All members discussed the part they are responsible to design. -Seow designed the overall layout of the building, the layout of the rooms, rough layout of lift, stairs and toilet. -Bernice designed the server room, the layout inside the labs, the interior of toilet. -Angela designed the layout inside hybrid	All members (16/10 – 18/10)

		classroom and the video conference room. -Karen designed the layout inside student lounge, study room and staff meeting room. -Angela and Karen do the enhancement on adding sofa at the open area. -Bernice and Seow do the labelling part. -All members do the final checking part.	
8	Duedate of floor plan	-All member agreed finish there respective parts together latest by Friday. -All members have a short discussion after the design is done.	All members (18/10)
9	Meeting ended	-At 0030, the meeting ended successfully.	



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UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2024/2025

SECR 1213: NETWORK COMMUNICATION

SECTION 01

TASK 2: INITIAL DESIGN - PRELIMINARY ANALYSIS

LECTURER: DR. MUHAMMAD ZAFRAN BIN MUHAMMAD ZALY SHAH

GROUP NAME: BEAUTIES

GROUP MEMBERS	MATRICS NUMBER
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BERNICE LOU MIN YUN	A23CS0056
KAREN YAM VEI XIN	A23CS0093
SEOW YEN ZHI	A23CS0177

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QUESTION AND ANSWER

1. What building materials will be used, and how might they impact the network?

Buildings using reinforced concrete, steel, or metal can significantly impact wireless signal strength, leading to signal attenuation. Building materials such as steel and concrete have high impact on the signal reception while brick has low impact towards the signal reception.[1] This requires strategic placement of access points and potential use of signal amplifiers to maintain strong Wi-Fi coverage.

2. What are the current and projected internet bandwidth requirements for the building?

The network needs to support for a high-speed internet especially in Cisco Network Lab, video conferencing in video conference room, online classes in hybrid classroom, IoT for Embedded Lab and high-speed internet in every place within the building as this Faculty has high demand on internet speed. For a faculty with over 1800 students and 140 staff members, an initial bandwidth of at least 10 Gbps is recommended to support high-density access and streaming, with future scalability for up to 15-20% growth.[2] This ensures smooth video conferencing, online classes, and IoT connectivity.

3. Will the network support both wired and wireless connectivity, and which wireless standard is preferred?

Yes, both wired (CAT 6A or higher) and wireless connectivity will be supported. The preferred wireless standard is Wi-Fi 6 (802.11ax) or higher, ensuring high-speed, low-latency connections in high-density environments.

4. How will the network be segmented to separate traffic between academic, student, and administrative areas?

Network segmentation can be achieved using VLANs, which separate traffic logically, and access control lists (ACLs) to restrict communication between segments. This setup provides enhanced security and prevents unauthorized access to sensitive administrative data.

5. Will the network infrastructure include cloud-based services for data storage and processing?

The network infrastructure is expected to include cloud-based services for data storage and processing. Utilizing platforms like Amazon Web Services (AWS) or Microsoft Azure can provide scalable resources for research projects and enhance collaboration among users. It's crucial to implement proper security measures, such as data encryption in transit and at rest, alongside strict access controls to protect sensitive information stored in the cloud.

6. Will interactive and collaborative technologies be used in the hybrid classroom?

Yes, hybrid classrooms will integrate smart boards, high-resolution video conferencing systems, and collaboration platforms (e.g., Zoom Rooms, Teams). Hybrid classroom should also integrate virtual conferencing platforms, camera and microphone should be utilize to engage in class.[3] This supports interactive learning with remote participants.

7. What power backup solutions will be available?

The building should be equipped with Uninterruptible Power Supply UPS systems for all critical networking equipment and backup generators for longer power outages.[4] UPS systems act as the linkage between the main power source and the critical equipment. When the is power failure, UPS instantly switches to its internal battery power supply. This can prevent shutting down of devices and avoiding data loss. This ensures minimal disruption during power failures.

8. Will an in-house IT team manage the network, or will third-party support be contracted?

Ideally, an in-house IT team familiar with the faculty's infrastructure should manage the network for quicker response and maintenance. However, third-party support for advanced troubleshooting and specialized services may be contracted.

9. What training will be provided to IT staff for managing and updating the new network system?

IT staff will receive regular training on new network management tools, cybersecurity protocols, and best practices through workshops, vendor certifications (e.g., Cisco, CompTIA), and simulation-based learning.

10. Will the building need specialized network segments for high-security activities, such as research or administrative tasks?

Yes, creating specialized network segments is essential for high-security activities.[5] Network segmentation allows separation of different types of traffic and prevents the characteristics in one segmented network from affecting another network. For example, separate VLANs (Virtual Local Area Networks) can be established for research and administrative tasks to ensure that sensitive data is isolated from general network traffic. This helps in enforcing stricter security policies, including more robust firewall rules and dedicated monitoring to prevent unauthorized access and data breaches.

11. What are the security requirements for different sections of the building (e.g., labs, student lounge, video conferencing room) to protect against unauthorized access?

- Labs: Security in the labs should include controlled access using key card systems or biometric scanners to ensure only authorized personnel can enter. Additionally, network segmentation should be implemented, separating lab networks from general student networks to protect sensitive data. Employing surveillance cameras and alarm systems in lab areas can enhance security.
- Student Lounge: This area can be more open, but it should still have a secure Wi-Fi network protected by WPA3 encryption.[6] WPA3 which launched in 2018 provides greater encryption and protection compared to WPA2. WPA3-Enterprise is used for organizational purposes. Guest access should be managed through a guest network that restricts access to critical resources and data.
- Video Conferencing Room: The video conferencing setup should use end-to-end encryption for all communications. Access controls should limit usage to specific users, and regular software updates should be performed to mitigate vulnerabilities in the conferencing tools.

12. What are the security protocols required to protect against network breaches?

Network security is the tools, technique and policies to detect and response to network intrusions and safeguard our digital assets.[7] Network security protocols such as Firewall, Antivirus, Intrusion Detection System (IDS) and VPN gateways. These protocols are essential to ensure the data safety in the building.

13. Is there a need for redundancy or backup connections in case of a primary network failure?

Outage can halt the whole system disturbing the automated process which leads to productivity lost. Redundant internet connection provides an alternative link when the primary connection fails.[8] Network redundancy ensures the network availability by duplicating only the critical parts. This is due to duplication of the whole network is costly.

14. What tools and techniques can be used for real-time monitoring and management of the network to ensure high availability and performance?

[9] There are several tools both free and paid version.

Free tools are such as Zabbix and Wazuh; while paid tools are Splunk and Sangfor.

15. What are the key requirements for a high-quality video conferencing room in terms of network connectivity and hardware?

[9] High bandwidth, and minimal latency.

FEASIBILITY

Feasibility Analysis:

Based on the requirements gathered, the project appears feasible with a clear understanding of network infrastructure needs and future scalability. The anticipated increase in students and staff aligns with the planned high-speed internet and robust security protocols, ensuring the network will be capable of supporting both current and future demands. Specialized equipment for the Cisco Network and Embedded Labs adds value to the project by meeting specific educational objectives in line with 4IR standards. The layout provides adequate facilities for user comfort, accessibility, and safety, which further supports the overall feasibility.

Technical Feasibility:

The technical feasibility discusses about the hardware should be able to support the internet bandwidth capacity and throughput. The design should be scalability as there is expected to have a significant growth of the users and requirement in the future. The project should be easy for further upgrades to increase the bandwidth without major disruptions. On the network security part, our project has implemented the strong security protocols including firewalls, antivirus, intrusion detection system (IDS) and VLANs. Regular security assessment can be conducted to update and improve the system.

Operational Feasibility:

For the operational part, training is vital to ensure that all members of the faculty are trained to operate the new system. The modification can be done with collecting their feedback after the testing phase. The project should also establish IT support team for handling queries and maintenance job. Regular maintenance should be scheduled to keep track the performance and ensure the security is able to keep updated.

Economic Feasibility:

The team will use the budget allocated on this project wisely by planning each expenses. With the budget of RM 1.5 million, we would spend the limited budget by prioritizing the critical and essential part. The critical components are the infrastructure hardware, security measure and essential requirement in the project.

Security:

The project incorporates advanced security features suitable for sensitive educational environments. Several strategies and precautions are taken to balance the security and user accessibility.

MEETING MINUTE**MEETING MINUTE**

DATE/TIME	5 November 2024		
LOCATION	MA7		
AGENDA	Discussion on preliminary analysis		
MEETING MC	Bernice Lou Min Yun		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Angela Lee Su Ing	1900	-	
Bernice Lou Min Yun	1900	-	
Karen Yam Vei Xin	1900	-	
Seow Yen Zhi	1900	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 2	- Bernice shared the information about task 2 and its rubric. - All members read and investigate the task carefully. - Solved the problem raised by group members	All members (05/11)

		together through discussion.	
2	Suggestion for questions	- All members discussed the possible questions that can be used in task 2. - Each member suggested 11 questions that can be used in task 2.	All members (05/11)
3	Filter and select the most related question	- Seow and Angela suggested preparing more than 10 questions and all members agreed with their suggestions.	All members (05/11)
4	Task distribution	- Karen distributed the task to all members equally. - All members had to do research to find a solution for all selected questions.	All members (05/11)

		- Bernice was assigned to complete the feasibility study. - Seow was assigned to check and arrange the references used by all members. - Angela and Karen were assigned to prepare for meeting minutes.	
5	Explain answer for each question	- Seow explained - Angela explained - Bernice explained - Karen explained	All members (05/11)
6	Review the answer for each question	- All members review the questions and the answer carefully - Karen suggested to drop the question which is not related to the syllabus - Bernice suggested to not	All members (05/11)

		use the question which have a general answer - Angela said that some questions are confusing - Seow highlighted the question that can be used - 11 questions are finalized	
7	Discuss on the feasibility assessment	- Bernice explained the feasibility assessment to all members - Karen ask if the feasibility assessment is in correct format - Angela suggested to continue with the current format as it covers several aspects	All members (05/11)
8	Meeting ended	- At 2300, the meeting ended after all	All members (05/11)

		discussion were done.	
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REFERENCES AND CITATION

Patel, H., & Patel, H. (2024, May 23). Network Infrastructure Design: your ultimate guide to planning and implementation. WPG Consulting. <https://wpgc.io/blog/network-infrastructure-design-guide-to-planning-and-implementation/>

M, D. (2024b, October 30). Best network management software Solution - Auvik Networks. Auvik. <https://www.auvik.com/wp-content/uploads/2022/03/Building-a-Modern-Educational-IT-Network.pdf>

[1] Building Materials and Methods (14 September 2024) – Sugar Marketing.
Retrieved from: <https://panrf.com/2021/12/building-materials-and-methods/> (Accessed on: 05 November 2024)

[2] Cisco Systems (n.d.) Campus LAN and WLAN Design Guide.
Retrieved from: Cisco's Website (Accessed on: 05 November 2024)

[3] Hybrid classroom: definition, importance, necessary equipment (02 May 2023).
Retrieved from: <https://dexonsystems.com/blog/hybrid-classroom> (Accessed on: 05 November 2024)

[4] Reliable Power Backup Solutions in UPS Systems (10 April 2024).
Retrieved from: <https://www.prostarsolar.net/article/reliable-power-backup-solutions-in-ups-systems.html> (Accessed on: 05 November 2024)

[5] Network Segment (n.d.). Retrieved from:
<https://www.sciencedirect.com/topics/computer-science/network-segment> (Accessed on: 05 November)

[6] Wireless Network Security: WEP, WPA, WPA2 & WPA3 Explained (29 April 2024) – Wi-Fi Protected Access III (WPA3). Retrieved from:
<https://www.esecurityplanet.com/trends/the-best-security-for-wireless->

[networks/#:~:text=Wi%2DFi%20Protected%20Access%20III,to%20this%20more%20modern%20standard.](#) (Accessed on: 05 November 2024)

[7] Network Security Devices (n.d.)

Retrieved from: <https://www.zenarmor.com/docs/network-security-tutorials/network-security-devices> (Accessed on: 05 November 2024)

[8] What is network redundancy, why it matters and how to build it (22 July 2024)

Retrieved from: <https://www.zenarmor.com/docs/network-security-tutorials/network-security-devices> (Accessed on: 05 November 2024)

[9] Dr. Muhammad Zafran bin Muhammad Zally Shah, personal communication on 22th November 2024.



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

Subject : SECR 1213 Network Communication

Section: 01

Task 3: Choosing the Appropriate LAN Devices

Lecturer : Dr. Muhammad Zafran bin Muhammad Zaly Shah

Date: 01/12/2024

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INTRODUCTION TO THE REQUIRED DEVICES

1. Routers

A router is a device that receives and sends data over the computer network. Router acts as a networking device that directs data packets between computer networks. It also acts as a gateway or traffic director that guides and directs network data by dividing network data into smaller packets. Router is able to analyze the incoming data packets, determine their destinations and by using routing tables and policies to select the most efficient path to forward them to the appropriate network. Routers are also able to manage traffic, prevent congestion and enhance the security by controlling the access. Router basically is important to enable the communications between devices on different networks and ensure data packets can be sent accurately and efficiently. Router is essential for ensuring a smooth data communication and directing traffic between different network segments which is required as mentioned previously in our design. This is to ensure that although there is network traffic in the faculty as there are many users, but the network is able to scale with the large amount of traffic.

2. Switches

Switch is a networking device that connects devices within a local area network which is known as LAN. Switch is also vital to facilitate efficient communication between them by forwarding data packets to and from those devices. Switch differs from router as switch only sends data to and single device but not to the networks of multiple devices. Switch is basically important for the communications between interconnecting devices. Switches are normally used in places which have many devices. In this case study, the large scale of devices required shows that the switch is needed for network communications. Switches can be used at OSI layer 2 (data link layer) or layer 3 (network layer). Most switches are the layer 2 switches which connect devices within their network by using the Ethernet cables.

3. Patch Panels

Patch panel is a mounted hardware component that is used in the network cabling systems. Patch panels in the local area network (LAN) contains ports to connect and manage the incoming and outgoing network cables, such as Ethernet or fiber optic. Patch panels are useful in cable management that keeps cables organized and accessible, enabling the flexible connectivity into network hardware that is located in a data center or an access or wiring closet. Patch panels are mounted into network racks, either above or below the switches. Patch panels here is more important for expanding the network without unnecessary clutter or signal loss in our daily usage in faculty.

4. Wireless Access Point Devices

Wireless access point (WAP) device is a networking device that allows the wireless-capable devices to connect to the wired network. WAP provides a simpler and easier way to connect the devices in the network rather than using wires and cables. It basically acts as a linkage bridge between the wireless devices and the wired network infrastructure. WAP is crucial for extending the wired network by enabling the Wi-Fi access. The allocation of multiple WAP in a strategic way can also be useful to increase the network's range. WAP can be used to extend signal range and strength of the wireless network to provide a complete wireless coverage that can get rid of the dead spot that is out of coverage especially in the large building in this current case study. WAP is necessary for extending the reach of a network, especially in the large building which comprises of two floors and multiple functional room.

5. Cables

Cable is one of the basic components in the computer networking that connects devices for data transmission. Cable in the networking refers to the physical medium that is used to connect devices for data transmission. Network cables are used to connect networking devices in a network. There are a few types of cables that are often used in networking such as twisted pair cable, coaxial cable, fiber optic cable, USB and HDMI cables. Twisted pair cable is commonly used in the Ethernet networking that includes CAT5e, CAT6 and CAT6A for a high-speed data transfer. Coaxial cable is used in the broadband internet and cable TV which is not primarily developed for computer network but more on general purposes. At the beginning of developing computer networking, this coaxial cable is used to build computer networks. Fiber optic cable transmits data using light for high-speed and long distance. Fiber optic cable can transmit data at the speed of 100Gbps up to distance of 40 kilometers. USB and HDMI cables are used for specific device connections and multimedia transfers.

6. Firewall

Firewall is a type of computer network security system that monitors and control the incoming and outgoing network traffic or within a private network based on the predefined security rules. Firewall selectively blocks and allows data packets to be transfer within the network which acts as barrier between trusted internal network and untrusted external networks. Firewall is effective in preventing unauthorized access, malware and cyberattacks. Firewall can be in hardware, software or even cloud-based. Firewall decides which network traffic that is allowed to pass through and which traffic is hazard and to be prohibited. In our project, we do install the hardware firewall that is use to perform inspection of both inbound and outbound networking traffic, enforcement of access controls and other security policies.

7. Computer

Computer is an electronic device that is used to manipulate data. Computer has the basic function of storing, retrieving and processing data. Computer consists of two main components which are hardware and software. Hardware refers to the physical structure of the computer regardless about the external part or internal part while software indicates the set of instructions that direct the hardware on what to do and how to do. A computer consists of computer case, motherboard, processor, power supply, storage device, monitor, keyboard and mouse.

8. Projector

Projector is an optical device that takes image or video signals from a source such as computer or Blu-ray player and projects them onto a larger surface such as projector screen. Projector acts as output device that projects the input onto a larger, flat and light colour screen. Projectors are commonly used in the educational institutions, offices and homes for multiple purposes such as presentations, lectures, entertainment and also immersive viewing experiences. The key features to indicate the quality of projectors are such as resolution, brightness, connectivity and portability. There are a variety of projectors such as LCD projectors that use liquid crystal display, DLP projectors that use digital light processing technology and LED projectors that are compact and energy-efficient.

9. Projector Screen

Projector screen is a surface that is specially designed to display images or videos that is projected by a projector. Projector screen provides a reflective and uniform surface that enhances the visibility and quality of the projected image which is a better option compared to a wall or other surfaces. Projector screen is particular on the material, aspect ratio and projector screen gain. These three features distinguish the quality of a good projector screen. Projector screen typically made from polyester, vinyl or other durable fabrics for better image contrast. The ratios of projector screen are such as 16:9, 4:3 or even 2.35:1. The gain of 1.0 basically means that the projector screen reflects light equally from all angles and higher gain value enhances brightness but also reduces the viewing angle.

10. Multifunctional Whiteboard

Multifunctional whiteboard is an advanced interactive whiteboard which integrated multiple digital features with the traditional functionality of the normal whiteboard. Multifunctional whiteboard are also known as smart whiteboard due to the features that is integrated in it which this multifunctional whiteboard is normally use in education and business. Multifunctional whiteboard can have features such as touchscreen interaction, digital writing and annotation, connectivity and integration to other devices, interactive features which supports the multimedia elements, storage and cloud access that allows users to save their progress to cloud.

11. Standing Microphone

Microphone is a transducer that converts the sound waves into electrical signals. Standing microphone is one of the microphone types that is mounted on a stand for use in various situations such as presentation and lectures. The standing microphone allows the microphone to be in a fixed position which allows speaker to present or conduct lecture in a lively way and interacts freely. The basic function of standing microphone is that it is hands-free use which is commonly apply in the conferences, meetings and presentations that allow clear and stable audio capture that allows speaker's voice to be projected and transmitted effectively throughout the room.

12. Video Camera

The basic function of a video camera is to capture the moving image and sound and later produce a signal to be displayed. Video camera records the visual and audio data to be stored and processed. Video camera has high demand on the lens and recording media. A good video camera focuses light to create a sharp image on the image sensor. Different storage has been used with the video camera such as SD cards, hard drives or even cloud storage.

13. Video Conference Terminal

Video conference terminal is a hardware device that is made up of a combination of multiple components such as camera, microphone, speaker, display device and transmission network. Video conference device is a specialized device that is used to facilitate the communication between several locations. Video conference terminal is useful for collaborative work, lectures and meetings that are held between two or more locations. Video conference terminal is equipped with hardware and software to ensure a high quality of video and audio transmission for a smooth communication.

14. IoT Hardware Board

IoT hardware board is specialized board designed for building and prototyping Internet of Things (IoT) applications. These boards are suitable to be equipped in the embedded lab. IoT hardware boards serve as the foundation for IoT projects to provide processing, communication and power capabilities. IoT hardware boards have programming interfaces that program the microcontroller via computer, power circuit with DC power, various input and output components.

15. Printer

Printer is an electronic device which produces a hard copy or printed version of electronic documents, images, or other digital content. Printer is for transferring information from a digital format to a tangible, paper-based format. Printer can connect and communicate with computers in several ways such as connected directly to the computer or indirectly via a network. A printer may be Ethernet enabled and be cabled to an Ethernet switch, or it may connect to a Wi-Fi (wireless) network, or both. Most common printer connection types are by USB (universal serial bus), cable (wired) or via Wi-Fi (wireless). It is widely used for a range of applications, from producing documents for personal use to handling high-volume printing in professional environments including homes, offices, educational institutions, and businesses.

16. Modem

Modem is a device that transforms digital information from your computer into analog signals that can transmit over wires (and vice versa) by modulating and demodulating electrical impulses sent through phone lines, coaxial cables, or other types of wiring. Most standalone modems have just two ports: one that connects to the outside world, and an Ethernet jack that connects to a computer or a router. Modem is able to have function such as modulation, demodulation and data transmission. Modem is essential for the internet connectivity and make it possible for digital devices to communicate with analog infrastructure which is known as compatibility.

17. Server

A server is computer hardware or software that provides functionality such as resources, services or programs for other programs or devices, known as "clients". It is a process that serves data for clients in a client-server 2 model, where communication is based on request and response. Server is the heart of computer operations and often being used in data centers. Servers are connected to one another via a local area network (LAN) or wide area network (WAN). Servers can provide various services such as sharing data or resources among multiple clients or performing computations for a client. They can be specialized for running specific types of servers, such as database servers, file servers, mail servers, web servers, and application servers. Servers facilitate communication between multiple clients and systems as part of the broader network architecture required for faculty usage.

18. Power Backup Supply System

Power backup supply system is designed to ensure there is continuous power ability when there is power outage and the main power supply fails to function. Power backup supply system is crucial in this project as our use case requires uninterrupted operation. Uninterruptible Power Supply UPS systems for all critical networking equipment and backup generators for longer power outages. UPS systems act as the linkage between the main power source and the critical equipment and instantly switches to its internal battery power supply when unexpected breakdown occurs. This can prevent shutting down of devices and avoiding data loss which ensures minimal disruption during power failures.

TABLE OF DEVICES

Devices	Description	Price (RM/unit)	Unit	Total Price (RM)
 Figure 3.2.1 Router	<u>TP-Link Archer AX73</u> Dual-band Wi-Fi 6 router: Supports the latest Wi-Fi 6 technology for faster speeds and better performance. Speeds of up to 5400 Mbps: Provides enough bandwidth for even the most demanding applications. Connects over 200 devices: Can handle many connected devices without sacrificing performance. HomeShield security: Protects your network from malware and other threats. Processor and memory: 1.5 GHz Triple-Core CPU, 512 MB of RAM, 1x Gigabit WAN port, and 4x Gigabit LAN ports. Processor and memory: Powered by a	484.00	6	2904.00

	1.7 GHz quad-core CPU with 512 MB RAM. Beamforming Technology: Enhances Wi-Fi coverage and focuses signals toward connected devices for a stronger and more reliable connection. Cooling Design: Features an advanced heat sink and vented panels for effective heat dissipation during heavy use.			
 Figure 3.2.2 Switches	<u>FS S3400-24T4FP</u> Power and Capacity: Up to 24 PoE+ ports with a total power budget of 370W, sufficient for powering high-demand devices. Reliable single power supply and cooling via 3 built-in fans. Performance: Advanced Layer 2 and Layer 3 functions, including ERPS, OSPF, RIP, and L2 multicast support. Security: Comprehensive security features like SSH, ACL, AAA, 802.1X, RADIUS, and TACACS+.	2033.00	7	14231.00

	Monitoring and Operation: Supports sFlow for enhanced network monitoring and analysis. Flexible operation through WEB, CLI, SNMP, and SSH interfaces. Dual-Firmware Support: Ensures a backup firmware version is available for safe upgrades and rollback in case of issues. Redundancy Features: Supports VRRP (Virtual Router Redundancy Protocol) for high availability and improved fault tolerance.			
 Figure 3.2.3 Patch Panel	<u>Linkbasic Cat6 Patch Panel</u> Design: Mounts to any standard 19" rack or cabinet, with a high-density 1U form factor to save space. Material: Made of black electrostatic powder-coated steel, corrosion-resistant with sturdy aluminum plate around RJ45 jacks. Wiring Compatibility: Supports T568A and	104.00	6	624.00

	T568B wiring patterns, meeting Cat 6 standards. Ease of Installation: Color-coded wiring for easy setup and accepts 22, 24, 26 AWG solid wires. Durability: Corrosion-resistant steel and fire self-extinguishing high-impact plastic for safe use. Cable Management Bar: Includes an integrated rear cable management bar to organize and secure cables efficiently. High Density: Offers support for up to 48 ports in a 1U configuration for larger network needs.			
 Figure 3.2.4 Wireless Access Point	<u>Ubiquiti UniFi 6 Lite</u> Wi-Fi Standard: 802.11a/b/g/n/ac/ax (Wi-Fi 6) Max Data Rate: 300 Mbps (2.4 GHz), 1.2 Gbps (5 GHz) Power: Powered via PoE (48V, 0.32A), Max. power consumption: 12W Max Clients: 300+	595.00	6	3570.00

	Security: WPA3, WPA2, WPA-PSK, WPA-Enterprise Mounting: Wall/ceiling (included) Certifications: CE, FCC, IC Network Interface: 1 GbE RJ45 port Environment Flexibility: Compact and sleek design suitable for modern interiors and tight spaces. Energy Efficient: Low power consumption for eco-conscious operations.			
 Figure 3.2.5 Cable	FS Cat6 Snagless Unshielded (UTP) Ethernet Network Patch Cable Cable Type: 20ft (6.1m) Cat6 24AWG UTP PVC CM Ethernet Patch Cable, Blue Standards: Exceeds ANSI/TIA-568.2-D Category 6 and ISO 11801 Class E Speed: Supports up to 1G/10G-T, 550 MHz	75.90/10.7m	300	2128.04

	Power: Compatible with PoE/PoE+/PoE++ (IEEE 802.3af/at/bt) Build: Stranded pure bare copper, 50μ" gold-plated RJ45 connectors Features: Strain relief, easy unplug clip protector, tested with Fluke Channel Test and wired T568B Noise Reduction: Twisted pair design minimizes crosstalk for better signal quality. Temperature Resistance: Operates in a wide temperature range for various environments. Enhanced Durability: Snaggles boot design protects connectors from damage during installation or removal.			
 Figure 3.2.6 Firewall	<u>FS NSG-5100 Next-Generation Firewall</u> Threat Detection: L2-L7 comprehensive detection and prevention Application Control: Identify, allow, block, or limit applications	21,849.48	2	43698.96

	VPN Services: Supports IPSec VPN, SSL VPN, and L2TP VPN Security Features: Includes IPS, URL filtering, antivirus, attack defense, and QoS (subscription required) Performance: Up to 2.5 Gbps firewall throughput Smart DNS: Inbound DNS for intelligent link load balancing High Availability: Supports failover clustering for uninterrupted service during hardware or network failures. Cloud Management: Offers remote management and monitoring capabilities via a cloud-based platform. Device Identification: Detects and classifies devices for more precise security policies.			
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Figure 3.2.7 Personal Computer (PC)

 Figure 3.2.7 Personal Computer (PC)	<u>HP EliteDesk 800 G9 Desktop</u> Processor: Up to 13th Gen Intel® Core™ i9 processors Memory: Up to 64GB DDR5 memory Storage: Up to 8TB NVMe SSD - Operating System: Windows 11 Pro, Windows 10 Pro Form Factor: Tower, Small Form Factor Security: HP Wolf Security for Business Manageability: HP Intelligent Management Suite Cooling Technology: Advanced thermal management system ensures efficient heat dissipation. Eco-Friendly: ENERGY STAR® certified, reducing power consumption and environmental impact. Upgradeable Design: Tool-less access for quick hardware	3895.00	130	506350.00
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	upgrades or replacements.			
 Figure 3.2.8 Projector	Epson EB-X06 XGA 3LCD Projector Technology: 3LCD for vibrant, true-to-life color brightness Resolution: Native XGA, 4:3 aspect ratio Brightness: 3,600 lumens Lamp Life: Up to 12,000 hours in ECO Mode Connectivity: HDMI, built-in 2W speaker Features: Easy setup, flexible positioning for classroom use. Keystone Correction: Automatic vertical and horizontal adjustments for a perfectly aligned image. Split-Screen Capability: Allows simultaneous projection of content from two sources. Compact and Lightweight: Easy to transport and set up in different locations.	1809.00	6	10854.00

 Figure 3.2.9 Projector Screen	Elite Screens Manual B Size: 100-inch diagonal with 16:9 aspect ratio, providing a viewing area of 49.2" H x 87.4" W. Material: MaxWhite, compatible with UHD/HD projectors, offering a 180° viewing angle and 1.1 gain. Features: Auto-lock mechanism for height adjustments, easy installation, and optional L brackets. Warranty: 2-Year/3-Year ENR-G Warranty and lifetime tech support. Black Backing Material: Blocks light penetration for sharper image quality. Smooth Retraction System: Ensures the screen retracts evenly and securely into the housing. Versatile Mounting Options: Compatible with wall, ceiling, or in-ceiling installations.	414.78	2	829.56
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Figure 3.2.10 Multifunctional Whiteboard

Figure 3.2.10 Multifunctional Whiteboard	<u>SMART Board</u> <u>MX075-V5</u> <u>Collaboration Display</u> <u>(MX275-V5)</u> Size: 75" 4K UHD LCD Touchscreen: 32 touchpoints with HyPr Touch Technology Processor: Octa-core, 8GB RAM, 64GB storage Connectivity: HDMI (4), USB (12), Gigabit Ethernet, Wi-Fi 6, Bluetooth 5.2 Audio: 20W speakers, 6-microphone array OS: Android 13, compatible with ChromeOS, Mac OS, Windows Power: 103W, Energy Star rated Warranty: 3 years Screen Sharing: Supports wireless screen mirroring for up to 9 devices simultaneously. Integrated Whiteboard App: Allows instant brainstorming without	16502.00	6	99012.00
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	requiring a connected device. Energy-Efficient Design: Auto-shutdown and eco-mode reduce power consumption when not in use.			
 Figure 3.2.11 Standing Microphone	<u>Audio-Technica AT2020</u> Type: Side-address condenser, cardioid polar pattern Frequency Response: 20Hz–20kHz Max SPL: 144dB Self Noise: 20dB Impedance: 100 ohms Connector: XLR Accessories: Stand mount, pouch Power: Requires +48V phantom power Features: Low-mass diaphragm for wide dynamic range, high SPL handling, low self-noise, pivoting mount for precise placement Wide Compatibility: Works seamlessly with audio interfaces, mixers, and	390.00	1	390.00

	portable recording systems. Compact Design: Lightweight and portable, ideal for field recording or podcasting. Shock Mount Compatibility: Optional shock mount for better isolation and vibration reduction.			
 Figure 3.2.12 Video Camera	<u>Canon Eos M50 Mark II</u> Sensor: 24MP APS-C CMOS Processor: DIGIC 8 Viewfinder: Electronic Display: Vari-angle Autofocus: Dual Pixel CMOS AF Shooting Speed: 10 fps continuous Video: 4K recording Connectivity: Wi-Fi, Bluetooth Features: Compact design, ideal for video and photography in various settings. Live Streaming Ready: Native YouTube streaming support and	2968.00	1	2968.00

	clean HDMI output for high-quality broadcasts. Silent Shooting Mode: Perfect for capturing moments in noise-sensitive environments. Eye and Face Detection AF: Improved focus tracking for sharp portraits and videos.			
 Figure 3.2.13 Video Conference Terminal	<u>Logitech Rally Plus</u> Imaging System: Ultra-HD Zoom: 15x HD zoom Pan/Tilt: ±90° Pan, +50° / -90° Tilt Presets: 3 camera presets Security: Kensington slot Indicators: Video mute/unmute LED Mount: Standard tripod thread Privacy Feature: Lens parks at -90° during sleep mode Warranty: 2 Years Logitech Manufacturer Warranty	16099.00	1	16099.00

	Features: Autofocus, motorized pan and tilt for smooth movement. 4K Video Processing: Supports up to 4K resolution for crystal-clear conferencing. Plug-and-Play Setup: Works with most video conferencing platforms, including Zoom, Microsoft Teams, and Google Meet. RightSound Technology: Enhances voice clarity while suppressing background noise.			
 Figure 3.2.14 IoT Hardware Board	<u>Raspberry Pi 4 Model B - 4GB</u> Processor: Broadcom BCM2711, Quad-core ARM Cortex-A72 @ 1.5GHz RAM: 4GB LPDDR4 Ports: 2 x USB 3.0, 2 x USB 2.0, 2 x micro HDMI (4Kp60), 27 x GPIO Networking: Gigabit Ethernet, 2.4GHz/5GHz Wi-Fi, Bluetooth 5.0 Storage: microSD	259.00	10	2590.00

	Power: 5V/3.0A via USB Type-C OS Support: Raspberry Pi OS, third-party OS Dual-Display Support: Drives two 4K monitors simultaneously. Expandable Storage: Supports USB-attached storage devices for additional capacity. Overclocking Capabilities: Allows advanced users to boost performance.			
 Figure 3.2.15 Printer	<u>HP LaserJet Pro MFP M428fdw</u> Print Speed: 40 ppm (up to 42 ppm with HP High Speed) Resolution: 1200 x 1200 dpi Print Technology: Laser Display: 2.7" color touchscreen Paper Capacity: 250-sheet input tray, 150-sheet output bin Connectivity: USB 2.0, Gigabit Ethernet, Wi-Fi,	2750.00	2	5500.00

	Apple AirPrint, Google Cloud Print Security: Secure Boot, encrypted data storage, SNMPv3 Power: 510W active, 7.5W ready Duplex Printing: Automatic two-sided printing saves time and paper. Fast Scanning: Dual-head scanner for simultaneous two-sided scanning. Mobile Printing: Supports HP Smart App, Apple AirPrint, and Google Cloud Print for easy printing from smartphones and tablets.			
 Figure 3.2.16 Modem	<u>NETGEAR CM1200 DOCSIS 3.1 Cable Modem</u> DOCSIS 3.1 Technology: Provides faster speeds, improved efficiency, and better latency for Gigabit internet. Multi-Gigabit Ethernet Ports: 4 Gigabit Ethernet ports for multiple wired devices.	847.30	2	1694.60

	Speeds up to 2 Gbps: Ideal for heavy users, gaming, and 4K streaming. Backward Compatibility: Works with DOCSIS 3.0 systems. IPv6 Support: Future-ready internet addressing. Advanced Security: Built-in DoS protection, firewall, and encryption for home network safety. Multi-IP Address Support: Ideal for households or businesses needing multiple static IPs. Passive Cooling Design: Operates quietly without fans. 24x8 Channel Bonding: Ensures better bandwidth management for stable internet.			
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Figure 3.2.17 Server

 Figure 3.2.17 Server	<u>Dell PowerEdge T550 Tower Server</u> Processor: Intel Xeon Silver 4309Y (8C/16T, 2.8 GHz, 12M Cache) Memory: 16GB 3200MT/s RDIMM Storage: 600GB 10K RPM SAS, 2.5" Hot-plug, 3.5" HYB CARR RAID: PERC H755 RAID Controller Networking: Dual port 1Gb + Quad port 1GbE Broadcom Adapter Power Supply: Dual, Hot-plug, Redundant 800W Management: iDRAC9 Enterprise Hot-Swap Drives: Supports up to 8 hot-swappable drives for quick replacements and upgrades. PCIe Gen 4 Support: Doubles data throughput for high-performance peripherals. Flexible Scalability: Modular design for memory and	18402.00	5	92010.00
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	storage expansions as needs grow.			
 Figure 3.2.18 Power Backup Supply System	<u>APC Back-UPS Pro 1000VA</u> 8 Total Outlets: 4 provide battery backup and surge protection, 4 provide surge protection only. Automatic Voltage Regulation (AVR): Ensures safe voltage levels without using battery power. Surge Protection: Ideal for computers, hard drives, and electronics. Intelligent Battery Management: Optimizes battery performance and life. Input Voltage Range: 88V - 148V for main operations LCD Display: Provides real-time information on battery and power status. User-Replaceable Batteries: Extend the UPS's lifespan with easy battery swaps. Data Protection: Includes software for automatic file saving during outages.	924.00	20	18480.00

Total	823933.16
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REFLECTION

Are you surprised by the prices? How were you surprised?

We are surprised by the price of RM 823933.16 which is 54.93% of our overall budget (RM1500000). This cost is unexpected because we never think that these network devices and equipment would be so costly since their options are not even the top or second-best choices. Even seemingly inexpensive items like cables surprised us with a total cost of RM 2128.04. Therefore, we were surprised by the overall cost of these items as we even haven't discussed the other additional cost such as keyboard.

Have you ever considered cost as a factor for choosing networking devices?

Yes, cost is definitely a key factor when choosing networking devices. While looking for the most suitable equipment, it is important to balance performance, reliability and budget. However, choice of lower-cost devices may sometimes compromise on scalability, security features or future-proofing capabilities which could affect the long-term operation of the network. So, we try to seek for the most suitable devices with affordable prices.

What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Router

We have chosen TP-Link Archer AX73 for this project. The major difference between TP-Link Archer AX73 with other different brands we compare is TP-Link Archer A7.

Features	TP-Link Archer AX73	TP-Link Archer A7
USB Ports	None	USB 2.0 port
Total Maximum Wi-Fi Speed	5400 Mbps (AX5400)	1750 Mbps (AC1750)

USB Port: The TP-Link Archer A7 features a USB 2.0 port, suitable for basic file sharing or printer access. In contrast, the TP-Link Archer AX73 lacks a USB port, which simplifies its hardware design but eliminates direct storage sharing capabilities.

Total Maximum Wi-Fi Speed: The TP-Link Archer AX73 supports Wi-Fi 6 (AX5400) with speeds of up to 5400 Mbps, making it ideal for demanding applications like 4K streaming and gaming. The TP-Link Archer A7 supports Wi-Fi 5 (AC1750) with speeds of up to 1750 Mbps, which is sufficient for general use but less effective in high-demand scenarios.

Switches

We have chosen FS S3400-24T4FP for this project. The major difference between FS S3400-24T4FP with other different brands we compare is TP-Link TL-SG3428.

Feature	FS S3400-24T4FP	TP-Link TL-SG3428
Packet Buffer Size	4 MB per port	1.5 MB per port
Cooling	3 built-in fans	no fans
Ability to Handle Traffic	High	Limited
Cooling Functionality	Strong	Weak

Packet Buffer Size:

FS S3400-24T4FP has a much larger packet buffer size of 4 MB per port, which provides better performance under heavy traffic conditions. This is especially important for high-traffic networks as larger buffers can store more packets, reducing packet loss and improving data flow.

TP-Link TL-SG3428 has a 1.5 MB buffer per port which is significantly smaller. This is adequate for small or medium-sized networks.

Cooling:

FS S3400-24T4FP has three built-in fans. This can ensure optimal performance in high-demand environments in order to prevent overheating and potential hardware damage.

TP-Link TL-SG3428 does not have fans which is less efficient in high-traffic conditions. Without fans, the switch may overheat in demanding scenarios, cause failing to perform optimally in large deployments.

Ability to Handle Traffic:

FS S3400-24T4FP supports advanced Layer 3 functions (such as OSPF, ERPS), which enables better handling of traffic in complex, multi-site networks.

TP-Link TL-SG3428 only supports Layer 2 functions, making it suitable for simpler networks without the need for advanced routing capabilities. It can handle traffic well for basic tasks but not as suited for enterprise-level or high-complexity networks.

Patch panel

We have chosen Linkbasic Cat6 Patch Panel for this project. The major difference between Linkbasic Cat6 Patch Panel with other different brands we compare is Generic No-Name Patch Panel.

Features	Linkbasic Cat6 Patch Panel	Generic No-Name Patch Panel
Space Efficiency	High-density 1U form factor, saving rack space	Might be bulkier or larger in form factor, taking up more space
Cable Management	Built-in cable management system	Lacks dedicated cable management features
Maximum Cable Support	Supports Cat 6 cables up to the highest performance standards	May not support the full range of high-speed network cables (e.g. Cat 6a or Cat 6)

The Linkbasic Cat6 Patch Panel is more efficient and versatile compared to Generic No-Name Patch Panel. It has a 1U high-density design that saves space in racks, making it ideal for environments with limited rack space. It also comes with a built-in cable management system, which helps keep cables organized, reducing clutter and making maintenance easier. Additionally, it supports Cat 6 and Cat 6a cables, ensuring high-speed performance and future-proofing for higher data transfer speeds.

In contrast, Generic No-Name Patch Panel may take up more space and often lacks dedicated cable management features, making it harder to keep cables organized. It may also only support lower-speed cables, limiting its ability to handle modern, high-performance networking needs.

Wireless Access Point

We have chosen Ubiquiti UniFi 6 Lite for this project. The major difference between Ubiquiti UniFi 6 Lite with other different brands we compare is TP-Link EAP225 V3.

Feature	Ubiquiti UniFi 6 Lite	TP-Link EAP225 V3
Total Maximum Speed	1.5 Gbps	1.3 Gbps
Max Clients Supported	300+	100+
Security	WPA3, WPA2, WPA-PSK, WPA-Enterprise	WPA2, WPA-PSK

Total Maximum Speed:

The UniFi 6 Lite offers 1.5 Gbps total speed, making it a better option for high-demand applications. The EAP225 V3 offers a 1.3 Gbps total speed, which is slower and suited for smaller setups or less demanding networks.

Client Capacity:

The UniFi 6 Lite supports up to 300+ clients, making it more suitable for environments with a large number of devices such as offices, campuses or high-density areas. The EAP225 V3 can support fewer clients (around 100), which is fine for smaller environments but limits scalability.

Security:

The UniFi 6 Lite features the latest WPA3 security, providing stronger encryption and protection for devices connected to the network. The TP-Link EAP225 V3 only supports WPA2 which is older but still secure for most setups.

Cable

We have chosen FS Cat6 Snagless Unshielded (UTP) Ethernet Network Patch Cable for this project. The major difference between FS Cat6 Snagless Unshielded (UTP) Ethernet Network Patch Cable with other different brands we compare is Mediabridge Ethernet Cable (Cat 6).

Feature	FS Cat6 Snagless UTP Ethernet Cable	Mediabridge Ethernet Cable (Cat 6)
Bandwidth	up to 550 MHz	up to 250 MHz
Speed	10/100/1000/10G Ethernet	10/100/1000 Mbps Ethernet

Bandwidth:

The FS Cat6 Snagless UTP Ethernet Cable offers a higher bandwidth of 550 MHz which is suitable for high-speed networking, especially for applications like 10 Gigabit Ethernet. In contrast, the Mediabridge Ethernet Cat 6 supports only 250 MHz which is less suitable for high-speed and high-demand environments.

Speed:

FS Cat6 Snagless UTP Ethernet Cable offers better support for 10G Ethernet (up to 10 Gigabits per second) due to its higher bandwidth. The Mediabridge Ethernet Cat 6 is more suited for typical home or small office use where 1 Gbps speeds are sufficient.

Firewall

We have chosen FS NSG-5100 Next-Generation Firewall for this project. The major difference between FS NSG-5100 Next-Generation Firewall with other different brands we compare is TP-Link SafeStream TL-ER7206 Firewall.

Feature	FS NSG-5100 Next-Generation Firewall	TP-Link SafeStream TL-ER7206 Firewall
Threat Detection	deep packet inspection	no deep packet inspection
Security	Includes IPS, antivirus, URL filtering, attack defense and QoS	Basic NAT, stateful inspection and simple firewall rules
Application Control	Advanced application control (allow/block/limit apps)	Basic application control, limited to basic features
User Authentication	Comprehensive support	Limited authentication support
Performance Throughput	Up to 2.5 Gbps	Up to 1 Gbps

Threat Detection and Security:

The FS NSG-5100 offers Layer 2 to Layer 7 (L2-L7) threat detection which enables it to handle complex traffic and detect various types of attacks such as malware and DDoS with advanced methods like deep packet inspection. The TP-Link TL-ER7206 offers basic stateful packet inspection (L2-L4) and lacks deep packet inspection which is less effective at identifying more sophisticated threats.

Application Control and User Authentication:

The FS NSG-5100 includes comprehensive application control and strong user authentication methods to support protocols like RADIUS, TACACS+ and 802.1X. This is crucial for ensuring that only authorized users and devices can access the network and for managing specific app-level restrictions. The TP-Link TL-ER7206 has basic user authentication and limited application control which is more suitable for small-scale deployments without complex security needs.

Performance and Throughput:

The FS NSG-5100 offers up to 2.5 Gbps throughput which is suitable for high-demand environments and larger networks. On the other hand, the TP-Link TL-ER7206 has a lower throughput of up to 1 Gbps which may be limiting for larger deployments or high-bandwidth applications.

Personal Computer (PC)

We have chosen HP EliteDesk 800 G9 Desktop for this project. The major difference between HP EliteDesk 800 G9 Desktop with other different brands we compare is Lenovo ThinkCentre M70q Desktop.

Feature	HP EliteDesk 800 G9 Desktop	Lenovo ThinkCentre M70q Desktop
Processor	13th Gen Intel® Core™ i9	Intel Core i5-10400T (6 cores, 12 threads)
Memory	64GB DDR5	16GB DDR4
Storage	8TB NVMe SSD	512GB SSD
Graphics	Integrated Intel UHD 770 or optional discrete GPU	Integrated Intel UHD Graphics 630

Processor and Performance:

The HP EliteDesk 800 G9 offers 13th Gen Intel Core i9 processor, which provides superior multi-core performance compared to the Intel Core i5-10400T in the Lenovo ThinkCentre M70q. The i9 processor will deliver better performance for more demanding applications, multitasking and professional-grade tasks, while the i5 in the Lenovo is adequate for everyday business tasks but not ideal for intensive workloads.

Memory and Storage:

The HP EliteDesk 800 G9 supports 64GB DDR5 memory and offers high storage options like 8TB NVMe SSD. This is a great option for heavy multitaskers and users requiring extensive data storage. In contrast, the Lenovo ThinkCentre M70q comes with 16GB of DDR4 memory and 512GB SSD storage which is more suited for light-to-medium office tasks and less demanding users.

Graphics:

The HP EliteDesk 800 G9 allows for an optional discrete GPU which can be useful for graphics-intensive applications like design, video editing or gaming whereas the Lenovo M70q relies on integrated Intel UHD Graphics 630 which is suitable for basic office applications but not for graphically demanding tasks.

Projector

We have chosen Epson EB-X06 XGA 3LCD Projector for this project. The major difference between Epson EB-X06 XGA 3LCD Projector with other different brands we compare is ViewSonic PA503S.

Feature	Epson EB-X06 XGA 3LCD Projector	ViewSonic PA503S
Resolution	XGA (1024 x 768)	SVGA (800 x 600)
Color Technology	3LCD technology	DLP technology

Resolution:

The Epson EB-X06 offers XGA resolution (1024 x 768) which provides better image clarity compared to the SVGA resolution (800 x 600) of the ViewSonic PA503S. XGA is better suited for professional presentations, educational use and displaying text clearly, while SVGA is typically sufficient for basic presentations and general use.

Color Technology:

The Epson EB-X06 uses 3LCD technology which is known for producing better color accuracy and brightness, particularly in terms of vibrant colors in bright environments. In contrast, the ViewSonic PA503S uses DLP technology which typically produces sharper images but with less accurate and vivid colors.

Projector Screen

We have chosen Elite Screens Manual B for this project. The major difference between Elite Screens Manual B with other different brands we compare is VIVO Manual Pull Down Screen.

Feature	Elite Screens Manual B	VIVO Manual Pull Down Screen
Screen Material	MaxWhite material	Matte white material
Viewing Angle	180°	160°
Ease of Installation	Auto-locking mechanism for height adjustments	Manual locking system

Screen Material:

The Elite Screens Manual B uses a MaxWhite material with a 1.1 gain. This can provide a better and brighter image with enhanced color accuracy, especially for HD or UHD projectors. The VIVO Manual Pull Down Screen uses a standard matte white material which works well for basic projection needs but does not provide the same color vibrancy or brightness as the Elite model.

Viewing Angle:

The Elite Screens offers a wider viewing angle of 180° which means it can deliver better quality visuals to a larger audience or from more seating positions. The VIVO screen offers a 160° viewing angle which may be acceptable for smaller groups but might have limitations in larger rooms.

Ease of Installation:

The Elite Screens Manual B comes with an auto-locking mechanism for easier height adjustment to make it more user-friendly. The VIVO model has a manual locking mechanism which may require more effort to adjust and can be less convenient in comparison.

Multifunctional Whiteboard

We have chosen SMART Board MX075-V5 for this project. The major difference between SMART Board MX075-V5 with other different brands we compare is Promethean ActivPanel Titanium.

Features	SMART Board MX075-V5	Promethean ActivPanel Titanium
Touchscreen	32 touchpoints, HyPr Touch.	20 touchpoints
Connectivity	HDMI (4), USB (12), Wi-Fi 6, Bluetooth 5.2.	HDMI (2), USB (6), no Bluetooth.
Power	Energy-efficient (103W)	Higher energy consumption (130W)

Touchpoints: SMART Board offers more touchpoints for enhanced collaboration.

Connectivity: SMART Board includes Bluetooth and more ports.

Energy efficiency: SMART Board is more energy-efficient.

Standing Microphone

We have chosen Audio-Technica AT2020 for this project. The major difference between Audio-Technica AT2020 with other different brands we compare is Shure SM58.

<u>Features</u>	Audio-Technica AT2020	Shure SM58
Type	Condenser, cardioid polar pattern	Dynamic, cardioid
Max SPL	144 dB	Not specified (generally lower for dynamics)
Connector	XLR	XLR

Sound sensitivity: AT2020 is better for capturing detailed sound.

Durability: SM58 is more robust and suited for live performances.

Application: AT2020 is ideal for studio recording; SM58 excels in live settings.

Video Camera

We have chosen Canon Eos M50 Mark II for this project. The major difference between Canon Eos M50 Mark II with other different brands we compare is Sony Alpha a6400.

Features	Canon Eos M50 Mark II	Sony Alpha a6400
Sensor	24MP APS-C CMOS	24.2MP APS-C CMOS
Video	4K recording	4K with better continuous recording
Connectivity	Wi-Fi, Bluetooth	Wi-Fi, Bluetooth

Video Recording: Sony offers longer 4K recording without overheating.

Focus: Canon has Dual Pixel AF, while Sony's real-time AF is faster.

Ease of Use: Canon is simpler for beginners; Sony caters to professionals.

Video Conference Terminal

We have chosen Logitech Rally Plus for this project. The major difference between Logitech Rally Plus with other different brands we compare is Poly Studio X50.

Features	Logitech Rally Plus	Poly Studio X50
Zoom and Imaging	15x HD Zoom, Ultra-HD imaging, advanced AI for framing	5x Zoom, 4K imaging, native support for video conferencing platforms
Connectivity	USB and proprietary Logitech Link	HDMI, USB, Ethernet

Zoom and Imaging: Logitech has higher zoom and AI features.

Platform Integration: Poly Studio is better for seamless meeting integration.

IoT Hardware Board

We have chosen Raspberry Pi 4 Model B for this project. The major difference between Raspberry Pi 4 Model B with other different brands we compare is Arduino MKR WiFi 1010.

Features	Raspberry Pi 4 Model B	Arduino MKR WiFi 1010
Processor	Quad-core 1.5 GHz	SAMD21 Cortex-M0
Ports	2 USB 3.0, 2 USB 2.0, 40 GPIO pins	Fewer GPIO pins (22)
Connectivity	Gigabit Ethernet, Wi-Fi	Wi-Fi, Bluetooth

Processing Power: Raspberry Pi is more powerful.

Ease of Use: Arduino is simpler for hardware prototyping.

Applications: Raspberry Pi is versatile for computing; Arduino excels in sensor-based IoT projects.

Printer

We have chosen HP LaserJet Pro MFP M428fdw for this project. The major difference between HP LaserJet Pro MFP M428fdw with other different brands we compare is Brother HL-L6200DW.

Features	HP LaserJet Pro MFP M428fdw	Brother HL-L6200DW
Speed	40 ppm	48 ppm
Connectivity	Wi-Fi, Ethernet, USB	Similar but lacks advanced touchscreen

Speed: Brother is faster.

Features: HP includes a touchscreen for ease of use.

Modem

We have chosen NETGEAR CM1200 for this project. The major difference between NETGEAR CM1200 with other different brands we compare is Arris SURFboard SB8200.

Features	NETGEAR CM1200	Arris SURFboard SB8200
Speed	Up to 2 Gbps	DOCSIS 3.1
Compatibility	Up to 2 Gbps	DOCSIS 3.1

NETGEAR includes multi-port Ethernet for direct device connections.

Performance: Both are comparable for speed and future-proofing.

Server

We have chosen Dell PowerEdge T550 for this project. The major difference between Dell PowerEdge T550 with other different brands we compare is HP ProLiant DL380 Gen10.

Features	Dell PowerEdge T550	HP ProLiant DL380 Gen10
Processor	Intel Xeon Silver 4309Y	Intel Xeon Gold 5220
Memory	16GB RDIMM	32GB RDIMM

Processor: HP offers better processing capabilities.

Cost: Dell is more cost-effective.

Power Backup Supply System

We have chosen APC Back-UPS Pro 1000VA for this project. The major difference between APC Back-UPS Pro 1000VA with other different brands we compare is CyberPower CP1000AVRLCD.

Features	APC Back-UPS Pro 1000VA	CyberPower CP1000AVRLCD
Outlets	8 (4 with battery backup)	10 (5 with battery backup)
User Interface	AVR, surge protection	LCD display for monitoring

Outlets: CyberPower offers more.

User Interface: CyberPower includes a detailed LCD display.

MEETING MINUTE

MEETING MINUTE

DATE/TIME	01 December 2024		
LOCATION	MA7		
AGENDA	Discussion on Floor Plan		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Angela Lee Su Ing		1300	-
Bernice Lou Min Yun		1300	-
Karen Yam Vei Xin		1300	-
Seow Yen Zhi		1300	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion on the requirement of the task.	-All members go through the instruction in the task 3. -Angela suggested to view the task 1 and task 2 to get some idea.	All members (01/12/2024)
2	Discussion on basic list of devices.	-Karen suggested to include the basic devices as mentioned in the case study.	All members (01/12/2024)
3	List the devices required.	-Seow in charge of listing all the devices required.	All members (01/12/2024)

		-Seow refers to several websites to get the appropriate type of devices required that is able to accomplish the requirements of FC.	
4	Table of devices.	-Bernice surveys for the devices listed previously with some specification. -Bernice surveys the price for each unit of device. -The team then together discuss the amount of each device that is required in this case study.	All members (01/12/2024)
5	Total devices.	-Seow reminds that the allocation for number of devices should be within the given budget which is RM 1.5M. -The team then revise and adjust the number of devices required.	All members (01/12/2024)
6	Reflection about is the team surprised by the price of device.	-Karen do research on the market price of the devices. -Karen presented the results of the survey and	All members (01/12/2024)

		the team discuss on the pricing. -The team is surprised that some devices vary by the cost but we also do have deeper research on the specification of devices.	
7	Reflection on cost consideration.	-The team do consider on the cost part when selecting device. -Bernice suggested to consider the devices with higher quality since our budget allocated is at the middle range compared to others. -The team aim to get the devices that is within the budget and its specification can meet our requirements.	All members (01/12/2024)
8	Comparison on same devices with different brands.	-Angela then continues with the further research on the same devices from different brands. -Angela compares the price of devices and the key features of each device.	All members (01/12/2024)

		-The team reviews whether the same devices from different brand is able to meet the basic requirement of the case study from the aspect of the key function.	
9	Meeting ended	-At 0000, the meeting ended after all discussion were done. -The team agrees to have short discussion to again review the devices whether they are able to meet the requirement of the case study.	All members (01/12/2024)

REFERENCES AND CITATION

1. *What is a router? - definition and uses* (2024) Cisco. Available at:
<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/what-is-a-router.html> (Accessed: 01 December 2024).
Product page: <https://shopee.com.my/TP-Link-Archer-AX73-AX5400-Dual-Band-Gigabit-Wireless-AX-WiFi-6-Router-Mercusys-MR70X-MR80X-MR90X-IBN-i.2794197.3278089197>
2. *What is a network switch? | switch vs. Router | Cloudflare*. Available at:
<https://www.cloudflare.com/learning/network-layer/what-is-a-network-switch/>
(Accessed: 01 December 2024).
Product page: <https://www.fs.com/sg/products/90131.html>
3. Froehlich, A., Rouse, M. and Gerwig, K. (2021) *What is a patch panel?*, *Search Networking*. Available at:
<https://www.techtarget.com/searchnetworking/definition/patch-panel> (Accessed: 01 December 2024).
Product page: <https://shopee.com.my/24-Port-Cat6-RJ45-19-1U-Rackmount-Patch-Panel-24Port-Cat5-Rack-Mount-UTP-i.76579217.2695563499>
4. *What is an access point?* (2024) Cisco. Available at:
<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/what-is-access-point.html#~what-is-an-access-point> (Accessed: 01 December 2024).
Product page: <https://shopee.com.my/Ubiquiti-U6-LITE-Access-Point-WiFi-6-LITE-Dual-band-2x2-MIMO-AP-2.4GHz-5GHz-Ceiling-Wall-mount-UBNT-UAP-SOCIAL-WIFI-i.76579217.9569497634>

5. *Network cable types and specifications* (no date) *ComputerNetworkingNotes*. Available at: <https://www.computernetworkingnotes.com/networking-tutorials/network-cable-types-and-specifications.html> (Accessed: 01 December 2024).
Product page: <https://www.fs.com/products/70611.html>
6. Kaspersky. (2020). *What is a Firewall? how firewalls work & types of firewalls*. Available at: <https://www.kaspersky.com/resource-center/definitions/firewall> (Accessed: 01 December 2024).
Product page: <https://www.fs.com/products/106549.html>
7. *Computer Basics: What is a Computer?* (no date). Available at: <https://edu.gcfglobal.org/en/computerbasics/what-is-a-computer/1/> (Accessed: 01 December 2024).
Product page: <https://www.hp.com/my-en/shop/hp-elite-tower-800-g9-desktop-pc-a3nd6pt.html?facetref=ec05aa8209ca7844>
8. *What is a projector?* (2024) *Computer Hope*. Available at: <https://www.computerhope.com/jargon/p/projector.htm> (Accessed: 01 December 2024).
Product page: <https://www.epson.com.my/Projectors/Corporate-and-Education/Epson-EB-X06-XGA-3LCD-Projector/p/V11H972052>
9. *multi-functional smart whiteboard 65 75 86 inches touchscreen interactive whiteboard for school* (no date) *Made*. Available at: <https://kep-tech.en.made-in-china.com/product/OnyrGQUJVWhL/China-Multi-Functional-Smart-Whiteboard-65-75-86-Inches-Touchscreen-Interactive-Whiteboard-for-School.html#:~:text=Real%2DTime%20Collaboration%20and%20Screen,to%20get%20audiences%20more%20engaged>. (Accessed: 01 December 2024).
Product page: <https://www.cdw.com/product/smart-mx075-v5-series-75-interactive-display/7976675>

10. *Microphone* (2024) *SVANTEK*. Available at: <https://svantek.com/academy/microphone/> (Accessed: 01 December 2024).
Product page: <https://www.musicbliss.com.my/products/audio-technica-at2020-malaysia-cardioid-condenser-microphone-at-2020>
11. *Video camera* (no date) *Video Camera - an overview | ScienceDirect Topics*. Available at: <https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/video-camera#:~:text=The%20term%20'video'%20in%20its,the%20standard%20television%20signal%20formats>. (Accessed: 01 December 2024).
Product page: [https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-\(3-YEARS-WARRANTY\)-i.46683178.1417788595](https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-(3-YEARS-WARRANTY)-i.46683178.1417788595)
12. (No date) *What is video conferencing? video conferencing definition | mitel*. Available at: <https://www.mitel.com/features-benefits/video-conferencing> (Accessed: 01 December 2024).
Product page: <https://www.logitech.com/en-us/products/video-conferencing/room-solutions/rally-ultra-hd-conferencecam.960-001225.html>
13. Author Cadence PCB Solutions *et al.* (2024) *An introduction to IOT development boards, Cadence*. Available at: <https://resources.pcb.cadence.com/blog/2022-an-introduction-to-iot-development-boards> (Accessed: 01 December 2024).
Product page: <https://my.element14.com/raspberry-pi/rpi4-modbp-4gb/raspberry-pi-4-model-b-4gb/dp/3051887>
14. *Color imageclass mf751cdw* (no date) *Shop Canon Color imageCLASS MF751Cdw | Canon U.S.A., Inc.* Available at: <https://www.usa.canon.com/shop/p/color-imageclass-mf751cdw?color=White&type=New> (Accessed: 01 December 2024).
Product page: <https://support.hp.com/us-en/drivers/hp-laserjet-pro-mfp-m428-m429-f-series/model/19204098>

15. *Modem definition & meaning* (no date) *Merriam-Webster*. Available at:
<https://www.merriam-webster.com/dictionary/modem> (Accessed: 01 December 2024).
Product page: <https://www.netgear.com/home/wifi/modems/cm1200/>
16. *What is a computer server* (no date) *What is a computer server | Lenovo US*. Available at: <https://www.lenovo.com/us/en/glossary/what-is-a-computer-server/?orgRef=https%253A%252F%252Fwww.google.com%252F>
(Accessed: 01 December 2024).
Product page: <https://www.dell.com/en-my/shop/ipovw/poweredge-t550>
17. Jon (2023) *What is a ups and how does it work?*, *Ormazabal*. Available at:
<https://www.ormazabal.com/en-gb/what-is-a-ups-and-how-does-it-work-2-2/>
(Accessed: 01 December 2024).
Product page: <https://www.amazon.com/APC-BR1000G-Back-UPS-Computer-Protector/dp/B077SR55QT>

Product Page

18. Router

<https://shopee.com.my/TP-Link-Archer-AX73-AX5400-Dual-Band-Gigabit-Wireless-AX-WiFi-6-Router-Mercusys-MR70X-MR80X-MR90X-IBN-i.2794197.3278089197>

19. Switches

<https://www.fs.com/sg/products/90131.html>

20. Patch Panel

<https://shopee.com.my/24-Port-Cat6-RJ45-19-1U-Rackmount-Patch-Panel-24Port-Cat5-Rack-Mount-UTP-i.76579217.2695563499>

21. Wireless Access Point

<https://shopee.com.my/Ubiquiti-U6-LITE-Access-Point-WiFi-6-LITE-Dual-band-2x2-MIMO-AP-2.4GHz-5GHz-Ceiling-Wall-mount-UBNT-UAP-SOCIAL-WIFI-i.76579217.9569497634>

22. Cable

<https://www.fs.com/products/70611.html>

23. Firewall

<https://www.fs.com/products/106549.html>

24. Personal Computer

<https://www.hp.com/my-en/shop/hp-elite-tower-800-g9-desktop-pc-a3nd6pt.html?facetref=ec05aa8209ca7844>

25. Projector
<https://www.epson.com.my/Projectors/Corporate-and-Education/Epson-EB-X06-XGA-3LCD-Projector/p/V11H972052>
26. Projector Screen
<https://shop.elitescreens.com/products/elite-screen-manual-b?variant=42697092202711>
27. Multifunctional Whiteboard
<https://www.cdw.com/product/smart-mx075-v5-series-75-interactive-display/7976675>
28. Standing Microphone
<https://www.musicbliss.com.my/products/audio-technica-at2020-malaysia-cardioid-condenser-microphone-at-2020>
29. Video Camera
[https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-\(3-YEARS-WARRANTY\)-i.46683178.1417788595](https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-(3-YEARS-WARRANTY)-i.46683178.1417788595)
30. Video Conference Terminal
<https://www.logitech.com/en-us/products/video-conferencing/room-solutions/rally-ultra-hd-conferencecam.960-001225.html>
31. IoT Hardware Board
<https://my.element14.com/raspberry-pi/rpi4-modbp-4gb/raspberry-pi-4-model-b-4gb/dp/3051887>

32. Printer

<https://support.hp.com/us-en/drivers/hp-laserjet-pro-mfp-m428-m429-f-series/model/19204098>

33. Modem

<https://www.netgear.com/home/wifi/modems/cm1200/>

34. Server

<https://www.dell.com/en-my/shop/ipovw/poweredge-t550>

35. Power Backup Supply System

<https://www.amazon.com/APC-BR1000G-Back-UPS-Computer-Protector/dp/B077SR55QT>



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

Subject : SECR 1213 Network Communication

Section: 01

Task 4: Making the Connections – LAN and WAN

Lecturer : Dr. Muhammad Zafran bin Muhammad Zaly Shah

Date: 23/12/2024

Group Name: Beauties

Members' Name & Matric Number:

Name	Matric Number
Angela Lee Su Ing	A23CS0047
Bernice Lou Min Yun	A23CS0056
Karen Yam Vei Xin	A23CS0093
Seow Yen Zhi	A23CS0177

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IDENTIFY THE WORK AREAS ON FLOOR PLAN

GROUND FLOOR

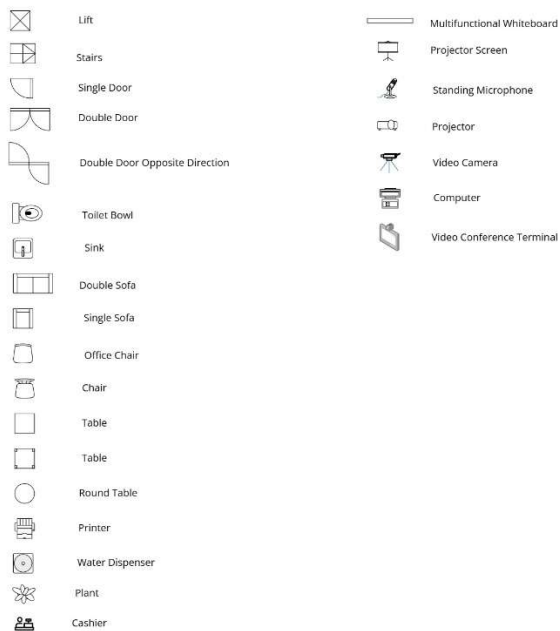
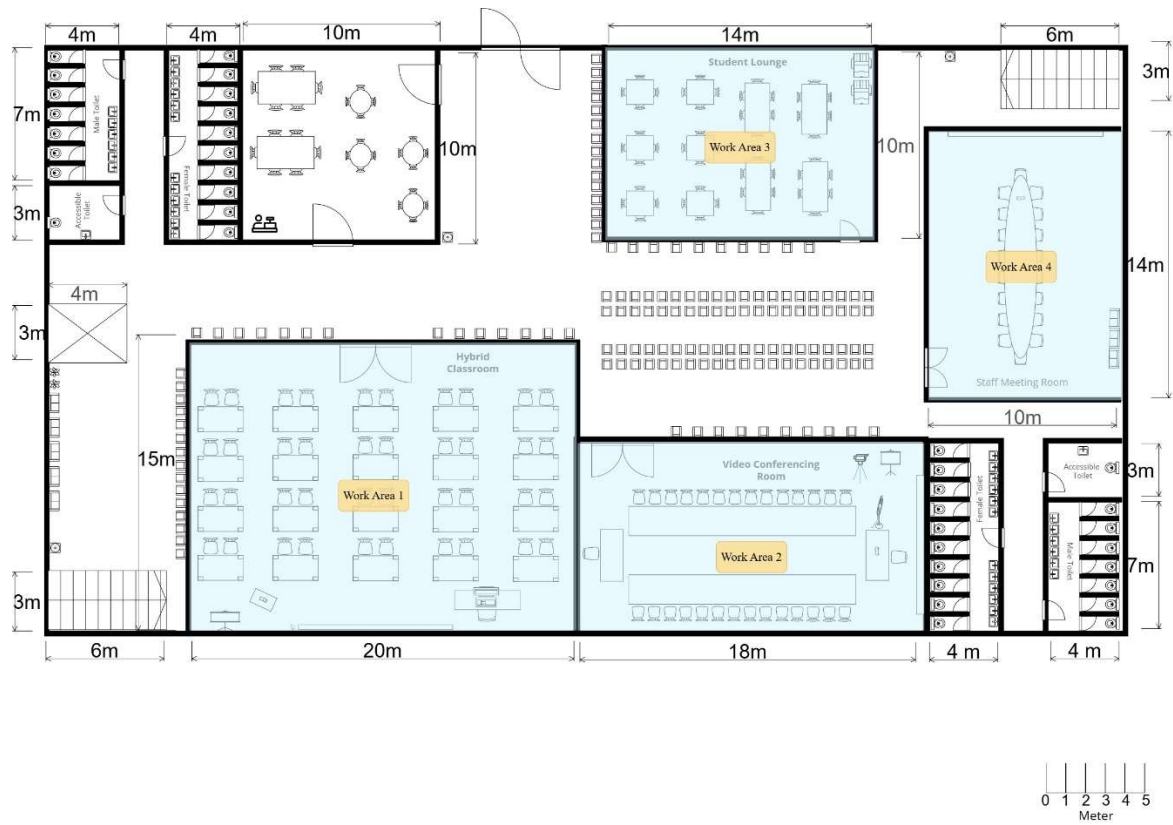


Figure 4.1.1 Work Area at Ground Floor

As shown in Figure 4.1.1 and Figure 4.1.2, there are total nine work areas in our designed project. There are four work areas located in ground floor and five work areas located in first floor.

Work Area 1 is the Hybrid Classroom with length of 20 meters and width of 15 meters. The Hybrid Classroom is designed to have best performance for teaching and learning. Hybrid Classroom is equipped with the multifunctional white board, projector screen, projector, video conference terminal and a computer for lecturer to assist the needs in the lecture.

Work Area 2 is the Video Conferencing Room with length of 18 meters and width of 10 meters. The Video Conferencing Room is designed to have multifunctional whiteboard, projector screen, projector, video camera and standing microphone. It can be seen that the video conference equipment in the Video Conferencing Room is more so that the video conference can be done smoothly.

Work Area 3 is the Student Lounge with length of 14 meters and width of 10 meters. The Student Lounge is equipped with 2 printers and multiple charging ports as it is designed for students to do their work and project discussions with their own laptop. The absence of computer in this work area is to be convenient for student's discussion to go on.

Work Area 4 is the Staff Meeting Room with length of 10 meters and width of 14 meter. The Staff Meeting Room is equipped with multifunctional whiteboard and projector. The meeting can be done when the staff's personal computer is connected to the projector or multifunctional whiteboard so that the content can be projected and been discussed in the meeting.

FIRST FLOOR

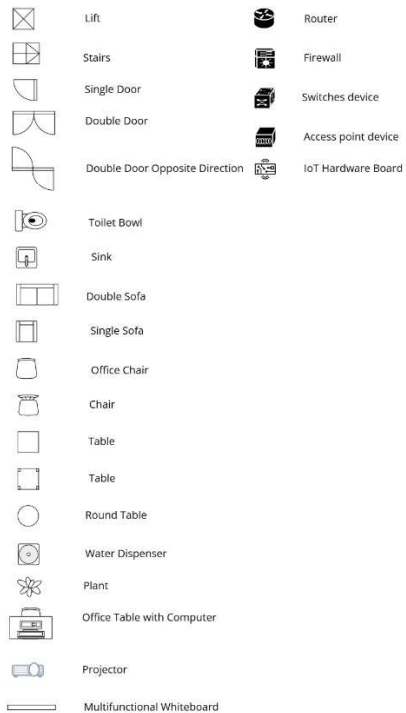
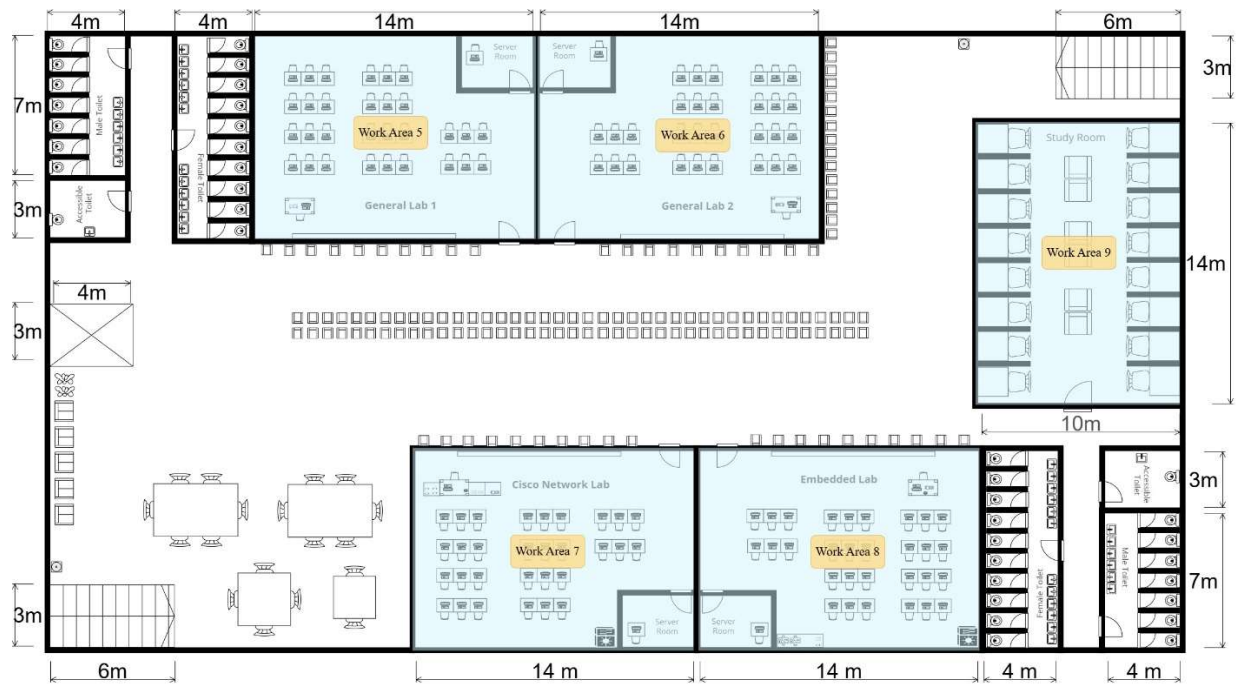


Figure 4.1.2 Work Area at First Floor

Work Area 5 and 6 are basically General Lab 1 and 2 with the exactly same design being mirrored. Each work area consists of one general lab and one server room. Each general lab has length of 14 meters and width of 10 meters. Each general lab consists of 31 computers with one projector and one multifunctional whiteboard. Each server room is equipped with computer and server. For the server to function well, routers and switches are also required.

Work Area 7 is the Cisco Network Lab and one server room with the length of 14 meters and width of 10 meters. Cisco Network Lab consist of 31 computers, multifunctional whiteboard, projector, router, switches device, access point devices, patch panels and firewall. The server room is equipped with computer and server. For the server to function well, routers and switches are also required.

Work Area 8 is the Embedded Lab and one server room with the length of 14 meters and width of 10 meters. Embedded Lab consists of 31 computers, multifunctional whiteboard, projector, router, switches device, access point devices, firewall and IoT hardware board. The server room is equipped with computer and server. For the server to function well, routers and switches are also required.

Work Area 9 is the study room with length of 10 meters and width of 14 meters. The study room provide a conducive workspace for students to have their self-study session. Study room should have desks and chairs that are suitable for study. Charging ports, modem and Wireless Access Point Devices are also essential.

THE DISTRIBUTION AND CONNECTION OF DEVICES

NETWORK DISTRIBUTION DIAGRAM

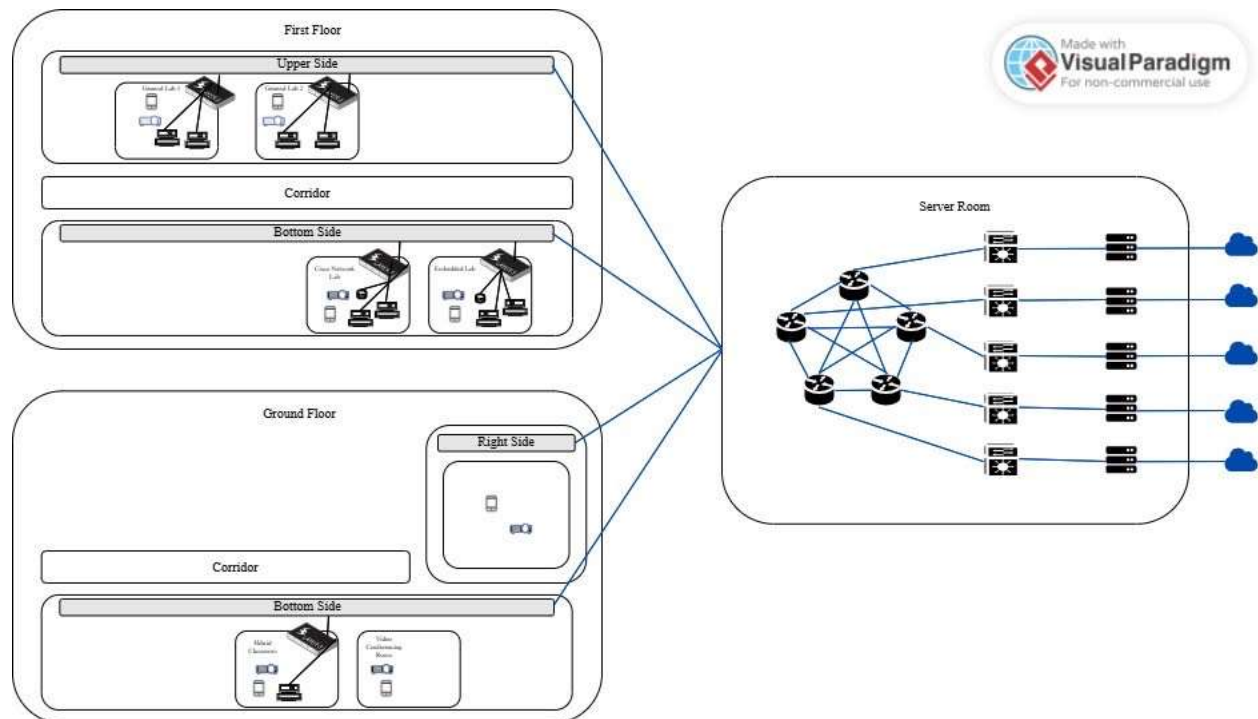


Figure 4.2.1.1: Network Distribution Diagram

The layout of network devices shown in the Network Distribution indicates the manner in which internet connection is designed in different rooms. Figure 4.2.1.1 shows the Network Distribution Diagram of upper and lower parts for the first floor and right and lower parts for the ground floor. Of particular interest is that all the rooms, aside from the rest area, the meeting room, the student lounge, and the staff meeting room all have a network switch installed.

A more detailed presentation of how the network distribution is within the first and the second floor is presented in the figures 4.2.1.2 below and the 4.2.1.3 below. Incidentally, one must note that the lines that connect the various parts of the building in the diagram do not run directly to the router in the server room. This intentional design choice plays an additional role in preventing the layout of an excessively abundant amount of material.

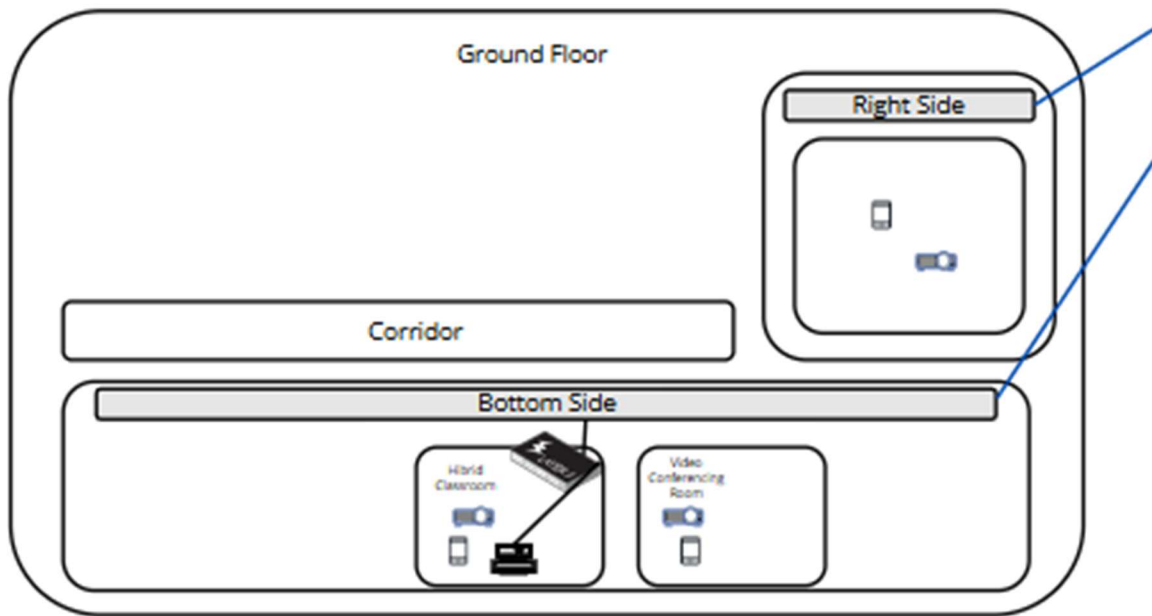


Figure 4.2.1.2: Network Distribution Diagram - Ground Floor

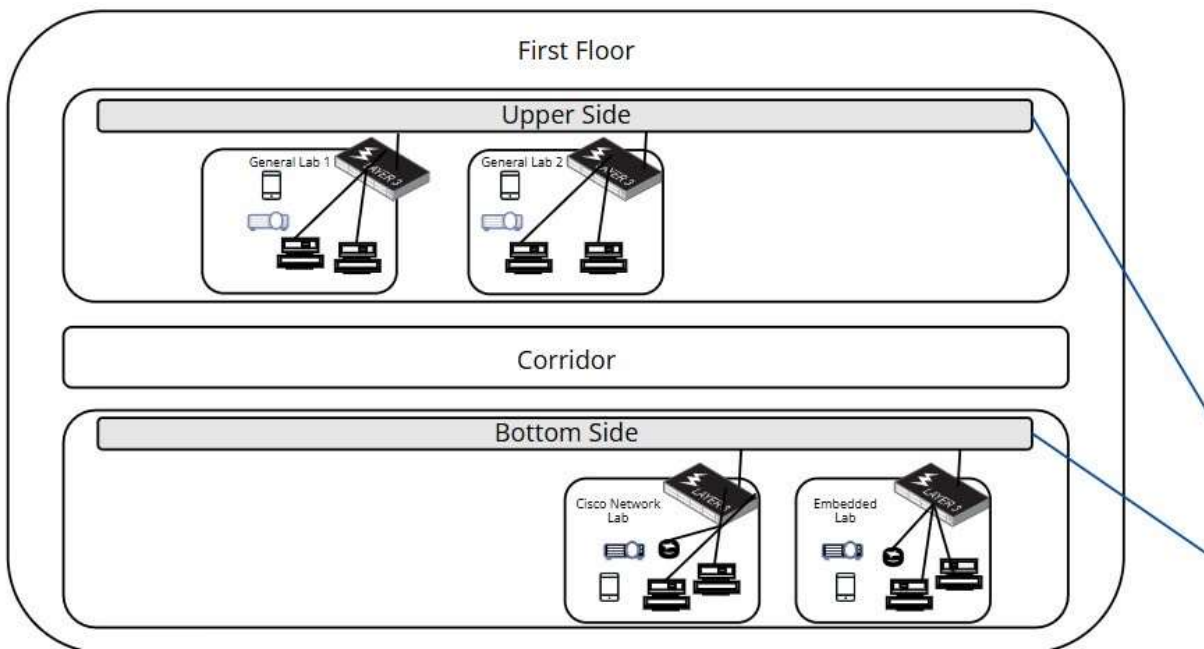


Figure 4.2.1.3: Network Distribution Diagram- First Floor

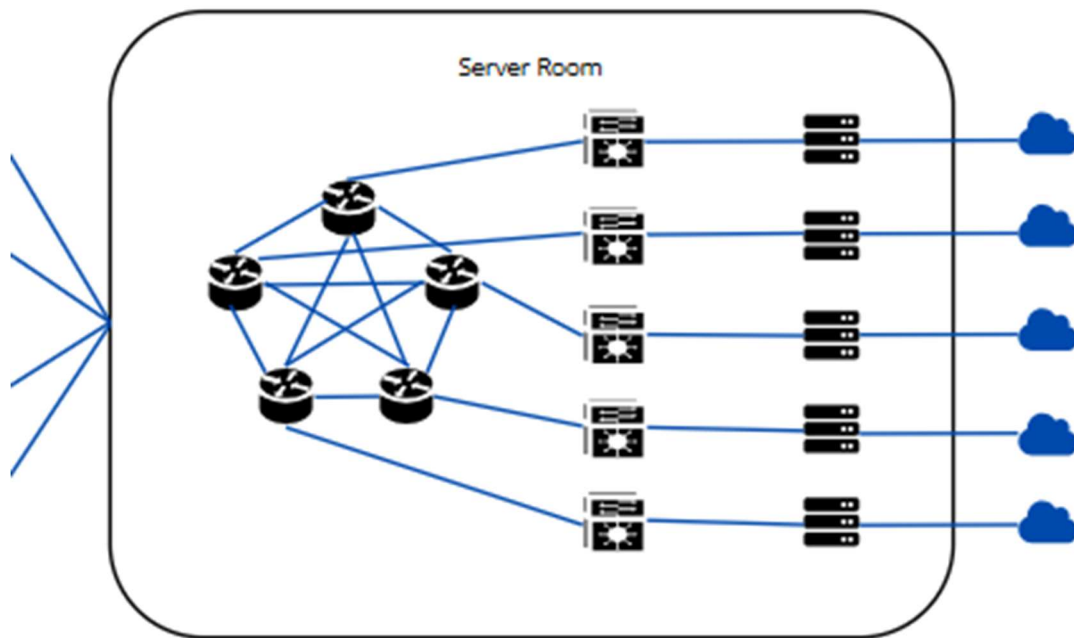


Figure 4.2.1.4: Network Distribution Diagram - Server Room

The next diagram refers to the network topology distribution in the server room with the emphasis on the router one. Five routers are currently incorporated into the setup but all are connected independently using fiber optic cable. This design also ensures that there is no total network failure in case one of the routers has developed a fault. To ensure increased reliability, the routers are connected in a mesh topology. In other words, each router is connected to all the other routers in the network. This means that there is always a spare router to plug into the net to continue with operation if another router is out of order. Also, most of the rooms are provided with just one switch to help in creating subnets thus making the subnets very stable. More importantly, what is shown is that one switch is actually connected to two routers, a backup connection in case of emergencies or mishaps.

ROOM LEVEL DISTRIBUTION DIAGRAM

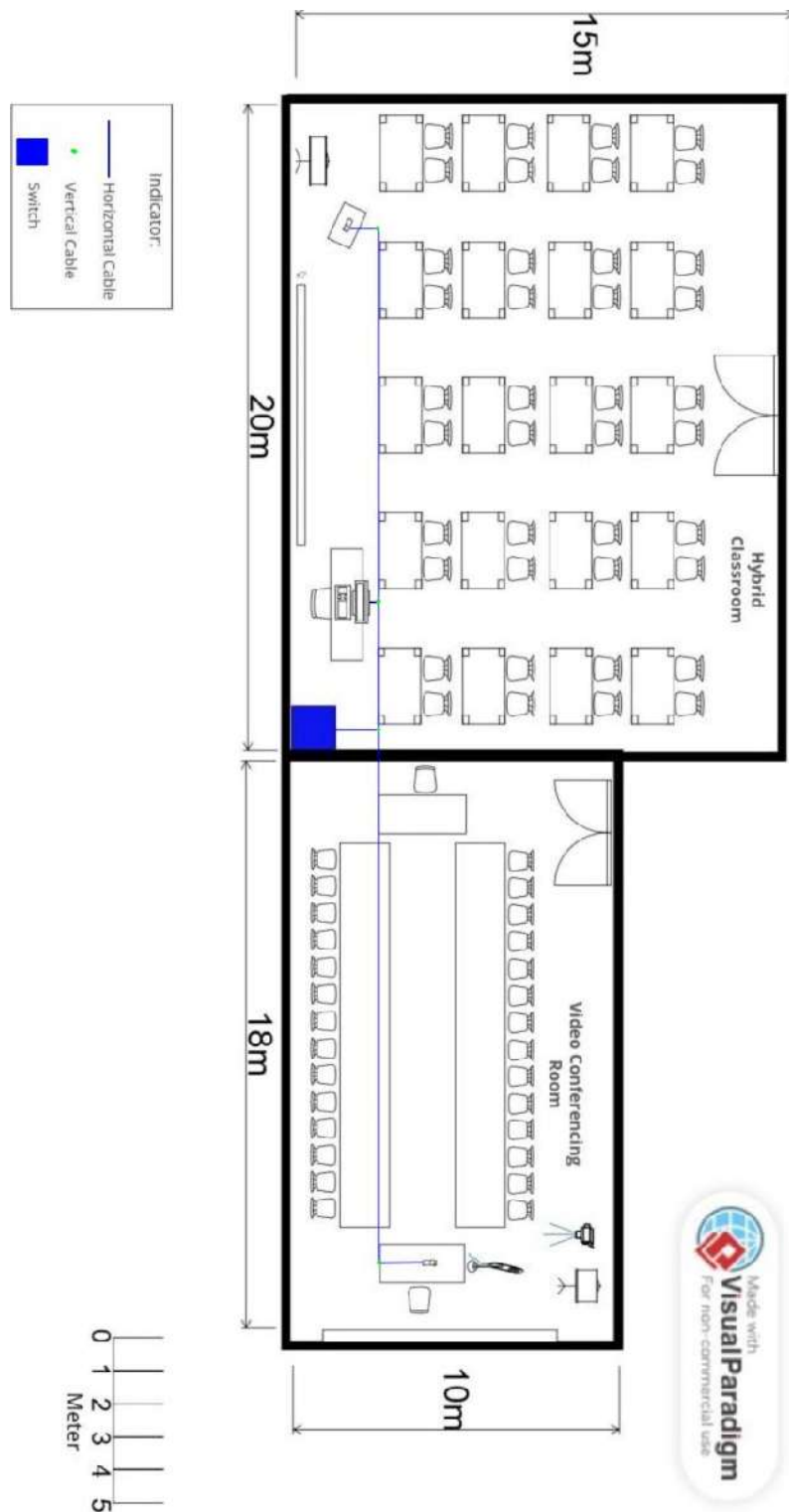


Figure 4.2.2.1: Hybrid Classroom and Video Conferencing Room

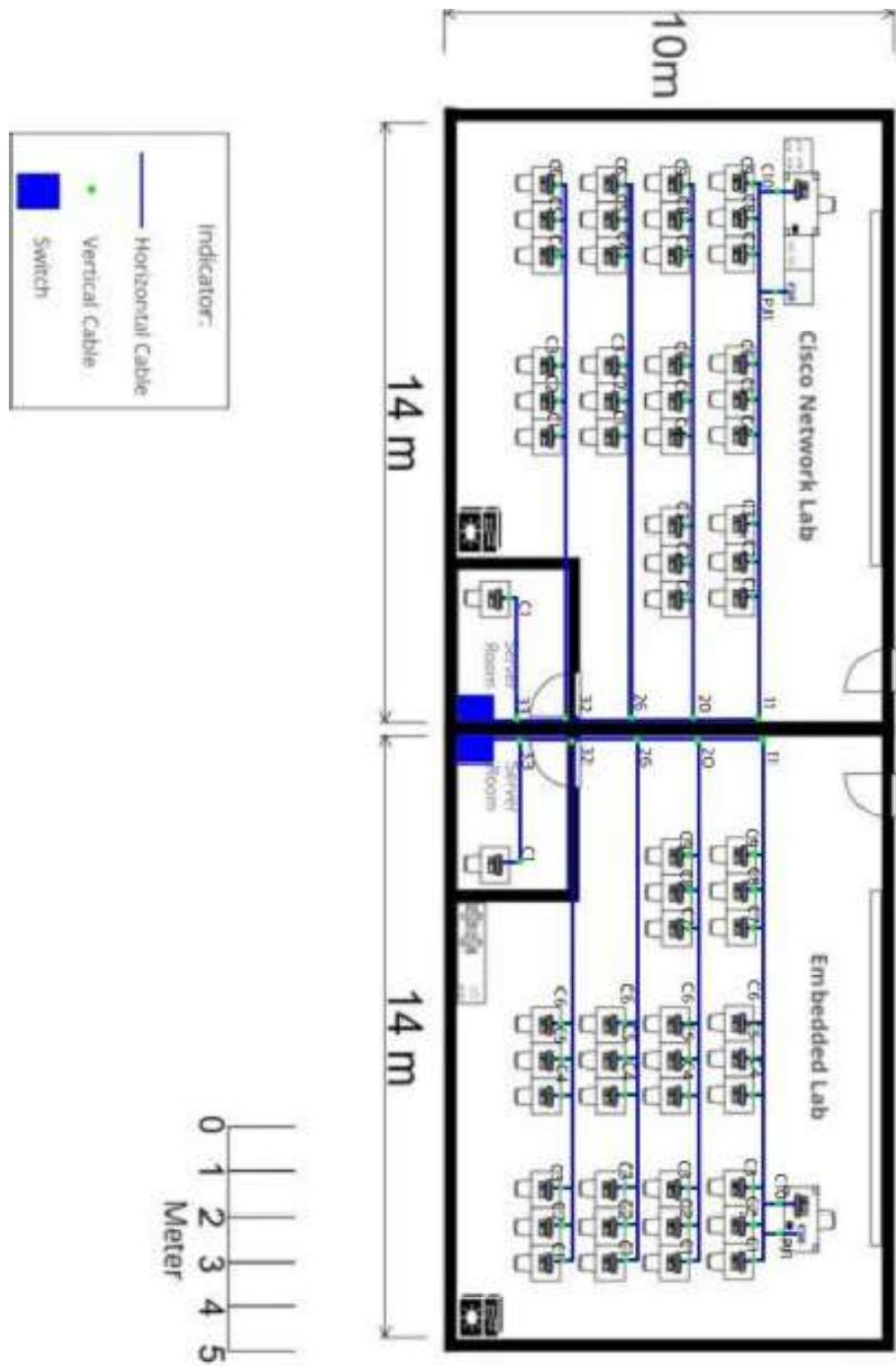


Figure 4.2.2.2: Cisco Network Lab and Embedded Lab

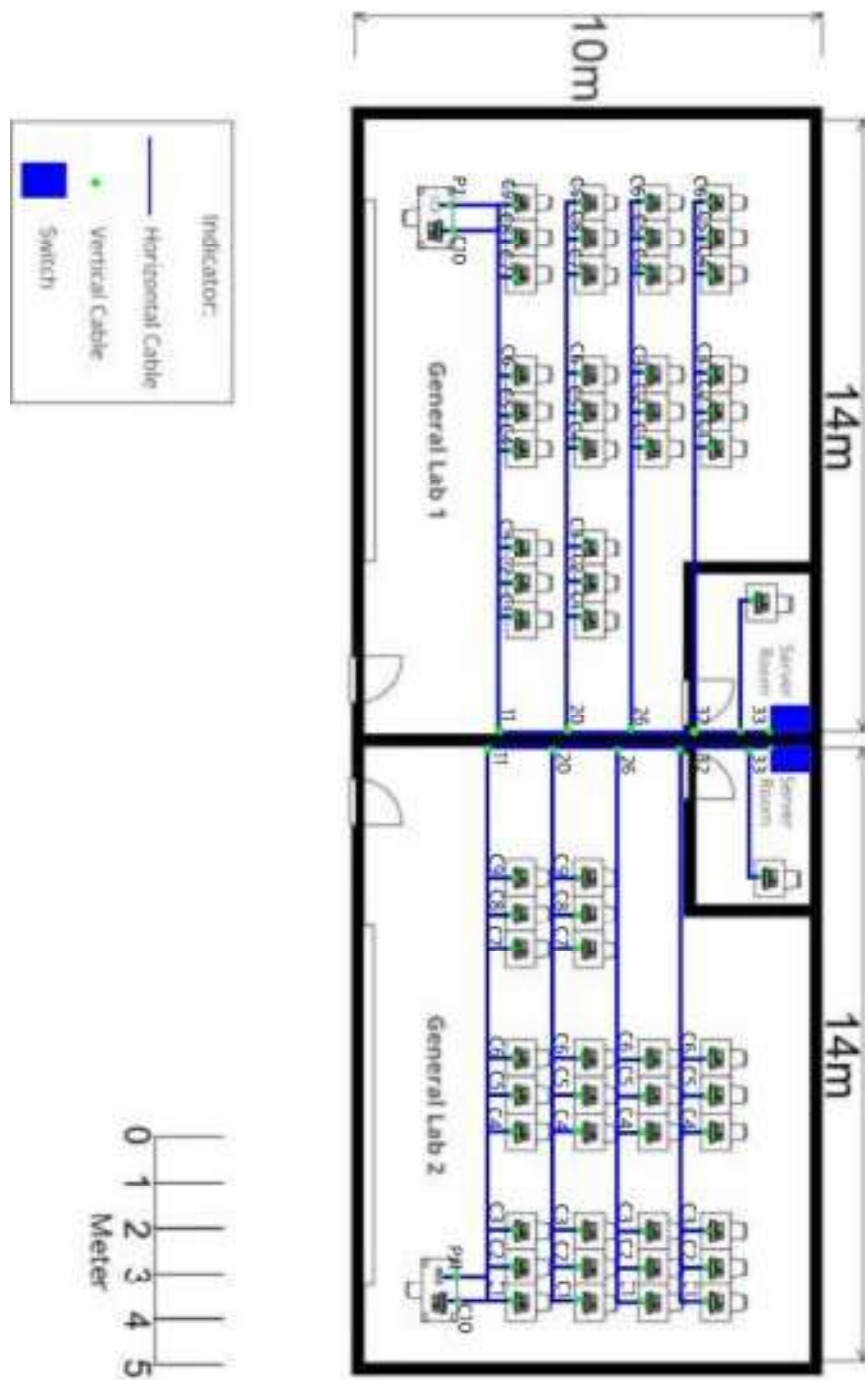


Figure 4.2.2.3: General Lab

FLOOR DIAGRAM

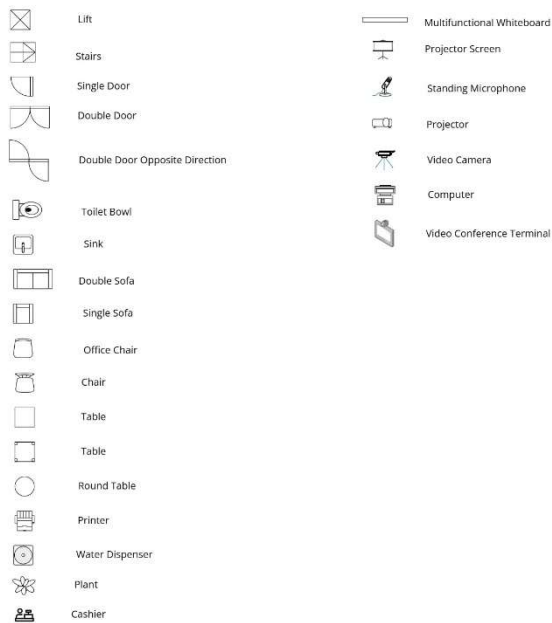
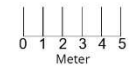
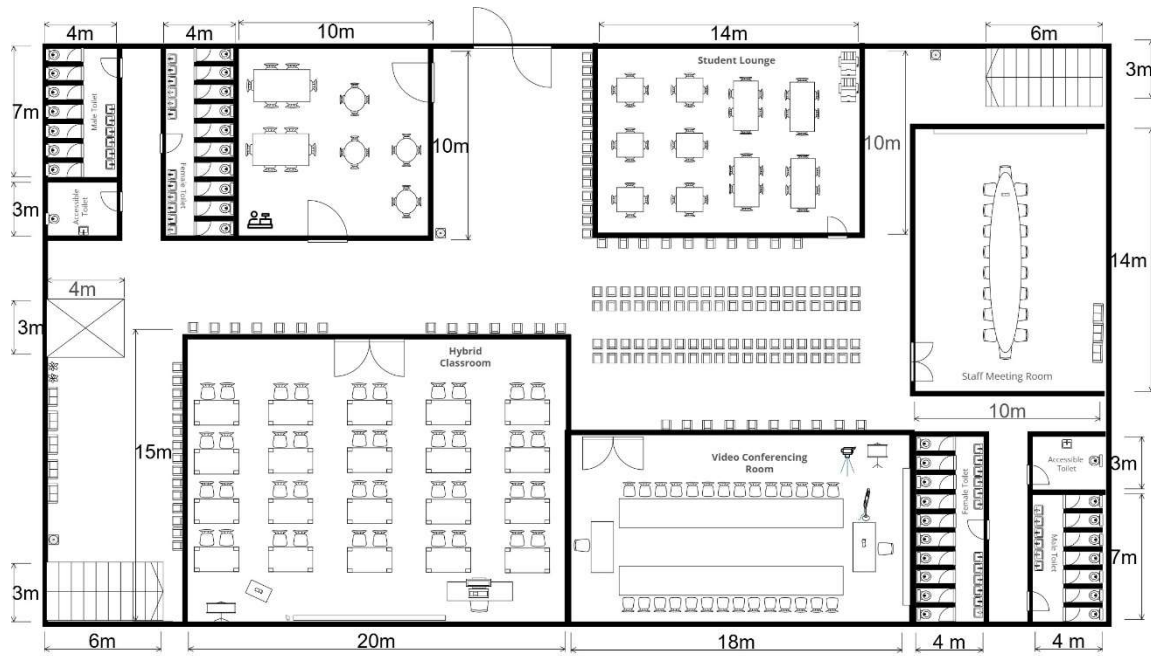


Figure 4.2.3.1: Floor Diagram of Ground Floor

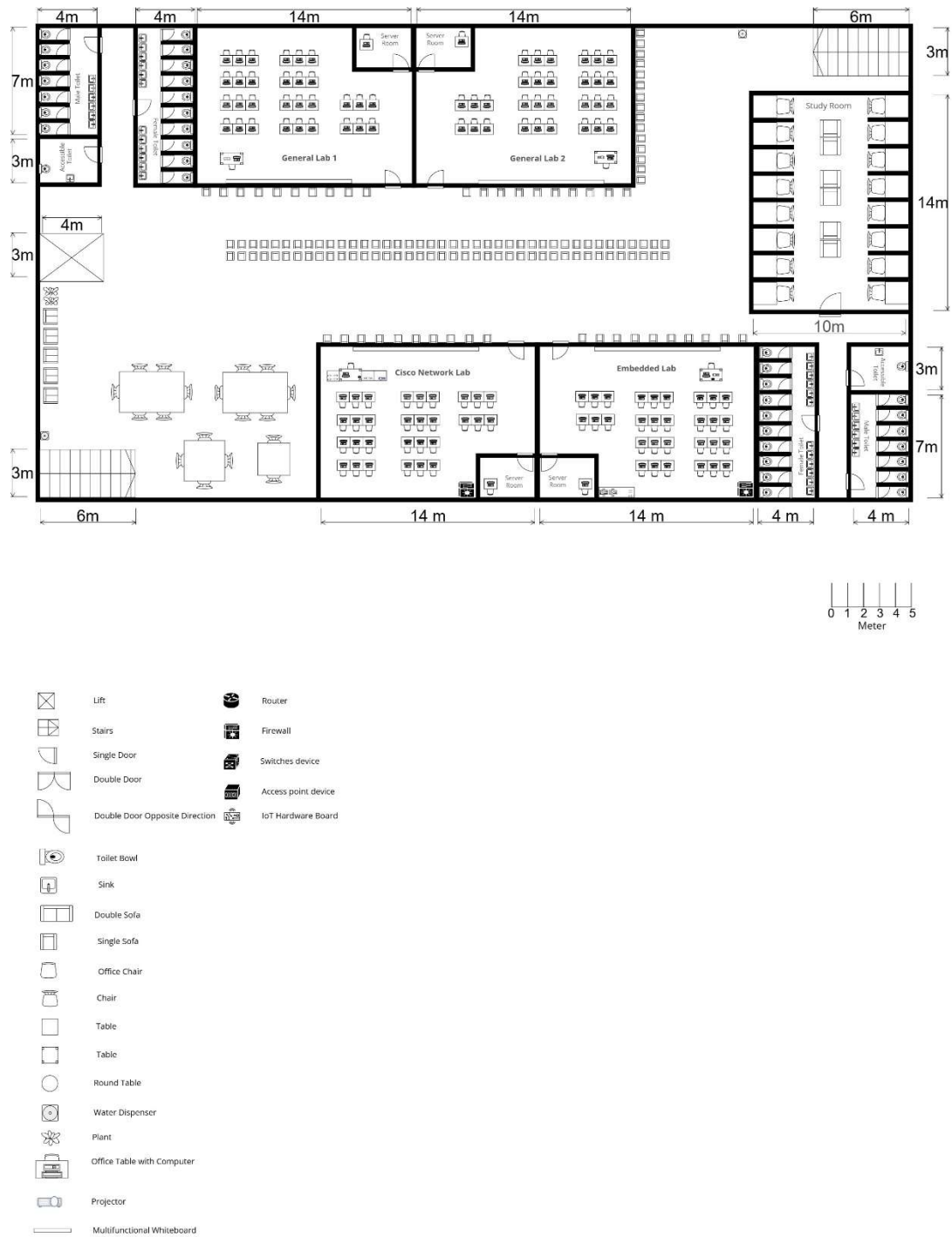


Figure 4.2.3.2 Floor Diagram of First Floor

NUMBER OF CONNECTIONS, PATCH CORDS AND SWITCH PORTS NEEDED

Area	Number of Switch Ports Needed
General Lab 1	32
General Lab 2	32
Cisco Network Lab	33
IoT Lab	33
Video Conference Room	3
Office	27
Meeting Room	1
Staff Meeting Room	1
Server Room	6
Total	168

Table 4.3.1 Number of Switch Ports Needed

Area	Number of Patch Cords Needed
General Lab 1	32
General Lab 2	32
Cisco Network Lab	33
IoT Lab	33
Video Conference Room	3
Office	27
Meeting Room	1
Staff Meeting Room	1
Server Room	6
Total	168

Table 4.3.2 Number of Patch Cords Needed

Area	Number of RJ45 Connector Needed
General Lab 1	64
General Lab 2	64
Cisco Network Lab	66
IoT Lab	66
Video Conference Room	6
Office	54
Meeting Room	2
Staff Meeting Room	2
Server Room	12
Total	336

Table 4.3.3 Number of RJ45 Connector Needed

IDENTIFY CABLE TYPES AND LENGTH

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
Room				
Hybrid Classroom	31.7	6.2	37.9	268.84
Video Conference Room	30.2	5.3	35.5	251.82
Staff Meeting Room	8.8	1.2	10.0	70.93
Ground Floor Room Total			83.4	591.59
First Floor				
Room				
General Lab 1	385.6	76.2	461.8	3275.76
General Lab 2	355.3	40.7	396.0	2809.01
Cisco Network Lab	328.2	15.3	343.5	2436.60
Embedded Lab	649.5	95.2	744.7	5282.50
Server Room	81.1	16.5	97.6	682.32
First Floor Room Total			2043.6	14486.19

Table 4.4.1: Number of Cable Types and Length

Ground Floor		
Path	Length (m)	Price of cables used (RM)
Server Room to Hybrid Classroom	29.9	212.09
Server Room to Video Conference Room	25.5	180.88
Ground Floor Room Total	55.4	392.97
Total - Ground Floor		
Total Length of Wire to Connect the Room (m)	55.4	392.97
Total Length of Wire in Horizontal and Vertical (m)	83.4	591.59
Total Length of Wire Needed (m)	138.8	984.56
First Floor		
Server Room to General Lab 1	64.8	459.66
Server Room to General Lab 2	64.8	459.66
Server Room to Cisco Network Lab	64.8	459.66
Server Room to Embedded Lab	64.8	459.66
First Floor Room Total	259.2	1838.64
Total - First Floor		
Total Length of Wire to Connect the Room (m)	259.2	1838.64
Total Length of Wire in Horizontal and Vertical (m)	2043.6	14486.19
Total Length of Wire Needed (m)	2302.8	16324.83
Total (Ground Floor + First Floor) (m)	2441.6	16717.80

Table 4.4.: Total Number of Cable Types and Length

In the preceding section, we outlined the distribution strategy and device connections. To optimize connectivity, careful consideration of cable types is essential. Following our group deliberation, we are opted for FS Cat6 Snagless Unshielded (UTP) Ethernet Network Patch Cable within the building.

CAT6 cables are selected for both horizontal and vertical computer cabling within rooms like the lab and video conference room. This choice is driven by CAT6's ability to support high-speed data transmission, reaching up to 10 gigabits, owing to its impressive bandwidth of 550 MHz per cable. The thicker copper wire in CAT6 facilitates efficient heat dissipation, making it apt for Power over Ethernet (PoE), thereby reducing cabling time and costs. CAT6 also excels in supporting Ethernet applications across the building. The specific type of CAT6 chosen is Unshielded Twisted Pair (UTP), known for minimizing crosstalk and reducing electromagnetic interference caused by natural or human-made sources.

Moreover, CAT6 cables are also used for interconnecting routers in the server room and extending the connection to the Internet Services Provider (ISP). Utilizing copper wire for data transmission, CAT6 cables support high-speed data transfer of up to 10Gbps over distances up to 55 meters without significant signal degradation. With a bandwidth of 550 MHz, CAT6 cables are well-suited for handling substantial data volumes in shorter runs. While they require signal boosters or switches for longer distances, CAT6 cables provide a cost-effective solution for reliable and fast data transmission within the building and to the ISP.

In summary, our building necessitates an estimated length of CAT6 cable to ensure seamless internal connectivity among devices within the building and a reliable linkage to the ISP. This choice balances cost-effectiveness with the performance requirements of high-speed data transmission.

MEETING MINUTE

MEETING MINUTE

DATE/TIME	20 December 2024		
LOCATION	MA7		
AGENDA	Discussion on Floor Plan		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Angela Lee Su Ing	1300		-
Bernice Lou Min Yun	1300		-
Karen Yam Vei Xin	1300		-
Seow Yen Zhi	1300		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 4	- All members go through the instruction in the task 4. - Solved the problem raised by group members together through discussion.	All members (20/12/2024)
2	Suggestion for network distribution	- Angela suggested the network distribution in the floor plan. - Seow and Bernice suggested the necessary details that need to be standardized.	All members (20/12/2024)

3	Task distribution	- Seow distributed the task to all members equally. - Karen was assigned to plan for the network distribution of IoT Lab. - Bernice was assigned to plan for the network distribution of Cisco Network Lab. - Angela was assigned to plan for the network distribution of the video conference room. - Seow was assigned to plan for the network distribution of the general lab.	All members (20/12/2024)
4	Finalize room level network distribution diagram	-Karen suggested making the wiring more visible by adding colour and thickening it. -Bernice suggested labeling the wiring. -The team then together discuss the amount of each device that is required in this case study. - Angela suggested to differentiate vertical and	All members (20/12/2024)

		horizontal cable by using different labeling. - All members share their ideas on how to improve the room level distribution diagram.	
5	Check the calculation of cable length needed	- Seow reminded about the correct calculation of cable length needed. - Bernice reminds that the allocation for number of devices should be within the given budget which is RM 1.5M. -The team then revise and adjust the number of devices required.	All members (20/12/2024)
6	Meeting ended	-At 1700, the meeting ended after all discussion were done. -The team agrees to have short discussion to again review the devices whether they are able to meet the requirement of the case study.	All members (20/12/2024)



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Subject : SECR 1213 Network Communication

Section: 01

Task 5: IP Addressing Scheme

Lecturer : Dr. Muhammad Zafran bin Muhammad Zaly Shah

Date: 13/01/2024

Group Name: Beauties

Members' Name & Matric Number:

Name	Matric Number
Angela Lee Su Ing	A23CS0047
Bernice Lou Min Yun	A23CS0056
Karen Yam Vei Xin	A23CS0093
Seow Yen Zhi	A23CS0177

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Distribution of Network Address for the Network	
Network Distribution for Different Work Area	3-4
Network Distribution for Workstation in Different Work Areas	5-9
Meeting Minute	10-11

NETWORK ADDRESS ASSIGNED

Group/Section	Network Address
1	192.16.0.0/8
2	192.17.0.0/8
3	192.18.0.0/8
4	192.19.0.0/8
5	192.20.0.0/8
6	192.21.0.0/8
7	192.22.0.0/8
8	192.23.0.0/8
9	192.24.0.0/8
10	192.25.0.0/8

Table above shows the list of network addresses assigned by the lecturer. Based on the allocated budget for our group which is RM 1.5M, our group is the Group 6. Thus, the network address assigned to our group by the lecturer is 192.21.0.0/8.

Network Address details:

IP Address (decimal)	192.21.0.0
IP Address (binary)	1100 0000. 0001 0101. 0000 0000. 0000 0000
Subnet Mask (decimal)	255.0.0.0
Subnet Mask (binary)	1111 1111. 0000 0000. 0000 0000.0000 0000

To further subnet this network address into smaller subnets, we subnet this /8 network into /15 network by having the first 15 bits as the network address and the remaining 17 bits as the host address. The reason /15 is used instead of /16 is that /16 might leave many unused IP Address. /15 subnet mask can allow a better scalability and address allocation by creating more manageable subnets.

Network Address details:

IP Address (decimal)	192.21.0.0
IP Address (binary)	1100 0000. 0001 0101. 0000 0000. 0000 0000
Subnet Mask (decimal)	255.254.0.0
Subnet Mask (binary)	1111 1111. 1111 1111. 0000 0000.0000 0000

Then, we performed the AND operation for the IP Address and the subnet mask to get the range of usable network address. We keep the first IP Address as the network address and the last IP Address as the broadcast address. Leaving the others to be the valid IP Address to be use.

Network Address details:

IP Address (decimal)	192.21.0.0
Broadcast Address (decimal)	192.21.255.255
Router Address (decimal)	192.21.0.1
Range of Usable IP Address (decimal)	192.21.0.2 – 192.21.255.254

DISTRIBUTION OF NETWORK ADDRESS FOR THE NETWORK

NETWORK DISTRIBUTION FOR DIFFERENT WORK AREA

Subnet	Network Portion	Subnet Bits	Host Portion		IP Address
0	1100 0000 0001 0101	000	00000 00000000	Network Address	192.21.0.0
			11111 11111111	Broadcast Address	192.21.31.255
1	1100 0000 0001 0101	001	00000 00000000	Network Address	192.21.32.0
			11111 11111111	Broadcast Address	192.21.63.255
2	1100 0000 0001 0101	010	00000 00000000	Network Address	192.21.64.0
			11111 11111111	Broadcast Address	192.21.95.255
3	1100 0000 0001 0101	011	00000 00000000	Network Address	192.21.96.0
			11111 11111111	Broadcast Address	192.21.127.255
4	1100 0000 0001 0101	100	00000 00000000	Network Address	192.21.128.0
			11111 11111111	Broadcast Address	192.21.159.255

Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Subnet Mask Address
Ground Floor					
0	Student Lounge & Staff Meeting Room	192.21.0.0	192.21.31.255	192.21.0.2 – 192.21.31.254	/19 255.255.224.0
1	Hybrid Classroom & Video Conferencing Room	192.21.32.0	192.21.63.255	192.21.32.2 – 192.21.63.254	/19 255.255.224.0
First Floor					
2	General Lab	192.21.64.0	192.21.95.255	192.21.64.2 - 192.21.95.254	/19 255.255.224.0
3	Cisco Network Lab & Embedded Lab	192.21.96.0	192.21.127.255	192.21.96.2 - 192.21.127.254	/19 255.255.224.0
4	Study Room	192.21.128.0	192.21.159.255	192.21.128.2 - 192.21.159.254	/19 255.255.224.0

NETWORK DISTRIBUTION FOR WORKSTATION IN DIFFERENT WORK AREAS

Ground floor

1. Hybrid classroom

Devices	IP address
PC 1	192.21.32.2
PJ1	192.21.32.3

2. Video conferencing room

Devices	IP address
PJ2	192.21.32.4

First floor

1. General lab 1

Devices	IP address
PC2	192.21.64.5
PC3	192.21.64.6
PC4	192.21.64.7
PC5	192.21.64.8
PC6	192.21.64.9
PC7	192.21.64.10
PC8	192.21.64.11
PC9	192.21.64.12
PC10	192.21.64.13
PC11	192.21.64.14
PC12	192.21.64.15
PC13	192.21.64.16
PC14	192.21.64.17
PC15	192.21.64.18
PC16	192.21.64.19
PC17	192.21.64.20
PC18	192.21.64.21
PC19	192.21.64.22
PC20	192.21.64.23
PC21	192.21.64.24
PC22	192.21.64.25
PC23	192.21.64.26
PC24	192.21.64.27
PC25	192.21.64.28
PC26	192.21.64.29
PC27	192.21.64.30
PC28	192.21.64.31
PC29	192.21.64.32
PC30	192.21.64.33
PC31	192.21.64.34
PC32	192.21.64.35
PC33	192.21.64.36
PJ3	192.21.64.37

2. General lab 2

Devices	IP address
PC34	192.21.64.38
PC35	192.21.64.39
PC36	192.21.64.40
PC37	192.21.64.41
PC38	192.21.64.42
PC39	192.21.64.43
PC40	192.21.64.44
PC41	192.21.64.45
PC42	192.21.64.46
PC43	192.21.64.47
PC44	192.21.64.48
PC45	192.21.64.49
PC46	192.21.64.50
PC47	192.21.64.51
PC48	192.21.64.52
PC49	192.21.64.53
PC50	192.21.64.54
PC51	192.21.64.55
PC52	192.21.64.56
PC53	192.21.64.57
PC54	192.21.64.58
PC55	192.21.64.59
PC56	192.21.64.60
PC57	192.21.64.61
PC58	192.21.64.62
PC59	192.21.64.63
PC60	192.21.64.64
PC61	192.21.64.65
PC62	192.21.64.66
PC63	192.21.64.67
PC64	192.21.64.68
PC65	192.21.64.69
PJ4	192.21.64.70

3. Cisco network lab

Devices	IP address
PC66	192.21.96.2
PC67	192.21.96.3
PC68	192.21.96.4
PC69	192.21.96.5
PC70	192.21.96.6
PC71	192.21.96.7
PC72	192.21.96.8
PC73	192.21.96.9
PC74	192.21.96.10
PC75	192.21.96.11
PC76	192.21.96.12
PC77	192.21.96.13
PC78	192.21.96.14
PC79	192.21.96.15
PC80	192.21.96.16
PC81	192.21.96.17
PC82	192.21.96.18
PC83	192.21.96.19
PC84	192.21.96.20
PC85	192.21.96.21
PC86	192.21.96.22
PC87	192.21.96.23
PC88	192.21.96.24
PC89	192.21.96.25
PC90	192.21.96.26
PC91	192.21.96.27
PC92	192.21.96.28
PC93	192.21.96.29
PC94	192.21.96.30
PC95	192.21.96.31
PC96	192.21.96.32
PC97	192.21.96.33
PJ5	192.21.96.34

4. Embedded lab

Devices	IP address
PC98	192.21.96.35
PC99	192.21.96.36
PC100	192.21.96.37
PC101	192.21.96.38
PC102	192.21.96.39
PC103	192.21.96.40
PC105	192.21.96.41
PC106	192.21.96.42
PC107	192.21.96.43
PC108	192.21.96.44
PC109	192.21.96.45
PC110	192.21.96.46
PC111	192.21.96.47
PC112	192.21.96.48
PC113	192.21.96.49
PC114	192.21.96.50
PC115	192.21.96.51
PC116	192.21.96.52
PC117	192.21.96.53
PC118	192.21.96.54
PC119	192.21.96.55
PC120	192.21.96.56
PC121	192.21.96.57
PC122	192.21.96.58
PC123	192.21.96.59
PC124	192.21.96.60
PC125	192.21.96.61
PC126	192.21.96.62
PC127	192.21.96.63
PC128	192.21.96.64
PC129	192.21.96.65
PC130	192.21.96.66
PJ6	192.21.96.67

MEETING MINUTE

MEETING MINUTE

DATE/TIME	13 January 2025		
LOCATION	MA7		
AGENDA	Discussion on Network Distribution		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Angela Lee Su Ing		1500	-
Bernice Lou Min Yun		1500	-
Karen Yam Vei Xin		1500	-
Seow Yen Zhi		1500	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 5	- All members go through the instruction in the task 5. -Solved the problem raised by group members together through discussion.	All members (13/01/2025)
2	Network address assigned	-Bernice clarifies the assigned network address by the lecturer. -Seow then partitioned the network address and calculate the range of usable network addresses.	All members (13/01/2025)

3	Distribution on network address for the network	-Angela then suggested that the usable network addresses to be distributed to different areas by step which first allocates them by work areas, next by workstations.	All members (13/01/2025)
4	Network distribution for different work areas	-All members the examine how the IP address can be allocated.	All members (13/01/2025)
5	Network distribution for workstations in different work areas	-Karen then work on allocating the IP Addresses based on the number of workstations in each work areas. -All Members then continue to identify and solve the problems faced.	All members (13/01/2025)
6	Meeting ended	-At 2200, the meeting ended after all discussion were done. -The team agrees to have short discussion to again review the devices whether they are able to meet the requirement of the case study.	All members (13/01/2025)

4.0 **CONCLUSION**

In conclusion, the Faculty of Computing's proposed expansion successfully addresses the need for modern, scalable and technologically advanced facilities to support its growing student and staff population. The project's strengths lie in its comprehensive design, scalability and focus on fostering collaboration through specialized labs, a hybrid classroom and high-performance connectivity. However, challenges such as budget constraints, potential delays, user adaptation and long-term maintenance need attention. To improve the project, prioritizing critical components within the budget, engaging stakeholders, providing training programs, integrating sustainable technologies and establishing a robust maintenance and feedback system are recommended. These measures will ensure the project meets its goals effectively and supports future growth and innovation.

5.0 **ADDITIONAL SUGGESTION**

There are several additional suggestions that can enhance the overall effectiveness and user experience of the network. First, consider implementing network redundancy to ensure reliability. This could involve setting up multiple connections for critical devices like routers and switches. By adopting a mesh topology, we can create a system where if one connection fails, others can seamlessly take over, minimizing downtime. Additionally, integrating load balancers will help distribute network traffic more evenly across servers, which is particularly beneficial during peak usage times.

Another area to focus on is cable management. A well-organized cabling system not only improves aesthetics but also facilitates easier troubleshooting and maintenance. Implementing a comprehensive labelling system for all cables and ports will make it simpler for anyone working with the network to identify connections quickly. Furthermore, using cable trays and organizers can help keep cables neatly arranged, reducing the risk of tangling or damage that could lead to signal loss.

Enhancing security measures is also crucial in today's digital landscape. Consider segmenting the network using VLANs (Virtual Local Area Networks) to isolate sensitive data and improve overall security. Additionally, deploying robust firewalls and intrusion detection systems can help monitor for unauthorized access, providing an extra layer of protection against potential threats.

Regular maintenance and monitoring of the network will ensure it operates at peak efficiency. Establishing a routine maintenance schedule for all devices will help catch any issues before they escalate into significant problems. Utilizing network monitoring tools can provide insights into performance metrics, helping you make informed decisions about upgrades or adjustments as needed.

Lastly, investing in training and documentation for users will enhance their experience with the network. Conducting training sessions can familiarize staff and students with the available resources and best practices for using the network effectively. Maintaining detailed documentation of the network configuration, including diagrams, and troubleshooting guides will be invaluable for future reference and maintenance efforts.

6.0 TEAM MEMBERS AND RESPONSIBILITIES

Group Name: Beauties		
Section: 01		
Group Members		
No.	Member's Information	Responsibilities
1	Name: Angela Lee Su Ing Matric No: A23CS0047 	Task 1: - Suggest student lounge and facilities. - Design lift and hybrid video room. Task 2: - Prepare questions. Task 3: - Write reflection and comparison. Task 4: - Identify cable types. Task 5: - Review overall work. - Do meeting minutes.
2	Name: Bernice Lou Min Yun Matric No: A23CS0056 	Task 1: - List user needs. - sketch layout. - design hybrid, server room, lab, and toilet. Task 2: - Prepare interview questions. - check feasibility. Task 3: - Create table of devices. Task 4: - Create network and room-level diagrams. Task 5:

		- Distribute subnets.
3	Name: Seow Yen Zhi Matric No: A23CS0177 	Task 1: - Suggest lab locations and stair concept. - design video room, layout. - Do meeting minutes. Task 2: - Check feasibility. - Prepare questions. Task 3: - Write introduction. - Do meeting minutes Task 4: - Identify work areas. - Do meeting minutes. Task 5: - Assign network addresses and distribute the network.
4	Name: Karen Yam Vei Xin Matric No: A23CS0093 	Task 1: - Suggest basic facilities and toilet location. - design video room, layout. - design stair, lounge, meeting, and study rooms. Task 2: - Review overall work. - Do meeting minutes. Task 3: - Write comparison. Task 4: - Count connections. Task 5: - Distribute workstations.

7.0 **REFERENCES**

Task 2

1. Patel, H., & Patel, H. (2024, May 23). Network Infrastructure Design: your ultimate guide to planning and implementation. WPG Consulting. <https://wpgc.io/blog/network-infrastructure-design-guide-to-planning-and-implementation/>
2. M, D. (2024b, October 30). Best network management software Solution - Auvik Networks. Auvik. <https://www.auvik.com/wp-content/uploads/2022/03/Building-a-Modern-Educational-IT-Network.pdf>
3. [1] Building Materials and Methods (14 September 2024) – Sugar Marketing.
Retrieved from: <https://panrf.com/2021/12/building-materials-and-methods/> (Accessed on: 05 November 2024)
4. [2] Cisco Systems (n.d.) Campus LAN and WLAN Design Guide.
Retrieved from: Cisco's Website (Accessed on: 05 November 2024)
5. [3] Hybrid classroom: definition, importance, necessary equipment (02 May 2023).
Retrieved from: <https://dexonsystems.com/blog/hybrid-classroom> (Accessed on: 05 November 2024)
6. [4] Reliable Power Backup Solutions in UPS Systems (10 April 2024).
Retrieved from: <https://www.prostarsolar.net/article/reliable-power-backup-solutions-in-ups-systems.html> (Accessed on: 05 November 2024)
7. [5] Network Segment (n.d.). Retrieved from:
<https://www.sciencedirect.com/topics/computer-science/network-segment> (Accessed on: 05 November)

8. [6] Wireless Network Security: WEP, WPA, WPA2 & WPA3 Explained (29 April 2024) – Wi-Fi Protected Access III (WPA3). Retrieved from: <https://www.esecurityplanet.com/trends/the-best-security-for-wireless-networks/#:~:text=Wi%2DFi%20Protected%20Access%20III,to%20this%20more%20modern%20standard>. (Accessed on: 05 November 2024)
9. [7] Network Security Devices (n.d.)
Retrieved from: <https://www.zenarmor.com/docs/network-security-tutorials/network-security-devices> (Accessed on: 05 November 2024)
10. [8] What is network redundancy, why it matters and how to build it (22 July 2024)
Retrieved from: <https://www.zenarmor.com/docs/network-security-tutorials/network-security-devices> (Accessed on: 05 November 2024)
11. [9] Dr. Muhammad Zafran bin Muhammad Zally Shah, personal communication on 22th November 2024.

Task 3

1. *What is a router? - definition and uses* (2024) Cisco. Available at:
<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/what-is-a-router.html> (Accessed: 01 December 2024).
Product page: <https://shopee.com.my/TP-Link-Archer-AX73-AX5400-Dual-Band-Gigabit-Wireless-AX-WiFi-6-Router-Mercusys-MR70X-MR80X-MR90X-IBN-i.2794197.3278089197>
2. *What is a network switch? | switch vs. Router | Cloudflare*. Available at:
<https://www.cloudflare.com/learning/network-layer/what-is-a-network-switch/>
(Accessed: 01 December 2024).
Product page: <https://www.fs.com/sg/products/90131.html>
3. Froehlich, A., Rouse, M. and Gerwig, K. (2021) *What is a patch panel?*, *Search Networking*. Available at:
<https://www.techtarget.com/searchnetworking/definition/patch-panel> (Accessed: 01 December 2024).
Product page: <https://shopee.com.my/24-Port-Cat6-RJ45-19-1U-Rackmount-Patch-Panel-24Port-Cat5-Rack-Mount-UTP-i.76579217.2695563499>
4. *What is an access point?* (2024) Cisco. Available at:
<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/what-is-access-point.html#~what-is-an-access-point> (Accessed: 01 December 2024).
Product page: <https://shopee.com.my/Ubiquiti-U6-LITE-Access-Point-WiFi-6-LITE-Dual-band-2x2-MIMO-AP-2.4GHz-5GHz-Ceiling-Wall-mount-UBNT-UAP-SOCIAL-WIFI-i.76579217.9569497634>

5. *Network cable types and specifications* (no date) *ComputerNetworkingNotes*. Available at: <https://www.computernetworkingnotes.com/networking-tutorials/network-cable-types-and-specifications.html> (Accessed: 01 December 2024).
Product page: <https://www.fs.com/products/70611.html>
6. Kaspersky. (2020). *What is a Firewall? how firewalls work & types of firewalls*. Available at: <https://www.kaspersky.com/resource-center/definitions/firewall> (Accessed: 01 December 2024).
Product page: <https://www.fs.com/products/106549.html>
7. *Computer Basics: What is a Computer?* (no date). Available at: <https://edu.gcfglobal.org/en/computerbasics/what-is-a-computer/1/> (Accessed: 01 December 2024).
Product page: <https://www.hp.com/my-en/shop/hp-elite-tower-800-g9-desktop-pc-a3nd6pt.html?facetref=ec05aa8209ca7844>
8. *What is a projector?* (2024) *Computer Hope*. Available at: <https://www.computerhope.com/jargon/p/projector.htm> (Accessed: 01 December 2024).
Product page: <https://www.epson.com.my/Projectors/Corporate-and-Education/Epson-EB-X06-XGA-3LCD-Projector/p/V11H972052>
9. *multi-functional smart whiteboard 65 75 86 inches touchscreen interactive whiteboard for school* (no date) *Made*. Available at: <https://kep-tech.en.made-in-china.com/product/OnyrGQUJVWhL/China-Multi-Functional-Smart-Whiteboard-65-75-86-Inches-Touchscreen-Interactive-Whiteboard-for-School.html#:~:text=Real%2DTime%20Collaboration%20and%20Screen,to%20get%20audiences%20more%20engaged>. (Accessed: 01 December 2024).
Product page: <https://www.cdw.com/product/smart-mx075-v5-series-75-interactive-display/7976675>

10. *Microphone* (2024) *SVANTEK*. Available at: <https://svantek.com/academy/microphone/> (Accessed: 01 December 2024).
Product page: <https://www.musicbliss.com.my/products/audio-technica-at2020-malaysia-cardioid-condenser-microphone-at-2020>
11. *Video camera* (no date) *Video Camera - an overview | ScienceDirect Topics*. Available at: <https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/video-camera#:~:text=The%20term%20'video'%20in%20its,the%20standard%20television%20signal%20formats>. (Accessed: 01 December 2024).
Product page: [https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-\(3-YEARS-WARRANTY\)-i.46683178.1417788595](https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-(3-YEARS-WARRANTY)-i.46683178.1417788595)
12. (No date) *What is video conferencing? video conferencing definition | mitel*. Available at: <https://www.mitel.com/features-benefits/video-conferencing> (Accessed: 01 December 2024).
Product page: <https://www.logitech.com/en-us/products/video-conferencing/room-solutions/rally-ultra-hd-conferencecam.960-001225.html>
13. Author Cadence PCB Solutions *et al.* (2024) *An introduction to IOT development boards, Cadence*. Available at: <https://resources.pcb.cadence.com/blog/2022-an-introduction-to-iot-development-boards> (Accessed: 01 December 2024).
Product page: <https://my.element14.com/raspberry-pi/rpi4-modbp-4gb/raspberry-pi-4-model-b-4gb/dp/3051887>
14. *Color imageclass mf751cdw* (no date) *Shop Canon Color imageCLASS MF751Cdw | Canon U.S.A., Inc.* Available at: <https://www.usa.canon.com/shop/p/color-imageclass-mf751cdw?color=White&type=New> (Accessed: 01 December 2024).
Product page: <https://support.hp.com/us-en/drivers/hp-laserjet-pro-mfp-m428-m429-f-series/model/19204098>

15. *Modem definition & meaning* (no date) *Merriam-Webster*. Available at:
<https://www.merriam-webster.com/dictionary/modem> (Accessed: 01 December 2024).
Product page: <https://www.netgear.com/home/wifi/modems/cm1200/>
16. *What is a computer server* (no date) *What is a computer server | Lenovo US*. Available at: <https://www.lenovo.com/us/en/glossary/what-is-a-computer-server/?orgRef=https%253A%252F%252Fwww.google.com%252F>
(Accessed: 01 December 2024).
Product page: <https://www.dell.com/en-my/shop/ipovw/poweredge-t550>
17. Jon (2023) *What is a ups and how does it work?*, *Ormazabal*. Available at:
<https://www.ormazabal.com/en-gb/what-is-a-ups-and-how-does-it-work-2-2/>
(Accessed: 01 December 2024).
Product page: <https://www.amazon.com/APC-BR1000G-Back-UPS-Computer-Protector/dp/B077SR55QT>

Product Page

18. Router

<https://shopee.com.my/TP-Link-Archer-AX73-AX5400-Dual-Band-Gigabit-Wireless-AX-WiFi-6-Router-Mercusys-MR70X-MR80X-MR90X-IBN-i.2794197.3278089197>

19. Switches

<https://www.fs.com/sg/products/90131.html>

20. Patch Panel

<https://shopee.com.my/24-Port-Cat6-RJ45-19-1U-Rackmount-Patch-Panel-24Port-Cat5-Rack-Mount-UTP-i.76579217.2695563499>

21. Wireless Access Point

<https://shopee.com.my/Ubiquiti-U6-LITE-Access-Point-WiFi-6-LITE-Dual-band-2x2-MIMO-AP-2.4GHz-5GHz-Ceiling-Wall-mount-UBNT-UAP-SOCIAL-WIFI-i.76579217.9569497634>

22. Cable

<https://www.fs.com/products/70611.html>

23. Firewall

<https://www.fs.com/products/106549.html>

24. Personal Computer

<https://www.hp.com/my-en/shop/hp-elite-tower-800-g9-desktop-pc-a3nd6pt.html?facetref=ec05aa8209ca7844>

25. Projector
<https://www.epson.com.my/Projectors/Corporate-and-Education/Epson-EB-X06-XGA-3LCD-Projector/p/V11H972052>
26. Projector Screen
<https://shop.elitescreens.com/products/elite-screen-manual-b?variant=42697092202711>
27. Multifunctional Whiteboard
<https://www.cdw.com/product/smart-mx075-v5-series-75-interactive-display/7976675>
28. Standing Microphone
<https://www.musicbliss.com.my/products/audio-technica-at2020-malaysia-cardioid-condenser-microphone-at-2020>
29. Video Camera
[https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-\(3-YEARS-WARRANTY\)-i.46683178.1417788595](https://shopee.com.my/Canon-Eos-M50-Mark-II-mark-2-Kit-15-45mm-original-and-new-EXTRA-BATTERY-ORIGINAL-(3-YEARS-WARRANTY)-i.46683178.1417788595)
30. Video Conference Terminal
<https://www.logitech.com/en-us/products/video-conferencing/room-solutions/rally-ultra-hd-conferencecam.960-001225.html>
31. IoT Hardware Board
<https://my.element14.com/raspberry-pi/rpi4-modbp-4gb/raspberry-pi-4-model-b-4gb/dp/3051887>

32. Printer

<https://support.hp.com/us-en/drivers/hp-laserjet-pro-mfp-m428-m429-f-series/model/19204098>

33. Modem

<https://www.netgear.com/home/wifi/modems/cm1200/>

34. Server

<https://www.dell.com/en-my/shop/ipovw/poweredge-t550>

35. Power Backup Supply System

<https://www.amazon.com/APC-BR1000G-Back-UPS-Computer-Protector/dp/B077SR55QT>

8.0 APPENDICES

FINANCIAL BUDGET

Devices	Price (RM/unit)	Unit	Total Price (RM)
 Figure 3.2.1 Router	484.00	6	2904.00
 Figure 3.2.2 Switches	2033.00	7	14231.00
 Figure 3.2.3 Patch Panel	104.00	6	624.00
	595.00	6	3570.00

Figure 3.2.4 Wireless Access Point			
 Figure 3.2.5 Cable	75.90/10.7m	300	2128.04
 Figure 3.2.6 Firewall	21,849.48	2	43698.96
 Figure 3.2.7 Personal Computer (PC)	3895.00	130	506350.00
 Figure 3.2.8 Projector	1809.00	6	10854.00

 Figure 3.2.9 Projector Screen	414.78	2	829.56
 Figure 3.2.10 Multifunctional Whiteboard	16502.00	6	99012.00
 Figure 3.2.11 Standing Microphone	390.00	1	390.00

 Figure 3.2.12 Video Camera	2968.00	1	2968.00
 Figure 3.2.13 Video Conference Terminal	16099.00	1	16099.00
 Figure 3.2.14 IoT Hardware Board	259.00	10	2590.00

 Figure 3.2.15 Printer	2750.00	2	5500.00
 Figure 3.2.16 Modem	847.30	2	1694.60
 Figure 3.2.17 Server	18402.00	5	92010.00

 Figure 3.2.18 Power Backup Supply System		924.00	20	18480.00
Total	RM 823 933.16			
Budget	RM 1500 000.00			
Remaining Budget	RM 676 066.84			
Budget Usage	$(823933.16/150000.00) \times 100\% = 54.93\%$			

The Financial Budget table above is basically from the table of devices from the previous task 3. As mentioned in the previous task, this is the list of devices required in our project. Each device purchased are based on our discussion in terms of allocation of devices with the minimum budget and the maximum function. Our team has high consideration on the function of the devices that are able to meet our requirement within the allocated budget.

Our team also take security precaution by reserve some of the budget for any emergency purposes just in case there are any unexpected obstacles happen in the future when our project is ongoing. With about 45% of budget left, we are able to allocate these funds in future infrastructure upgrading to benefits the Faculty of Computing in future. Some of the remaining budget is use for user training, this is to ensure that the user in Faculty of Computing can fully leverage the potential and function of the new technology installed. Additionally, a portion of budget is estimated to be used in the future software updates. This is because a system maintenance is way more costly than the system development itself. The budget strategy reflects the team's commitment in creating a cost-effective, secure and future-proof network solution for the Faculty of Computing, ensuring in both immediate and long-term benefits.

MEETING MINUTE

Meeting Minute Task 1

MEETING MINUTE

DATE/TIME	16 October 2024		
LOCATION	MA7		
AGENDA	Discussion on Floor Plan		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Angela Lee Su Ing		1930	-
Bernice Lou Min Yun		1930	-
Karen Yam Vei Xin		1930	-
Seow Yen Zhi		1930	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion on the case study given. Understanding the requirement in the case study.	-All members go through the case study. -Bernice listed out the important user needs.	All members (16/10)
2	Discussion on the basic facilities to be inserted in the design.	-Karen suggested that besides meeting the user needs, a building must have some basic facilities such as stairs and toilet.	All members (16/10)
3	Discussion on the allocation of the rooms.	-Angela suggested to place the student lounge should be placed at the ground floor. -Seow suggested the 4 labs should all be located at the first floor.	All members (16/10)
4	Sketching on floorplan	-All members give their opinion and discussed on the sketched floor plan. -Bernice sketch the layout using ipad.	All members (16/10)

		-Karen suggested that the toilet should be located at two ends of the building to be user friendly. -Seow then continue to suggest that the stairs should also have that concept for safety purposes. -Angela suggested that the empty space in the open area should be equipped with tables, chairs and sofa for the lecturers, students and staffs. -Bernice then suggested to place a staff meeting room at the ground floor. -Karen then suggested to only build one lift to save the budget. -Seow suggested to place create a study room at first floor. -Angela suggested to have a café at the ground floor.	
5	Enhancement on floor plan sketching	-After the basic sketching, the team proceed with discussion on the size of the rooms and facilities in the building. -Bernice and Seow have the idea that the hybrid classroom and video conference room should be designed larger to fit more users in the future. -Angela suggested the size of the lift to be able to fit a certain amount of people. -Karen then suggested the size of the stairs that are	All members (16/10)

		able to meet the safety requirement. -Bernice suggested that there should be more female toilet compared to male toilet. -Seow suggested that the lift can be located near to the toilet. -Karen suggested should have server room. -Angela then suggested to share one server room for two labs to ensure the speed of the network.	
6	Designing floorplan in software	-All members explore on the tools available online to design the floorplan. -Seow suggested to get a software that all members can access the design at the same time to improve the efficiency in designing. -All members spent some time testing on different tools. -Angela and Bernice manage to get the online tools that meets our requirement. -Karen suggested to calculate the scale before starting to design. -Seow suggested scale 1m to 10mm.	All members (16/10)
7	Distribution of task in designing floor plan	-All members discussed the part they are responsible to design. -Seow designed the overall layout of the building, the layout of the	All members (16/10 – 18/10)

		rooms, rough layout of lift, stairs and toilet. -Bernice designed the server room, the layout inside the labs, the interior of toilet. -Angela designed the layout inside hybrid classroom and the video conference room. -Karen designed the layout inside student lounge, study room and staff meeting room. -Angela and Karen do the enhancement on adding sofa at the open area. -Bernice and Seow do the labelling part. -All members do the final checking part.	
8	Duedate of floor plan	-All member agreed finish there respective parts together latest by Friday. -All members have a short discussion after the design is done.	All members (18/10)
9	Meeting ended	-At 0030, the meeting ended successfully.	

Meeting Minute Task 2

MEETING MINUTE

DATE/TIME	5 November 2024		
LOCATION	MA7		
AGENDA	Discussion on preliminary analysis		
MEETING MC	Bernice Lou Min Yun		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Angela Lee Su Ing	1900	-	
Bernice Lou Min Yun	1900	-	
Karen Yam Vei Xin	1900	-	
Seow Yen Zhi	1900	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 2	- Bernice shared the information about task 2 and its rubric. - All members read and investigate the task carefully. - Solved the problem raised by group members together through discussion.	All members (05/11)

2	Suggestion for questions	- All members discussed the possible questions that can be used in task 2. - Each member suggested 11 questions that can be used in task 2.	All members (05/11)
3	Filter and select the most related question	- Seow and Angela suggested preparing more than 10 questions and all members agreed with their suggestions.	All members (05/11)
4	Task distribution	- Karen distributed the task to all members equally. - All members had to do research to find a solution for all selected questions. - Bernice was assigned to	All members (05/11)

		complete the feasibility study. - Seow was assigned to check and arrange the references used by all members. - Angela and Karen were assigned to prepare for meeting minutes.	
5	Explain answer for each question	- Seow explained - Angela explained - Bernice explained - Karen explained	All members (05/11)
6	Review the answer for each question	- All members review the questions and the answer carefully - Karen suggested to drop the question which is not related to the syllabus - Bernice suggested to not use the question	All members (05/11)

		which have a general answer - Angela said that some questions are confusing - Seow highlighted the question that can be used - 11 questions are finalized	
7	Discuss on the feasibility assessment	- Bernice explained the feasibility assessment to all members - Karen ask if the feasibility assessment is in correct format - Angela suggested to continue with the current format as it covers several aspects	All members (05/11)
8	Meeting ended	- At 2300, the meeting ended after all discussion were done.	All members (05/11)

Meeting Minute Task 3**MEETING MINUTE**

DATE/TIME	01 December 2024		
LOCATION	MA7		
AGENDA	Discussion on Floor Plan		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Angela Lee Su Ing	1300		-
Bernice Lou Min Yun	1300		-
Karen Yam Vei Xin	1300		-
Seow Yen Zhi	1300		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion on the requirement of the task.	-All members go through the instruction in the task 3. -Angela suggested to view the task 1 and task 2 to get some idea.	All members (01/12/2024)
2	Discussion on basic list of devices.	-Karen suggested to include the basic devices as mentioned in the case study.	All members (01/12/2024)
3	List the devices required.	-Seow in charge of listing all the devices required.	All members (01/12/2024)

		-Seow refers to several websites to get the appropriate type of devices required that is able to accomplish the requirements of FC.	
4	Table of devices.	-Bernice surveys for the devices listed previously with some specification. -Bernice surveys the price for each unit of device. -The team then together discuss the amount of each device that is required in this case study.	All members (01/12/2024)
5	Total devices.	-Seow reminds that the allocation for number of devices should be within the given budget which is RM 1.5M. -The team then revise and adjust the number of devices required.	All members (01/12/2024)
6	Reflection about is the team surprised by the price of device.	-Karen do research on the market price of the devices. -Karen presented the results of the survey and	All members (01/12/2024)

		the team discuss on the pricing. -The team is surprised that some devices vary by the cost but we also do have deeper research on the specification of devices.	
7	Reflection on cost consideration.	-The team do consider on the cost part when selecting device. -Bernice suggested to consider the devices with higher quality since our budget allocated is at the middle range compared to others. -The team aim to get the devices that is within the budget and its specification can meet our requirements.	All members (01/12/2024)
8	Comparison on same devices with different brands.	-Angela then continues with the further research on the same devices from different brands. -Angela compares the price of devices and the key features of each device.	All members (01/12/2024)

		-The team reviews whether the same devices from different brand is able to meet the basic requirement of the case study from the aspect of the key function.	
9	Meeting ended	-At 0000, the meeting ended after all discussion were done. -The team agrees to have short discussion to again review the devices whether they are able to meet the requirement of the case study.	All members (01/12/2024)

Meeting Minute Task 4**MEETING MINUTE**

DATE/TIME	20 December 2024		
LOCATION	MA7		
AGENDA	Discussion on Floor Plan		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Angela Lee Su Ing		1300	-
Bernice Lou Min Yun		1300	-
Karen Yam Vei Xin		1300	-
Seow Yen Zhi		1300	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 4	- All members go through the instruction in the task 4. - Solved the problem raised by group members together through discussion.	All members (20/12/2024)
2	Suggestion for network distribution	- Angela suggested the network distribution in the floor plan. - Seow and Bernice suggested the necessary details that need to be standardized.	All members (20/12/2024)

3	Task distribution	- Seow distributed the task to all members equally. - Karen was assigned to plan for the network distribution of IoT Lab. - Bernice was assigned to plan for the network distribution of Cisco Network Lab. - Angela was assigned to plan for the network distribution of the video conference room. - Seow was assigned to plan for the network distribution of the general lab.	All members (20/12/2024)
4	Finalize room level network distribution diagram	-Karen suggested making the wiring more visible by adding colour and thickening it. -Bernice suggested labeling the wiring. -The team then together discuss the amount of each device that is required in this case study. - Angela suggested to differentiate vertical and	All members (20/12/2024)

		horizontal cable by using different labeling. - All members share their ideas on how to improve the room level distribution diagram.	
5	Check the calculation of cable length needed	- Seow reminded about the correct calculation of cable length needed. - Bernice reminds that the allocation for number of devices should be within the given budget which is RM 1.5M. -The team then revise and adjust the number of devices required.	All members (20/12/2024)
6	Meeting ended	-At 1700, the meeting ended after all discussion were done. -The team agrees to have short discussion to again review the devices whether they are able to meet the requirement of the case study.	All members (20/12/2024)

Meeting Minute Task 5**MEETING MINUTE**

DATE/TIME	13 January 2025		
LOCATION	MA7		
AGENDA	Discussion on Network Distribution		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Angela Lee Su Ing	1500	-	
Bernice Lou Min Yun	1500	-	
Karen Yam Vei Xin	1500	-	
Seow Yen Zhi	1500	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 5	- All members go through the instruction in the task 5. -Solved the problem raised by group members together through discussion.	All members (13/01/2025)
2	Network address assigned	-Bernice clarifies the assigned network address by the lecturer. -Seow then partitioned the network address and calculate the range of usable network addresses.	All members (13/01/2025)

3	Distribution on network address for the network	-Angela then suggested that the usable network addresses to be distributed to different areas by step which first allocates them by work areas, next by workstations.	All members (13/01/2025)
4	Network distribution for different work areas	-All members the examine how the IP address can be allocated.	All members (13/01/2025)
5	Network distribution for workstations in different work areas	-Karen then work on allocating the IP Addresses based on the number of workstations in each work areas. -All Members then continue to identify and solve the problems faced.	All members (13/01/2025)
6	Meeting ended	-At 2200, the meeting ended after all discussion were done.	All members (13/01/2025)

Meeting Minute Task 6**MEETING MINUTE**

DATE/TIME	14 January 2025		
LOCATION	MA7		
AGENDA	Discussion on Documentation and Reflection		
MEETING MC	Seow Yen Zhi		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Angela Lee Su Ing		0900	-
Bernice Lou Min Yun		0900	-
Karen Yam Vei Xin		0900	-
Seow Yen Zhi		0900	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 6	- All members go through the instruction in the task 6.	All members (14/01/2025)
2	Division of tasks	-Angela leads the team to divide the tasks among the member.	All members (14/01/2025)
3	Abstract and introduction	-Bernice is in charged in this task.	All members (14/01/2025)
4	Project background and conclusion	-Angela is responsible for this part.	All members (14/01/2025)
5	Additional suggestion and team member responsibilities	-Karen is the main person in charged for this part. -Angela and Bernice refers to the previous tasks to help	All members (14/01/2025)

		out in listing down our job scope in the previous tasks.	
6	Table of figure, compilation of task, references and appendices	-Seow is mainly responsible in compiling and organizing the report.	All members (14/01/2025)
7	Meeting ended	-At 1500, the meeting ended after all discussion were done.	All members (14/01/2025)

FEEDBACKS AND COMMENTS FROM LECTURER ON PREVIOUS TASKS

Task 1:

user Group Name: 2M Fit description: 1M 30Total workstations per lab: 2M Scale: 1M Clear and readable: 1M Creativity: 1M Total: 8M	user Make the labels for each room and labs clearer. Add devices that are related to hybrid class rooms and video conferencing rooms. Internet Camera, IP Phone etc. Do your research.	user Differentiate the Cisco Network Lab and Embedded Labs with the General Labs. Study how those specialized labs look like.
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Task 2:

Lack interview with FC representatives:

[Comments](#)

Task 3:

Enough research: 1M - Lack comparison between similar brands of each device.

References: 0.5M Need to include product page as well as references.

LAN Devices: 1.5M Selected the wrong device for router.

Characteristics: 2M

Surprise: 1M

Reflect: 1M

Major differences: 1M

[View comments](#)

Task 4:

Connections identified - 1 (Show the connection to the Wifi routers for rooms that provide Wifi connectivity)

Cable length - 2

Choices, suitable - 2

Sketch - 3

Scale - 1

Total: 9

Task 5:

FIGURE

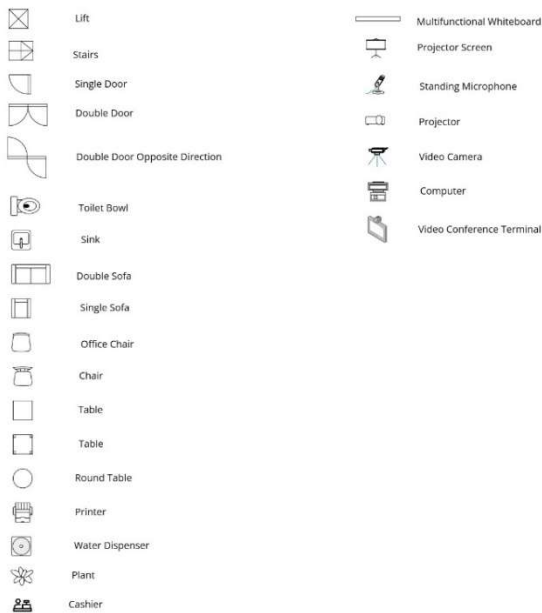
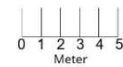
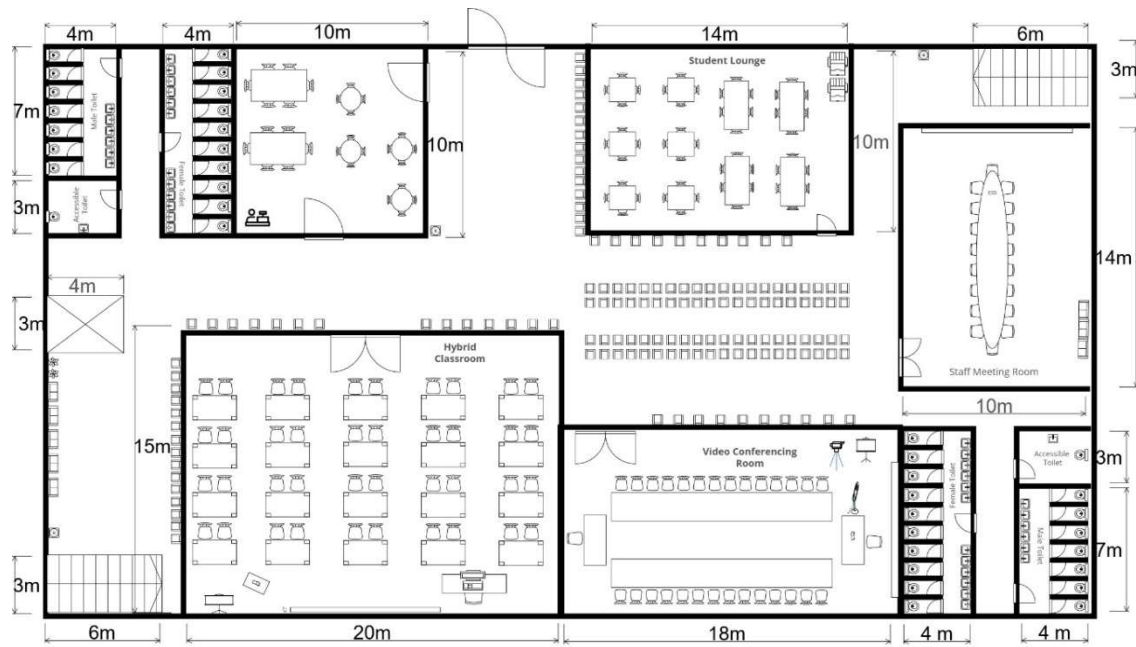


Figure 1.1: Floor Plan of Ground Floor

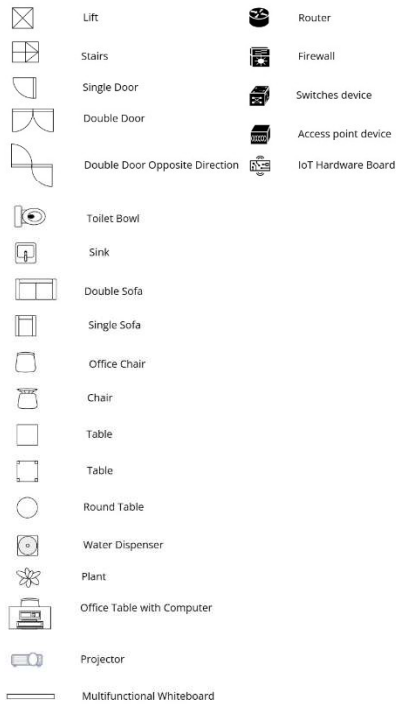
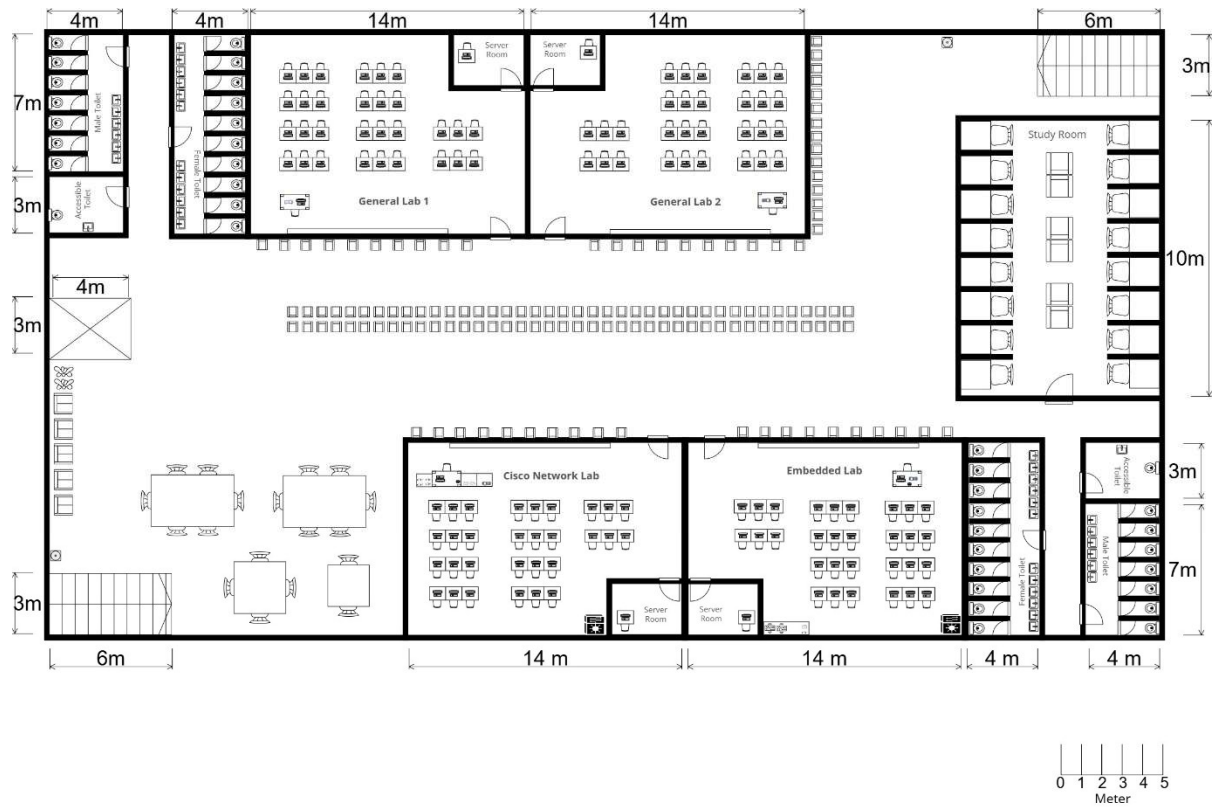


Figure 1.2: Floor Plan of First Floor



Figure 3.2.1 Router



Figure 3.2.2 Switches



Figure 3.2.3 Patch Panel



Figure 3.2.4 Wireless Access Point



Figure 3.2.5 Cable



Figure 3.2.6 Firewall



Figure 3.2.7 Personal Computer (PC)



Figure 3.2.8 Projector



Figure 3.2.9 Projector Screen



Figure 3.2.10 Multifunctional Whiteboard



Figure 3.2.11 Standing Microphone



Figure 3.2.12 Video Camera



Figure 3.2.13 Video Conference Terminal



Figure 3.2.14 IoT Hardware Board



Figure 3.2.15 Printer



Figure 3.2.16 Modem



Figure 3.2.17 Server



Figure 3.2.18 Power Backup Supply System

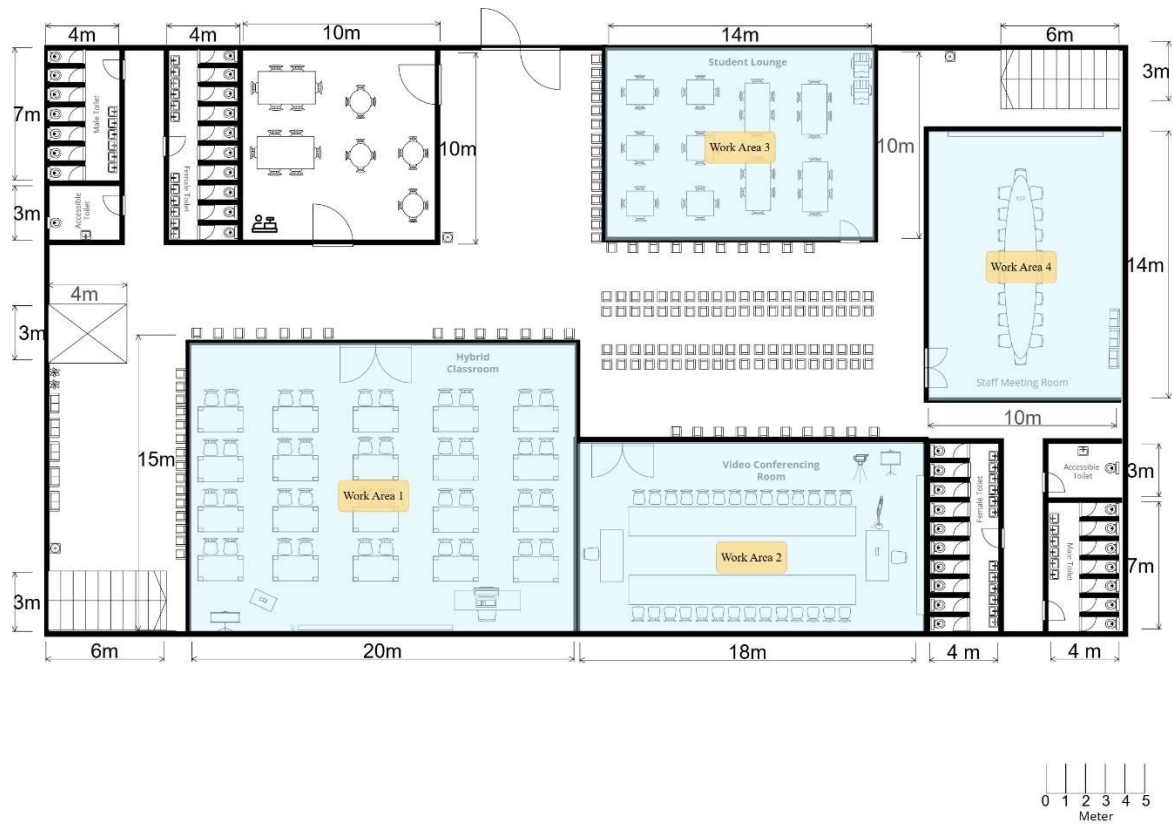


Figure 4.1.1 Work Area at Ground Floor

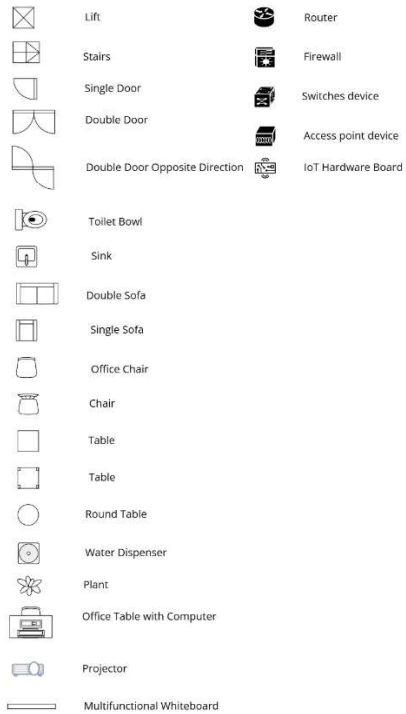
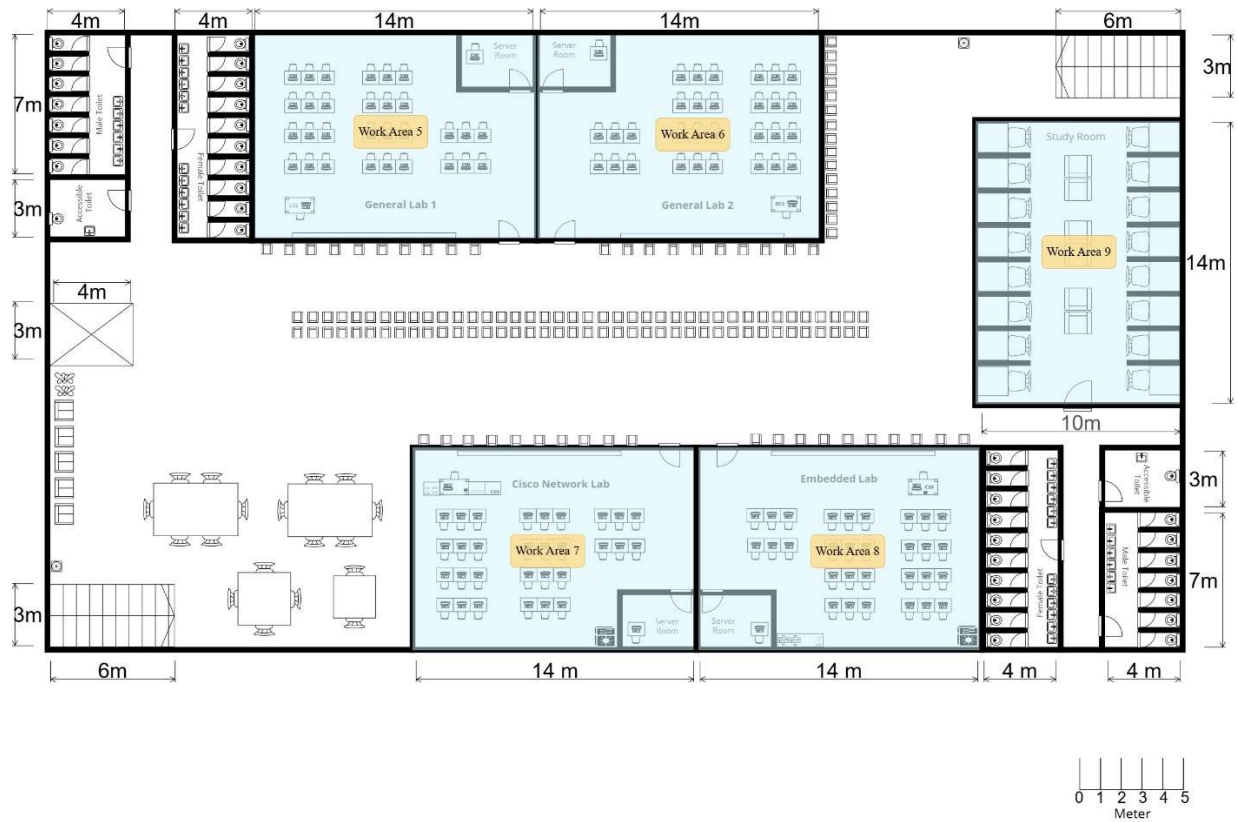


Figure 4.1.2 Work Area at First Floor

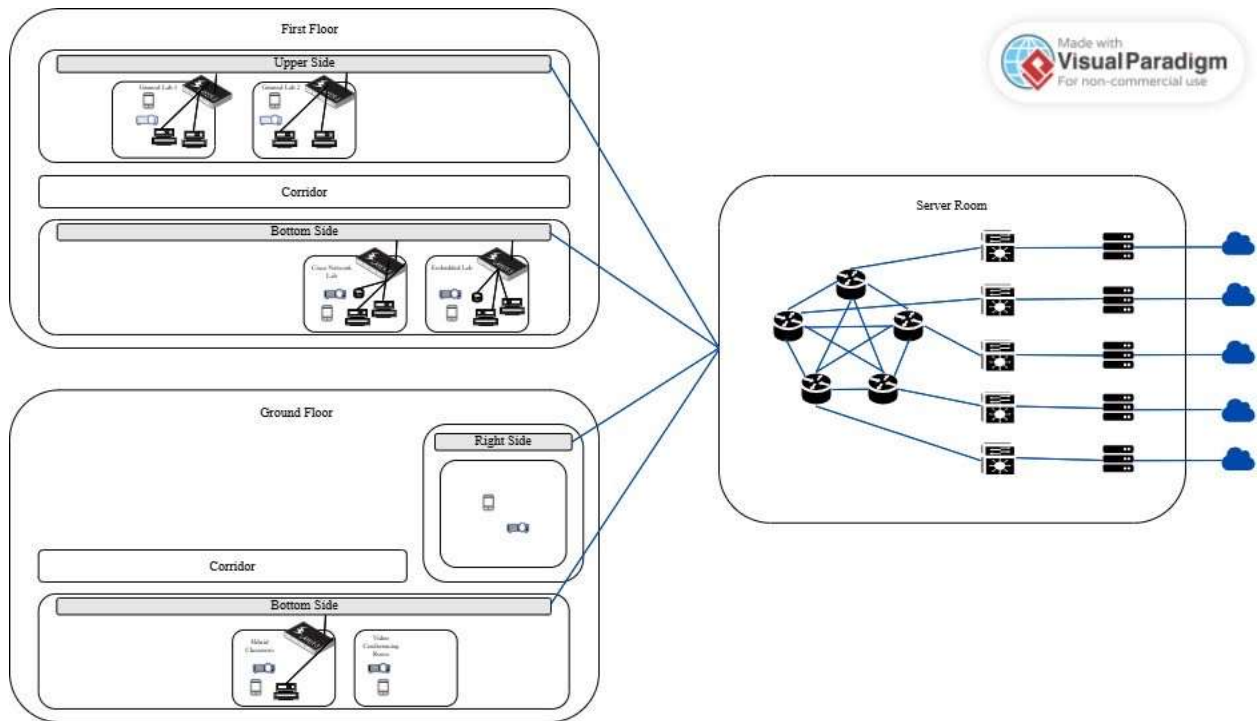


Figure 4.2.1.1: Network Distribution Diagram

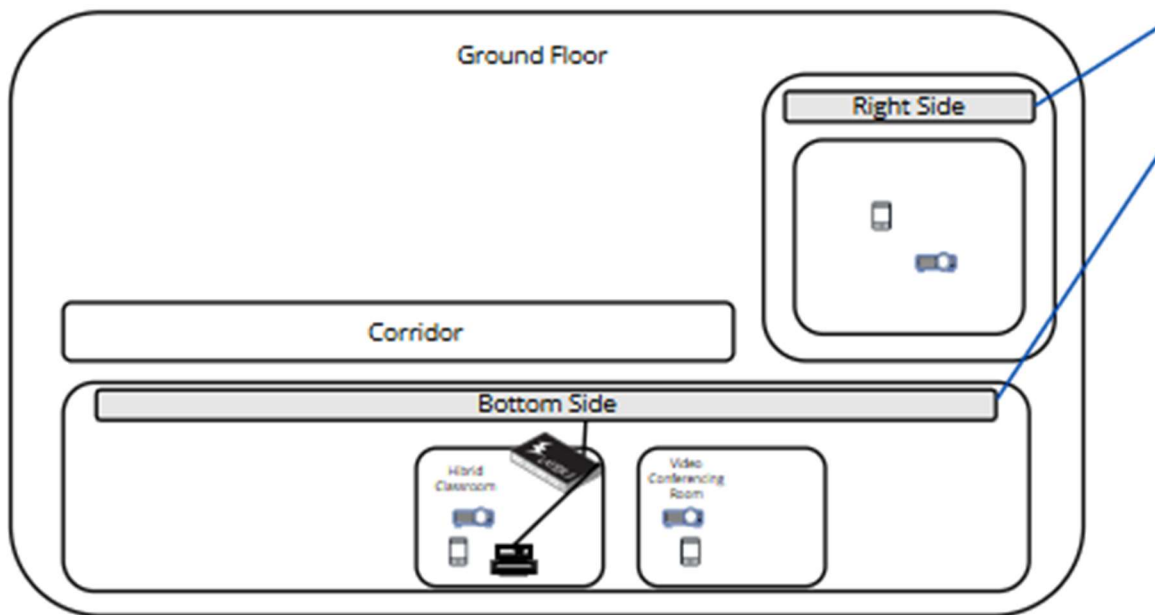


Figure 4.2.1.2: Network Distribution Diagram - Ground Floor

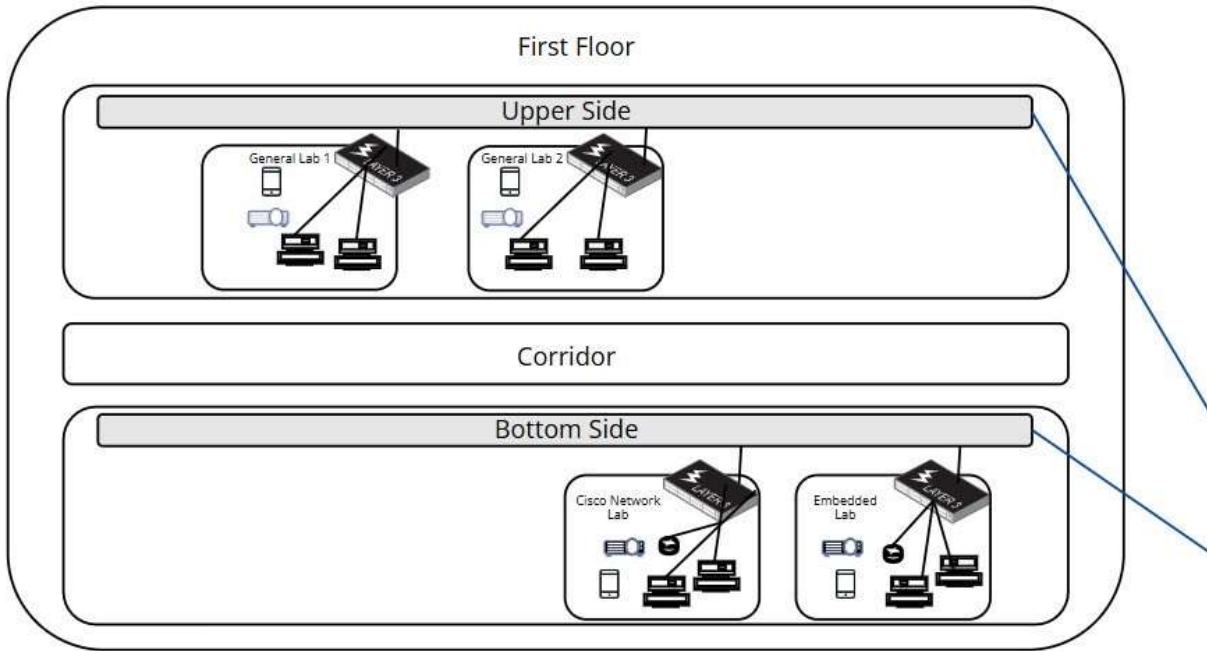


Figure 4.2.1.3: Network Distribution Diagram- First Floor

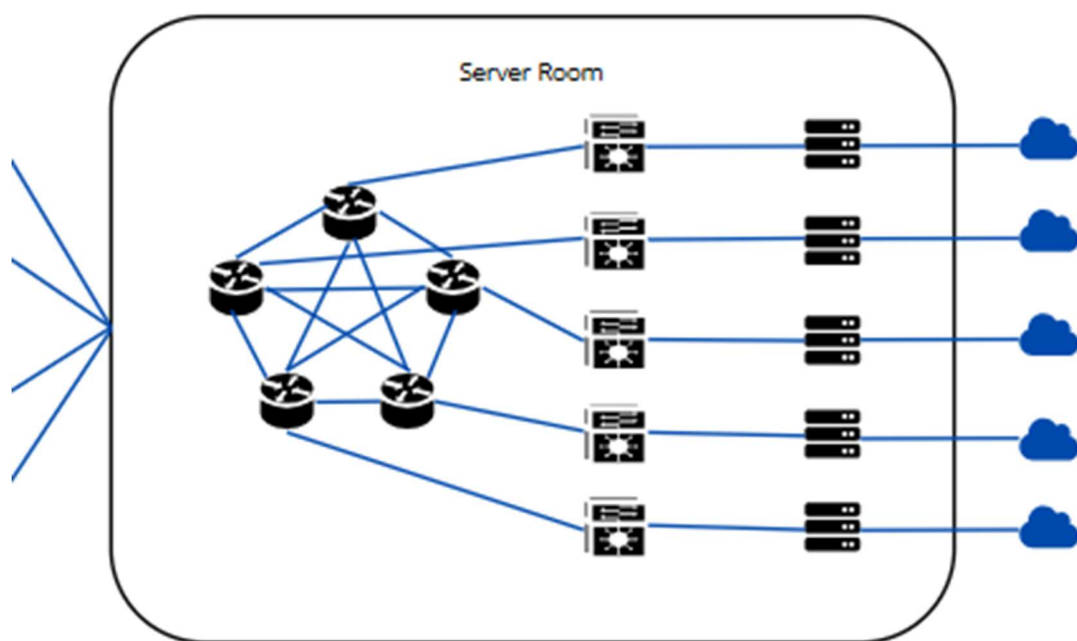


Figure 4.2.1.4: Network Distribution Diagram - Server Room

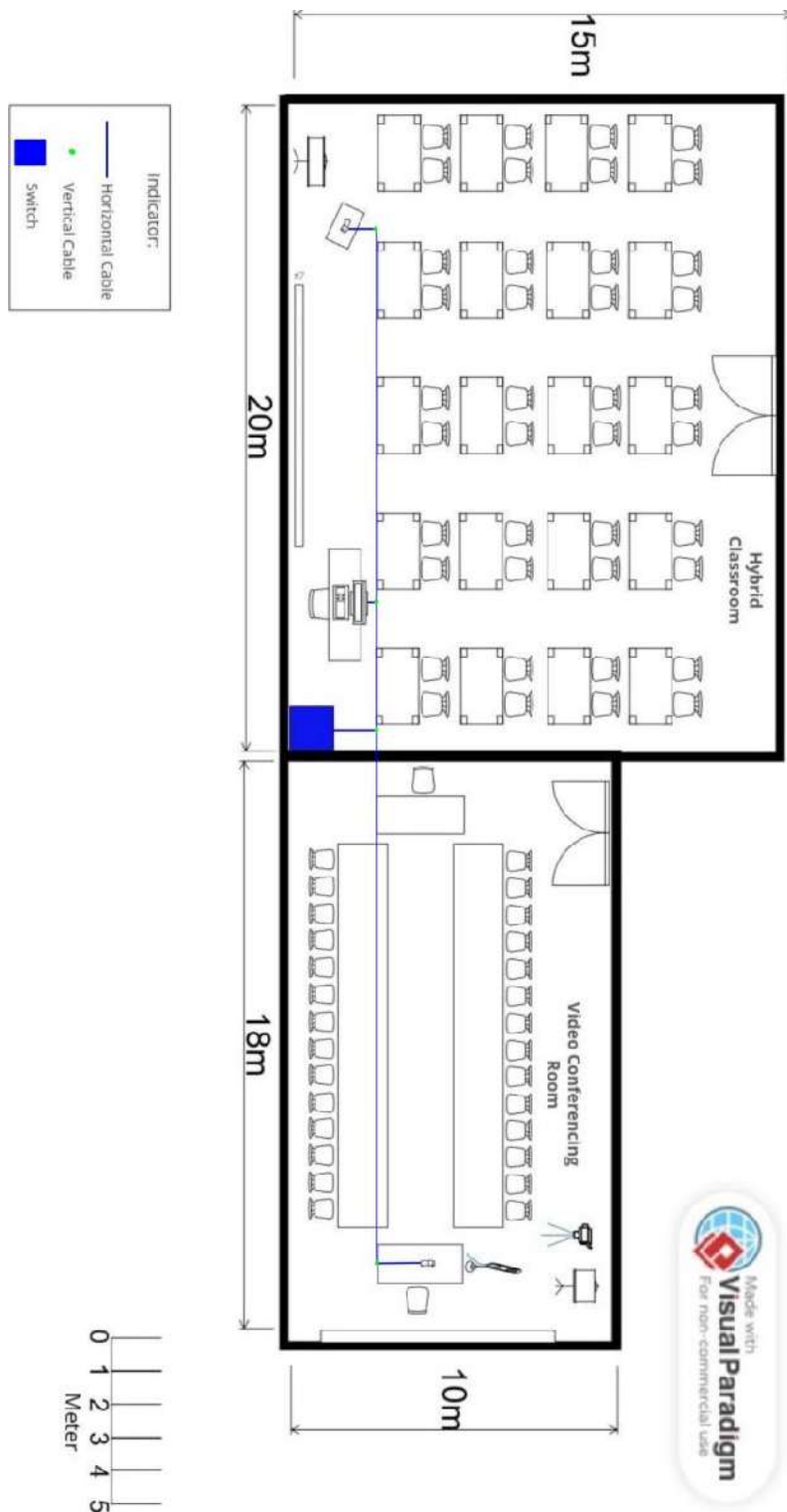


Figure 4.2.2.1: Hybrid Classroom and Video Conferencing Room

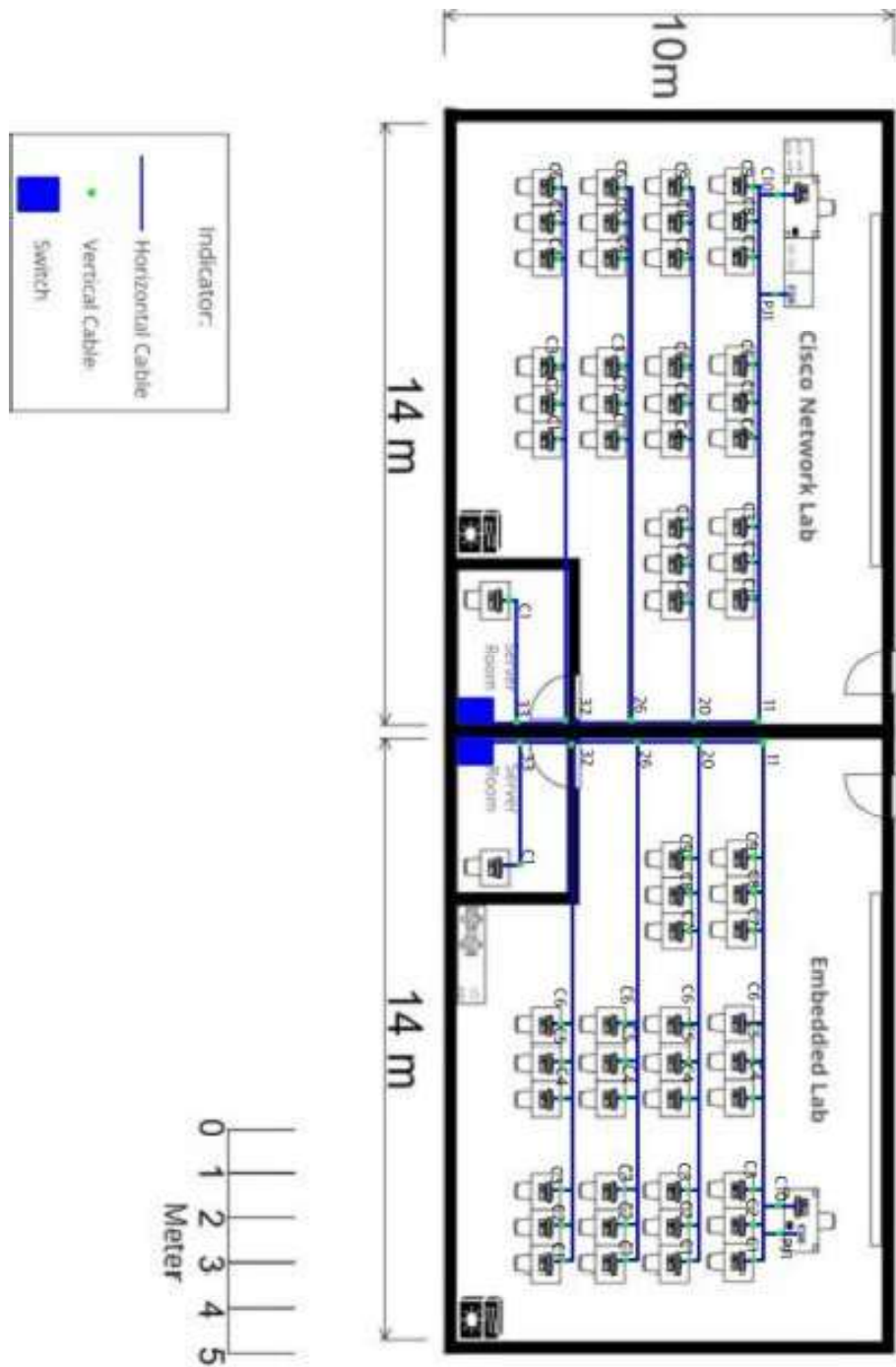


Figure 4.2.2.2: Cisco Network Lab and Embedded Lab

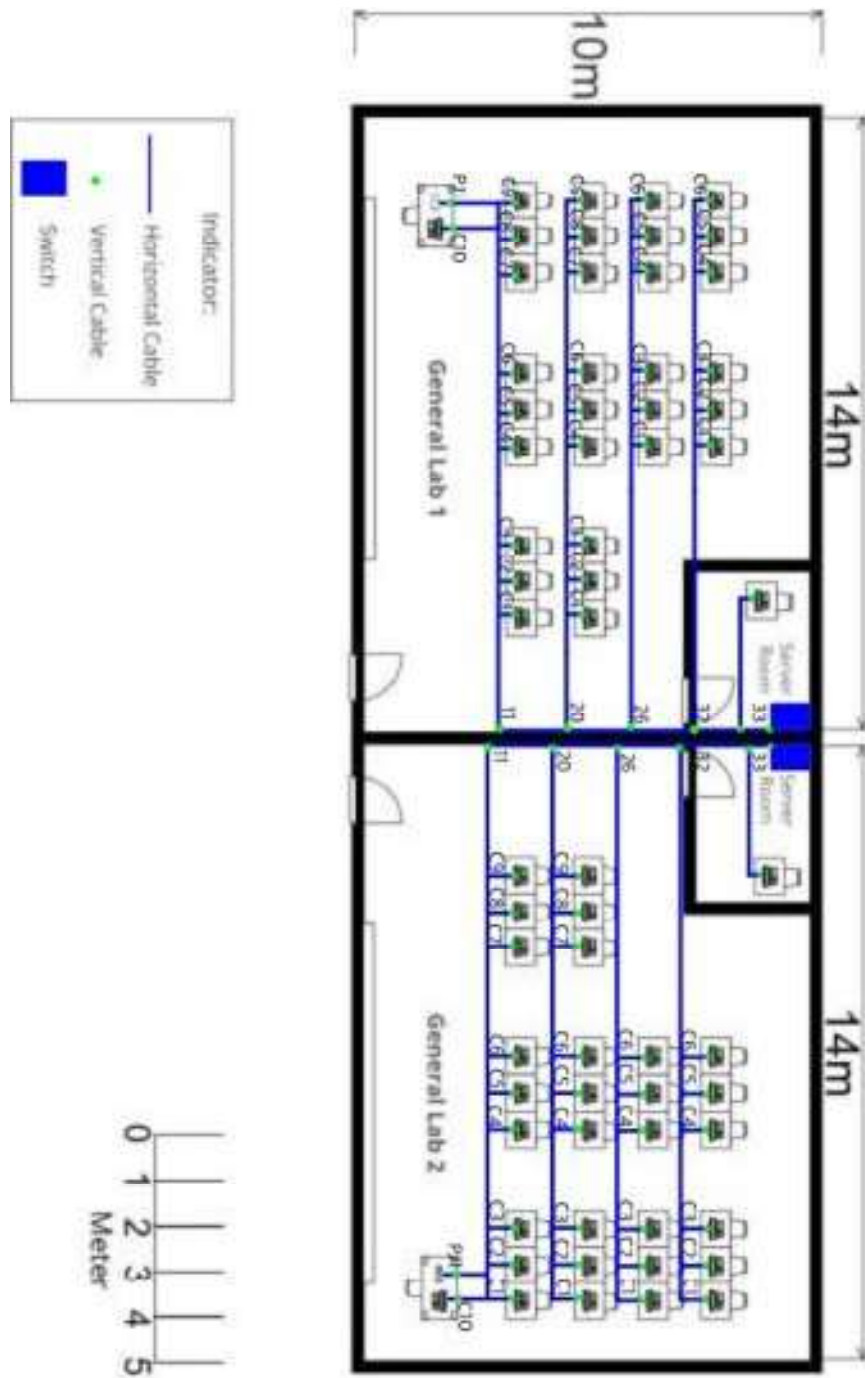


Figure 4.2.2.3: General Lab

FLOOR DIAGRAM

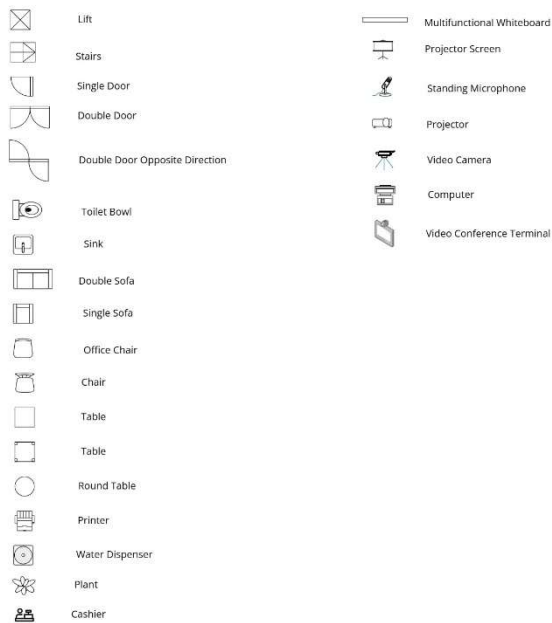
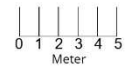
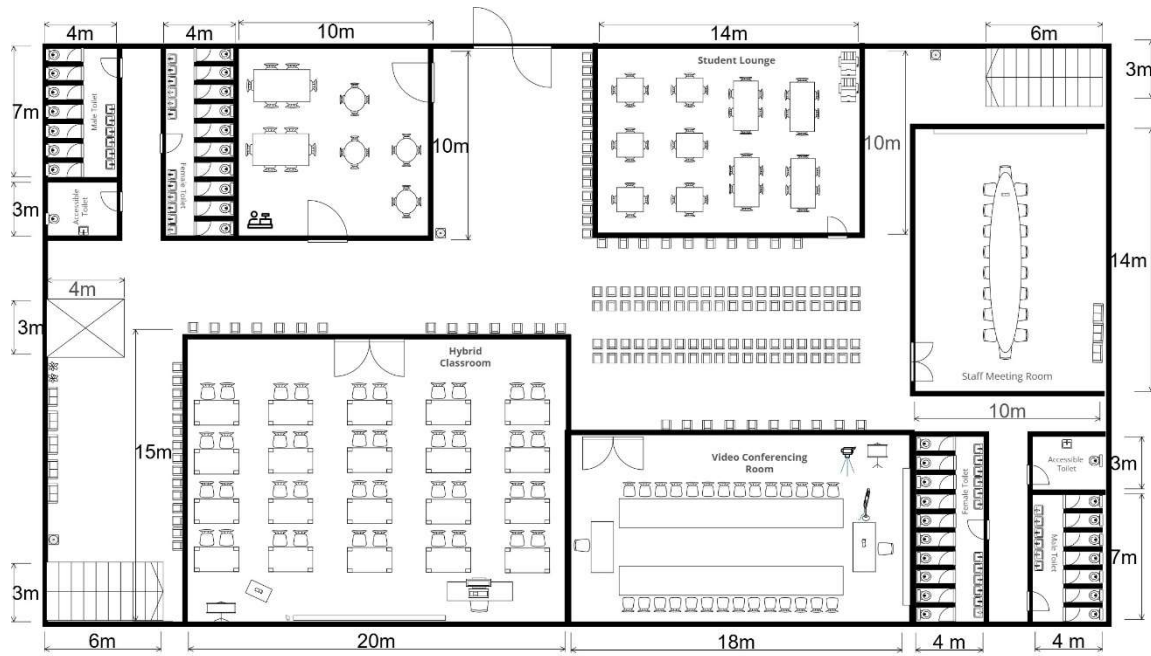


Figure 4.2.3.1: Floor Diagram of Ground Floor

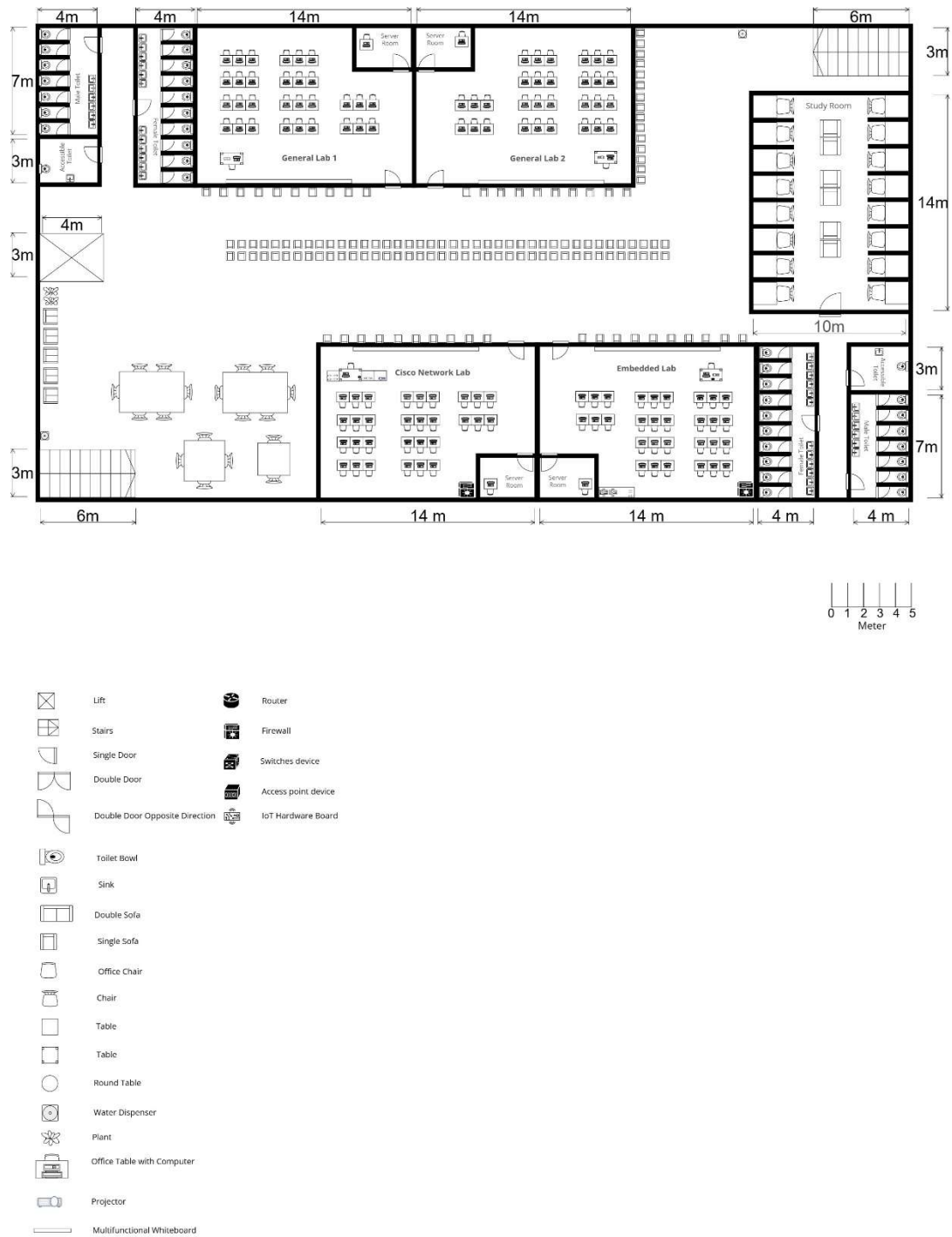


Figure 4.2.3.2 Floor Diagram of First Floor

TABLE

Table of Devices

Devices	Description	Price (RM/unit)	Unit	Total Price (RM)
 Figure 3.2.1 Router	<u>TP-Link Archer AX73</u> Dual-band Wi-Fi 6 router: Supports the latest Wi-Fi 6 technology for faster speeds and better performance. Speeds of up to 5400 Mbps: Provides enough bandwidth for even the most demanding applications. Connects over 200 devices: Can handle many connected devices without sacrificing performance. HomeShield security: Protects your network from malware and other threats. Processor and memory: 1.5 GHz Triple-Core CPU, 512 MB of RAM, 1x Gigabit WAN port, and 4x Gigabit LAN ports. Processor and memory: Powered by a	484.00	6	2904.00

	1.7 GHz quad-core CPU with 512 MB RAM. Beamforming Technology: Enhances Wi-Fi coverage and focuses signals toward connected devices for a stronger and more reliable connection. Cooling Design: Features an advanced heat sink and vented panels for effective heat dissipation during heavy use.			
 Figure 3.2.2 Switches	<u>FS S3400-24T4FP</u> Power and Capacity: Up to 24 PoE+ ports with a total power budget of 370W, sufficient for powering high-demand devices. Reliable single power supply and cooling via 3 built-in fans. Performance: Advanced Layer 2 and Layer 3 functions, including ERPS, OSPF, RIP, and L2 multicast support. Security: Comprehensive security features like SSH, ACL, AAA, 802.1X, RADIUS, and TACACS+.	2033.00	7	14231.00

	Monitoring and Operation: Supports sFlow for enhanced network monitoring and analysis. Flexible operation through WEB, CLI, SNMP, and SSH interfaces. Dual-Firmware Support: Ensures a backup firmware version is available for safe upgrades and rollback in case of issues. Redundancy Features: Supports VRRP (Virtual Router Redundancy Protocol) for high availability and improved fault tolerance.			
 Figure 3.2.3 Patch Panel	<u>Linkbasic Cat6 Patch Panel</u> Design: Mounts to any standard 19" rack or cabinet, with a high-density 1U form factor to save space. Material: Made of black electrostatic powder-coated steel, corrosion-resistant with sturdy aluminum plate around RJ45 jacks. Wiring Compatibility: Supports T568A and	104.00	6	624.00

	T568B wiring patterns, meeting Cat 6 standards. Ease of Installation: Color-coded wiring for easy setup and accepts 22, 24, 26 AWG solid wires. Durability: Corrosion-resistant steel and fire self-extinguishing high-impact plastic for safe use. Cable Management Bar: Includes an integrated rear cable management bar to organize and secure cables efficiently. High Density: Offers support for up to 48 ports in a 1U configuration for larger network needs.			
 Figure 3.2.4 Wireless Access Point	<u>Ubiquiti UniFi 6 Lite</u> Wi-Fi Standard: 802.11a/b/g/n/ac/ax (Wi-Fi 6) Max Data Rate: 300 Mbps (2.4 GHz), 1.2 Gbps (5 GHz) Power: Powered via PoE (48V, 0.32A), Max. power consumption: 12W Max Clients: 300+	595.00	6	3570.00

	Security: WPA3, WPA2, WPA-PSK, WPA-Enterprise Mounting: Wall/ceiling (included) Certifications: CE, FCC, IC Network Interface: 1 GbE RJ45 port Environment Flexibility: Compact and sleek design suitable for modern interiors and tight spaces. Energy Efficient: Low power consumption for eco-conscious operations.			
 Figure 3.2.5 Cable	FS Cat6 Snagless Unshielded (UTP) Ethernet Network Patch Cable Cable Type: 20ft (6.1m) Cat6 24AWG UTP PVC CM Ethernet Patch Cable, Blue Standards: Exceeds ANSI/TIA-568.2-D Category 6 and ISO 11801 Class E Speed: Supports up to 1G/10G-T, 550 MHz	75.90/10.7m	300	2128.04

	Power: Compatible with PoE/PoE+/PoE++ (IEEE 802.3af/at/bt) Build: Stranded pure bare copper, 50μ" gold-plated RJ45 connectors Features: Strain relief, easy unplug clip protector, tested with Fluke Channel Test and wired T568B Noise Reduction: Twisted pair design minimizes crosstalk for better signal quality. Temperature Resistance: Operates in a wide temperature range for various environments. Enhanced Durability: Snaggles boot design protects connectors from damage during installation or removal.			
 Figure 3.2.6 Firewall	<u>FS NSG-5100 Next-Generation Firewall</u> Threat Detection: L2-L7 comprehensive detection and prevention Application Control: Identify, allow, block, or limit applications	21,849.48	2	43698.96

	VPN Services: Supports IPSec VPN, SSL VPN, and L2TP VPN Security Features: Includes IPS, URL filtering, antivirus, attack defense, and QoS (subscription required) Performance: Up to 2.5 Gbps firewall throughput Smart DNS: Inbound DNS for intelligent link load balancing High Availability: Supports failover clustering for uninterrupted service during hardware or network failures. Cloud Management: Offers remote management and monitoring capabilities via a cloud-based platform. Device Identification: Detects and classifies devices for more precise security policies.			
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Figure 3.2.7 Personal Computer (PC)

 Figure 3.2.7 Personal Computer (PC)	<u>HP EliteDesk 800 G9 Desktop</u> Processor: Up to 13th Gen Intel® Core™ i9 processors Memory: Up to 64GB DDR5 memory Storage: Up to 8TB NVMe SSD - Operating System: Windows 11 Pro, Windows 10 Pro Form Factor: Tower, Small Form Factor Security: HP Wolf Security for Business Manageability: HP Intelligent Management Suite Cooling Technology: Advanced thermal management system ensures efficient heat dissipation. Eco-Friendly: ENERGY STAR® certified, reducing power consumption and environmental impact. Upgradeable Design: Tool-less access for quick hardware	3895.00	130	506350.00
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	upgrades or replacements.			
 Figure 3.2.8 Projector	Epson EB-X06 XGA 3LCD Projector Technology: 3LCD for vibrant, true-to-life color brightness Resolution: Native XGA, 4:3 aspect ratio Brightness: 3,600 lumens Lamp Life: Up to 12,000 hours in ECO Mode Connectivity: HDMI, built-in 2W speaker Features: Easy setup, flexible positioning for classroom use. Keystone Correction: Automatic vertical and horizontal adjustments for a perfectly aligned image. Split-Screen Capability: Allows simultaneous projection of content from two sources. Compact and Lightweight: Easy to transport and set up in different locations.	1809.00	6	10854.00

 Figure 3.2.9 Projector Screen	Elite Screens Manual B Size: 100-inch diagonal with 16:9 aspect ratio, providing a viewing area of 49.2" H x 87.4" W. Material: MaxWhite, compatible with UHD/HD projectors, offering a 180° viewing angle and 1.1 gain. Features: Auto-lock mechanism for height adjustments, easy installation, and optional L brackets. Warranty: 2-Year/3-Year ENR-G Warranty and lifetime tech support. Black Backing Material: Blocks light penetration for sharper image quality. Smooth Retraction System: Ensures the screen retracts evenly and securely into the housing. Versatile Mounting Options: Compatible with wall, ceiling, or in-ceiling installations.	414.78	2	829.56
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Figure 3.2.10 Multifunctional Whiteboard

Figure 3.2.10 Multifunctional Whiteboard	<u>SMART Board</u> <u>MX075-V5</u> <u>Collaboration Display</u> <u>(MX275-V5)</u> Size: 75" 4K UHD LCD Touchscreen: 32 touchpoints with HyPr Touch Technology Processor: Octa-core, 8GB RAM, 64GB storage Connectivity: HDMI (4), USB (12), Gigabit Ethernet, Wi-Fi 6, Bluetooth 5.2 Audio: 20W speakers, 6-microphone array OS: Android 13, compatible with ChromeOS, Mac OS, Windows Power: 103W, Energy Star rated Warranty: 3 years Screen Sharing: Supports wireless screen mirroring for up to 9 devices simultaneously. Integrated Whiteboard App: Allows instant brainstorming without	16502.00	6	99012.00
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	requiring a connected device. Energy-Efficient Design: Auto-shutdown and eco-mode reduce power consumption when not in use.			
 Figure 3.2.11 Standing Microphone	<u>Audio-Technica AT2020</u> Type: Side-address condenser, cardioid polar pattern Frequency Response: 20Hz–20kHz Max SPL: 144dB Self Noise: 20dB Impedance: 100 ohms Connector: XLR Accessories: Stand mount, pouch Power: Requires +48V phantom power Features: Low-mass diaphragm for wide dynamic range, high SPL handling, low self-noise, pivoting mount for precise placement Wide Compatibility: Works seamlessly with audio interfaces, mixers, and	390.00	1	390.00

	portable recording systems. Compact Design: Lightweight and portable, ideal for field recording or podcasting. Shock Mount Compatibility: Optional shock mount for better isolation and vibration reduction.			
 Figure 3.2.12 Video Camera	<u>Canon Eos M50 Mark II</u> Sensor: 24MP APS-C CMOS Processor: DIGIC 8 Viewfinder: Electronic Display: Vari-angle Autofocus: Dual Pixel CMOS AF Shooting Speed: 10 fps continuous Video: 4K recording Connectivity: Wi-Fi, Bluetooth Features: Compact design, ideal for video and photography in various settings. Live Streaming Ready: Native YouTube streaming support and	2968.00	1	2968.00

	clean HDMI output for high-quality broadcasts. Silent Shooting Mode: Perfect for capturing moments in noise-sensitive environments. Eye and Face Detection AF: Improved focus tracking for sharp portraits and videos.			
 Figure 3.2.13 Video Conference Terminal	<u>Logitech Rally Plus</u> Imaging System: Ultra-HD Zoom: 15x HD zoom Pan/Tilt: ±90° Pan, +50° / -90° Tilt Presets: 3 camera presets Security: Kensington slot Indicators: Video mute/unmute LED Mount: Standard tripod thread Privacy Feature: Lens parks at -90° during sleep mode Warranty: 2 Years Logitech Manufacturer Warranty	16099.00	1	16099.00

	Features: Autofocus, motorized pan and tilt for smooth movement. 4K Video Processing: Supports up to 4K resolution for crystal-clear conferencing. Plug-and-Play Setup: Works with most video conferencing platforms, including Zoom, Microsoft Teams, and Google Meet. RightSound Technology: Enhances voice clarity while suppressing background noise.			
 Figure 3.2.14 IoT Hardware Board	<u>Raspberry Pi 4 Model B - 4GB</u> Processor: Broadcom BCM2711, Quad-core ARM Cortex-A72 @ 1.5GHz RAM: 4GB LPDDR4 Ports: 2 x USB 3.0, 2 x USB 2.0, 2 x micro HDMI (4Kp60), 27 x GPIO Networking: Gigabit Ethernet, 2.4GHz/5GHz Wi-Fi, Bluetooth 5.0 Storage: microSD	259.00	10	2590.00

	Power: 5V/3.0A via USB Type-C OS Support: Raspberry Pi OS, third-party OS Dual-Display Support: Drives two 4K monitors simultaneously. Expandable Storage: Supports USB-attached storage devices for additional capacity. Overclocking Capabilities: Allows advanced users to boost performance.			
 Figure 3.2.15 Printer	<u>HP LaserJet Pro MFP M428fdw</u> Print Speed: 40 ppm (up to 42 ppm with HP High Speed) Resolution: 1200 x 1200 dpi Print Technology: Laser Display: 2.7" color touchscreen Paper Capacity: 250-sheet input tray, 150-sheet output bin Connectivity: USB 2.0, Gigabit Ethernet, Wi-Fi,	2750.00	2	5500.00

	Apple AirPrint, Google Cloud Print Security: Secure Boot, encrypted data storage, SNMPv3 Power: 510W active, 7.5W ready Duplex Printing: Automatic two-sided printing saves time and paper. Fast Scanning: Dual-head scanner for simultaneous two-sided scanning. Mobile Printing: Supports HP Smart App, Apple AirPrint, and Google Cloud Print for easy printing from smartphones and tablets.			
 Figure 3.2.16 Modem	<u>NETGEAR CM1200 DOCSIS 3.1 Cable Modem</u> DOCSIS 3.1 Technology: Provides faster speeds, improved efficiency, and better latency for Gigabit internet. Multi-Gigabit Ethernet Ports: 4 Gigabit Ethernet ports for multiple wired devices.	847.30	2	1694.60

	Speeds up to 2 Gbps: Ideal for heavy users, gaming, and 4K streaming. Backward Compatibility: Works with DOCSIS 3.0 systems. IPv6 Support: Future-ready internet addressing. Advanced Security: Built-in DoS protection, firewall, and encryption for home network safety. Multi-IP Address Support: Ideal for households or businesses needing multiple static IPs. Passive Cooling Design: Operates quietly without fans. 24x8 Channel Bonding: Ensures better bandwidth management for stable internet.			
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Figure 3.2.17 Server

 Figure 3.2.17 Server	<u>Dell PowerEdge T550 Tower Server</u> Processor: Intel Xeon Silver 4309Y (8C/16T, 2.8 GHz, 12M Cache) Memory: 16GB 3200MT/s RDIMM Storage: 600GB 10K RPM SAS, 2.5" Hot-plug, 3.5" HYB CARR RAID: PERC H755 RAID Controller Networking: Dual port 1Gb + Quad port 1GbE Broadcom Adapter Power Supply: Dual, Hot-plug, Redundant 800W Management: iDRAC9 Enterprise Hot-Swap Drives: Supports up to 8 hot-swappable drives for quick replacements and upgrades. PCIe Gen 4 Support: Doubles data throughput for high-performance peripherals. Flexible Scalability: Modular design for memory and	18402.00	5	92010.00
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	storage expansions as needs grow.			
 Figure 3.2.18 Power Backup Supply System	<u>APC Back-UPS Pro 1000VA</u> 8 Total Outlets: 4 provide battery backup and surge protection, 4 provide surge protection only. Automatic Voltage Regulation (AVR): Ensures safe voltage levels without using battery power. Surge Protection: Ideal for computers, hard drives, and electronics. Intelligent Battery Management: Optimizes battery performance and life. Input Voltage Range: 88V - 148V for main operations LCD Display: Provides real-time information on battery and power status. User-Replaceable Batteries: Extend the UPS's lifespan with easy battery swaps. Data Protection: Includes software for automatic file saving during outages.	924.00	20	18480.00

Total	823933.16
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Table of Number of Connections

Area	Number of Switch Ports Needed
General Lab 1	32
General Lab 2	32
Cisco Network Lab	33
IoT Lab	33
Video Conference Room	3
Office	27
Meeting Room	1
Staff Meeting Room	1
Server Room	6
Total	168

Table 4.3.1 Number of Switch Ports Needed

Area	Number of Patch Cords Needed
General Lab 1	32
General Lab 2	32
Cisco Network Lab	33
IoT Lab	33
Video Conference Room	3
Office	27
Meeting Room	1
Staff Meeting Room	1
Server Room	6
Total	168

Table 4.3.2 Number of Patch Cords Needed

Area	Number of RJ45 Connector Needed
General Lab 1	64
General Lab 2	64
Cisco Network Lab	66
IoT Lab	66
Video Conference Room	6
Office	54
Meeting Room	2
Staff Meeting Room	2
Server Room	12
Total	336

Table 4.3.3 Number of RJ45 Connector Needed

Table of Cable Type and Length

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
Room				
Hybrid Classroom	31.7	6.2	37.9	268.84
Video Conference Room	30.2	5.3	35.5	251.82
Staff Meeting Room	8.8	1.2	10.0	70.93
Ground Floor Room Total			83.4	591.59
First Floor				
Room				
General Lab 1	385.6	76.2	461.8	3275.76
General Lab 2	355.3	40.7	396.0	2809.01
Cisco Network Lab	328.2	15.3	343.5	2436.60
Embedded Lab	649.5	95.2	744.7	5282.50
Server Room	81.1	16.5	97.6	682.32
First Floor Room Total			2043.6	14486.19

Table 4.4.1: Number of Cable Types and Length

Ground Floor		
Path	Length (m)	Price of cables used (RM)
Server Room to Hybrid Classroom	29.9	212.09
Server Room to Video Conference Room	25.5	180.88
Ground Floor Room Total	55.4	392.97
Total - Ground Floor		
Total Length of Wire to Connect the Room (m)	55.4	392.97
Total Length of Wire in Horizontal and Vertical (m)	83.4	591.59
Total Length of Wire Needed (m)	138.8	984.56
First Floor		
Server Room to General Lab 1	64.8	459.66
Server Room to General Lab 2	64.8	459.66
Server Room to Cisco Network Lab	64.8	459.66
Server Room to Embedded Lab	64.8	459.66
First Floor Room Total	259.2	1838.64
Total - First Floor		
Total Length of Wire to Connect the Room (m)	259.2	1838.64
Total Length of Wire in Horizontal and Vertical (m)	2043.6	14486.19
Total Length of Wire Needed (m)	2302.8	16324.83
Total (Ground Floor + First Floor) (m)	2441.6	16717.80

Table 4.4.: Total Number of Cable Types and Length